FSA Exercise Book

Workbook edition — with space to write and a worked example for every type

Financial Statement Analysis · Prof. Stefano Bianchi
134 exercises (128 by topic plus six full papers) · 67 new worked examples
Exercise numbers are unchanged: solutions in the companion file *FSA 6 — Exercise Solutions*

HOW EVERY TYPE IS LAID OUT

① **Worked example — solved.** New data, full method, every intermediate figure.

②③ **Your turn.** Two exercises from the book, each followed by its own writing space.

How to use this book

Every exercise is written the way the exam writes it: a block of data, then *Required*. The book is now organised by **type of exercise**. Each type opens with a **worked example** on new figures — read it, understand the order of the steps — and then gives you **two exercises from the original book** to work on your own, in the space provided. Only then open the solutions file.

Three rules carried over from the lecturer's own answers, because they are worth marks on their own:

- When the question asks for a textbook ratio, name the substitution before computing — *"Operating profit margin (instead of Profit margin)"* — then the formula, then the assumptions, then the arithmetic.
- State the tax rate you assumed (30 per cent unless the question gives one) and what you treated as operating.
- If only one balance sheet date is given, write *"Can we calculate the average? No, we can't"* and use the year-end figure.

Use 360 days. Finish every quantitative answer with two or three lines of comment.

The writing space

Grid

For calculations, schedules and statements. Draw your own columns; one square is 5 mm.

Ruled lines

For comments, explanations and written answers. The number of lines is the length the question asks for.

Blank tables

Pre-drawn answer tables (plans, ratios, horizontal and common-size columns): fill in the empty cells.

Answer boxes

Multiple choice and true/false: write the letter, or tick T or F.

The space is sized to the question — two lines for a one-line answer, most of a page for a statement of cash flows. Whatever is left at the bottom of a page is added to the last exercise on it, so you always have room to finish.

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SECTION A

Statements, formats and classification

Ten drills on what goes where. These come straight from Exercise Session 1 and from the accounting-topics class, and they are the cheapest marks on the paper.

TYPE A.1

Current or non-current: classifying balance sheet items

The twelve-month test, contra accounts and equity items.

① Worked example → ② A1 → ③ A2

WORKED EXAMPLE · SOLVED

Classify ten balance sheet items

Classify each item of Ostro Srl as a current or non-current asset, a current or non-current liability, equity, or a contra account — and say in a few words why.

Item	Item
Raw materials	Trade receivables due in 60 days
Software licences (5-year)	Loan to a supplier, repayable in 3 years
Prepaid rent for the next six months	Unearned revenue (customer deposits)
Bank loan, final instalment in 4 years	Warranty provision, used next year
Allowance for doubtful accounts	Legal reserve

Step 1 — Apply the twelve-month test

RULE An item is **current** if it will be realised, consumed or settled within twelve months (or the operating cycle, if longer). Everything else is **non-current**. Contra accounts are neither assets nor liabilities: they reduce the account they belong to.

Step 2 — Classify

Item	Classification	Why
Raw materials	**Current asset**	consumed within the operating cycle
Trade receivables, 60 days	**Current asset**	collected within twelve months
Software licences (5-year)	**Non-current asset**	intangible, used over five years
Loan to a supplier, 3 years	**Non-current asset**	collected after twelve months
Prepaid rent, six months	**Current asset**	service received within twelve months
Unearned revenue	**Current liability**	owed in goods or services next year
Bank loan, 4 years	**Non-current liability**	the instalment due next year is a current liability
Warranty provision, next year	**Current liability**	provision expected to be used within twelve months
Allowance for doubtful accounts	**Contra-asset**	deducted from receivables, not a liability
Legal reserve	**Equity**	retained profit set aside by law, not a liability

Comment

Look for three traps: the **current portion** of a long-term loan, **contra accounts** (accumulated depreciation, allowance, treasury shares) and **equity reserves**, which are never liabilities.

A1 Current / non-current classification CLASSIFICATION

YOUR TURN

Classify each of the following balance sheet items as a current or a non-current asset, or as a current or a non-current liability.

Item	Classification	Item	Classification
Inventories		Marketable securities	
Goodwill		Deferred tax assets (long-term)	
Short-term investments		Accrued expenses	
Prepaid insurance		Provisions	
Accounts receivable		Income tax payables	
Land		Long-term borrowings	
Construction in progress		Accounts payable	
Investments in associates		Current portion of long-term borrowings	
Machinery		Bonds	
Cash and cash equivalents		Advances received from customers	

✎ NOTES

A2 Classification including equity items CLASSIFICATION

YOUR TURN

Classify each item below. Three of them are neither assets nor liabilities — say what they are.

Item	Classification	Item	Classification
Patents		Short-term derivative liabilities	
Cash at bank		Employee benefit liabilities	
Property, plant and equipment		Dividends payable	
Current income tax assets		Long-term obligations under leases	
Trademarks		Bank overdraft	
Merchandise inventory		Notes payable due in 20 months	
Financial assets held to maturity (8 years)		Deferred tax liabilities (long-term)	
Notes receivable due in 5 months		Retained earnings	
Accumulated depreciation		Share premium (paid-in capital in excess of par)	
Treasury shares		Wages payable	

✎ THE ITEMS THAT ARE NEITHER ASSETS NOR LIABILITIES

TYPE A.2

Operating versus financing, and the classified balance sheet

Sorting every balance into its block, and splitting liabilities by the activity that created them.

① Worked example → ② A3 → ③ A7

WORKED EXAMPLE · SOLVED

A classified balance sheet, with operating and financing liabilities

The balances of Velia Srl at 31 December, in thousands, are listed below. Prepare a classified balance sheet in horizontal format and mark each liability as operating (O) or financing (F).

Cash	60	Accounts payable	140
Accounts receivable	210	Accrued wages	25
Inventories	180	Unearned revenue	30
Prepaid expenses	15	Bank overdraft	40
Equipment	900	Bank loan, due in 5 years	350
Accumulated depreciation	(300)	Share capital	400
Trademarks	75	Retained earnings	155

Step 1 — Sort every balance into its block

Assets: non-current (equipment net of accumulated depreciation, trademarks) and current (inventories, receivables, prepaid, cash). Financing side: equity, non-current liabilities, current liabilities. Accumulated depreciation is **not** a liability: it is netted against equipment.

Step 2 — Lay out the two sides and check they balance

Assets	**€ 000**	**Equity and liabilities**	**€ 000**
Non-current assets			
Equipment	900	Share capital	400
Less accumulated depreciation	(300)	Retained earnings	155
Equipment, net	600	**Total equity**	**555**
Trademarks	75	Bank loan, 5 years **(F)**	350
Total non-current assets	**675**	**Total non-current liabilities**	**350**
Current assets / current liabilities			
Inventories	180	Accounts payable **(O)**	140
Accounts receivable	210	Accrued wages **(O)**	25
Prepaid expenses	15	Unearned revenue **(O)**	30
Cash	60	Bank overdraft **(F)**	40
Total current assets	**465**	**Total current liabilities**	**235**
Total assets	**1,140**	**Total equity and liabilities**	**1,140**

Step 3 — Why operating versus financing matters

Operating liabilities (suppliers, employees, customers, tax authority) arise from running the business and carry no explicit interest; financing liabilities (overdraft, loans, bonds, leases) are borrowed money. In chapter 8 the first group is netted against operating assets to give **NOA**, the second goes into **NFO**; mixing them distorts RNOA, leverage and the net financial rate.

Comment

Total assets 1,140 = total equity and liabilities 1,140: always close with the check.

A3 Operating versus financing liabilities CLASSIFICATION

YOUR TURN

Chapter 3 splits liabilities by the activity that creates them, and Chapter 8 then uses that split to build net operating assets. Classify each item as an operating or a financing liability, and say in one line why the distinction matters.

Liability	Operating or financing
Accounts payable	
Bank loan, 5 years	
Unearned revenue	
Bonds payable	
Taxes payable	
Current portion of long-term debt	
Accrued wages	
Finance lease obligation	
Postretirement liabilities	
Short-term notes payable to a bank	

✎ WHY THE DISTINCTION MATTERS

A7 Build a classified balance sheet FORMAT

YOUR TURN

From the balances below, prepare a classified balance sheet in horizontal format for Delaro NV at 31 December. All amounts in thousands.

Cash	140	Accounts payable	310
Accounts receivable, net	420	Accrued liabilities	95
Inventories	515	Income taxes payable	60
Prepaid expenses	35	Bank loan, due in 4 years	500
Equipment	1,180	Bonds payable	300
Accumulated depreciation	(390)	Share capital	700
Patents	180	Retained earnings	115

✎ CLASSIFIED BALANCE SHEET — DELARO NV, 31 DECEMBER

Assets	€ 000	Equity and liabilities	€ 000

✎ EXTRA SPACE

TYPE A.3

From T-format to multi-step

Find net income as the balancing figure, then rebuild the statement layer by layer.

① Worked example → ② A4 → ③ A5

WORKED EXAMPLE · SOLVED

From T-format to multi-step

The income statement of Sorel Srl is presented in T-format. Rewrite it in the multi-step (two-step) format.

Left side / debit		Right side / credit	
Cost of sales	240	Revenue	380
Marketing expenses	30	Rental income (spare warehouse)	6
General and administrative	42		
Depreciation expense	18		
Interest expense	12		
Income tax expense	17		

Step 1 — Find net income: the balancing figure of the T

Total credits	380 + 6	**386**
Total debits	240 + 30 + 42 + 18 + 12 + 17	**359**
Net income	386 − 359	**27**

Step 2 — Rebuild it in steps

Sorel Srl — Income statement	€
Revenue	380
Cost of sales	(240)
Gross profit	**140**
Marketing expenses	(30)
General and administrative expenses	(42)
Depreciation expense	(18)
Operating profit	**50**
Other income: rental income	6
Interest expense	(12)
Income before tax	**44**
Income tax expense	(17)
Net income	**27**

Comment

Only sales and cost of sales sit above gross profit. Rental income is not part of sales, so it goes below operating profit as other income (state your choice). The two routes must give the same net income: 27.

A4 From T-format to multi-step FORMAT

YOUR TURN

The income statement of Avelli Srl for the year is presented in T-format. Rewrite it in the two-step (multi-step) horizontal format.

Left side / debit		Right side / credit	
Cost of sales	128	Revenue	186
R&D expenses	11		
SG&A expenses	24		
Depreciation expense	6		
Interest expense	5		
Income tax expense	4		

✎ NET INCOME CHECK

✎ MULTI-STEP STATEMENT

Avelli Srl — Income statement	€

✎ EXTRA SPACE

A5 Multi-step with a second revenue line FORMAT

YOUR TURN

Rewrite the following T-format statement of Kenwood Ltd in two-step format. Rental revenue is not part of sales.

Left side / debit		Right side / credit	
Cost of goods sold	412	Sales	690
Marketing expenses	58	Rental revenue	9
SG&A expenses	96		
Depreciation expense	37		
Interest expense	18		
Income tax expense	23		

✎ NET INCOME CHECK

✎ MULTI-STEP STATEMENT

Kenwood Ltd — Income statement	€

✎ EXTRA SPACE

TYPE A.4

Multi-step and single-step from a list

Placing each line in the right layer, and knowing which format the analyst needs.

① Worked example → ② A6 → ③ A9

WORKED EXAMPLE · SOLVED

Multi-step and single-step from a scrambled list

The balances of Tavola SpA for the year, in thousands, are listed in no particular order. Prepare the income statement in multi-step and in single-step format, and say which one an analyst needs.

Net sales	860	Cost of goods sold	510
Selling expenses	95	Administrative expenses	70
Interest expense	14	Loss on sale of equipment	8
Income tax expense	49	Depreciation	25

Step 1 — Multi-step: one subtotal per layer

Tavola SpA — multi-step	€ 000
Net sales	860
Cost of goods sold	(510)
Gross profit	**350**
Selling expenses	(95)
Administrative expenses	(70)
Depreciation	(25)
Operating profit	**160**
Loss on sale of equipment	(8)
Interest expense	(14)
Income before tax	**138**
Income tax expense	(49)
Net income	**89**

✎ EXTRA SPACE

Step 2 — Single-step: all revenues, then all expenses

Tavola SpA — single-step	€ 000
Net sales	860
Expenses	
Cost of goods sold	510
Selling expenses	95
Administrative expenses	70
Depreciation	25
Loss on sale of equipment	8
Interest expense	14
Income tax expense	49
Total expenses	**(771)**
Net income	**89**

Comment

The analyst needs the **multi-step** format: it shows gross profit and operating profit, the inputs of gross margin, operating margin, NOPAT and times interest earned. The single-step format reaches the same 89 but hides them.

✎ EXTRA SPACE

A6 Build the statement from a scrambled list FORMAT

YOUR TURN

The following balances, in thousands, belong to Mirani SpA for the year ended 31 December. Prepare a multi-step income statement.

Interest expense	26	Net sales	1,240
Income tax expense	61	Cost of goods sold	806
Selling expenses	142	Administrative expenses	95
Depreciation and amortisation	48	Gain on sale of equipment	12

✎ MULTI-STEP STATEMENT

Mirani SpA — Income statement	€ 000

✎ EXTRA SPACE

A9 Single-step and multi-step, same data FORMAT

YOUR TURN

Pell Company reports net sales 940, cost of goods sold 611, selling and administrative expenses 188, interest expense 21 and income tax expense 36. Prepare the income statement in both the single-step and the multi-step format, and say in two lines which one a financial analyst needs and why.

✎ SINGLE-STEP

Single-step	€

✎ MULTI-STEP

Multi-step	€

✎ WHICH ONE THE ANALYST NEEDS, AND WHY

TYPE A.5

Missing subtotals and errors

Recovering figures from the identities, and testing every line against its layer.

① Worked example → ② A8 → ③ A10

WORKED EXAMPLE · SOLVED

Missing subtotals, and a statement with errors

(a) Complete the statement of Lodi Spa: net sales ?, cost of goods sold (720), gross profit 480, operating expenses ?, operating profit 180, interest expense (30), income before tax ?, income tax expense (30%) ?, net income ?.

(b) A junior analyst prepared the statement below for Brera Srl (tax rate 30 per cent). Find the errors and redraft it.

Net sales	900
Cost of goods sold	(540)
Gross profit	**360**
Administrative expenses	(120)
Dividends paid	(40)
Operating profit	**200**
Interest expense	(20)
Income before tax	**180**
Income tax expense (30%)	(60)
Net income	**120**

Step 1 — (a) Walk the statement, one identity at a time

Net sales	Gross profit + COGS = 480 + 720	**1,200**
Operating expenses	Gross profit − operating profit = 480 − 180	**300**
Income before tax	180 − 30	**150**
Income tax expense	30% × 150	**45**
Net income	150 − 45	**105**

Step 2 — (b) Test every line against its layer

- **Error 1** — *Dividends paid* are a distribution of profit to shareholders, not an expense: they go to the statement of changes in equity, never in the income statement.
- **Error 2** — the tax charge (60) is 30% of operating profit; tax is computed on **income before tax**.

✎ EXTRA SPACE

Step 3 — Redraft

Brera Srl — corrected	€
Net sales	900
Cost of goods sold	(540)
Gross profit	**360**
Administrative expenses	(120)
Operating profit	**240**
Interest expense	(20)
Income before tax	**220**
Income tax expense (30% × 220)	(66)
Net income	**154**

Comment

Checklist: only sales and cost of sales above gross profit; only operating expenses between gross and operating profit; interest and non-operating gains or losses below it; tax on income before tax; no dividends anywhere.

A8 Fill the missing subtotals FORMAT

YOUR TURN

Complete the income statement of Hoven Industries. Each question mark is recoverable from the others.

Hoven Industries	Given	Your figure
Net sales	?	
Cost of goods sold	(1,452)	
Gross profit	968	
Operating expenses	?	
Operating profit	311	
Interest expense	(44)	
Income before tax	?	
Income tax expense (30%)	?	
Net income	?	

✎ WORKING

A10 Find the four errors FORMAT

YOUR TURN

A junior analyst has prepared the statement below. Identify the errors and redraft the statement correctly.

Net sales	1,500
Gain on disposal of a warehouse	40
Cost of goods sold	(900)
Gross profit	**640**
Selling expenses	(210)
Interest expense	(30)
Income tax expense	(60)
Operating profit	**340**
Net income	**340**

✎ THE FOUR ERRORS

✎ REDRAFT

Corrected statement	€

✎ EXTRA SPACE

SECTION B

Depreciation, amortisation and disposals

The depreciation-method spreadsheet of Exercise Session 1 in four shapes, plus the plans and the age measures from chapter 4.

TYPE B.1

Straight-line plans and disposal at the end of life

Annual charge, the plan table, and gain or loss when the asset is sold at the end.

① Worked example → ② B1 → ③ B2

WORKED EXAMPLE · SOLVED

A straight-line plan, then a disposal at the end of life

On 1 January Year 1, Pisani Spa buys a machine for €50,000. Useful life 4 years, residual value €6,000. (a) Prepare the straight-line depreciation plan. (b) At the end of Year 4 the machine is sold — in scenario (a) for €7,500, in scenario (b) for €4,200. Compute the gain or loss in each scenario.

Step 1 — The annual charge

Annual depreciation = (Cost − Residual value) ÷ Useful life = (50,000 − 6,000) ÷ 4 = **€11,000**

Step 2 — The plan

Year	Annual depreciation	Accumulated depreciation	Book value
1 Jan Year 1	—	—	50,000
Year 1	11,000	11,000	39,000
Year 2	11,000	22,000	28,000
Year 3	11,000	33,000	17,000
Year 4	**11,000**	**44,000**	**6,000**

Step 3 — Disposal at the end of life

At the end of its life the book value equals the residual value, €6,000.

(a) Sold for €7,500	7,500 − 6,000	**Gain €1,500**
(b) Sold for €4,200	4,200 − 6,000	**Loss €1,800**

Comment

Gain or loss = proceeds − book value. A gain means the residual value was estimated too low (depreciation was too high); a loss means the opposite. Neither is cash: the cash is the full proceeds.

✎ EXTRA SPACE

B1 Straight-line depreciation plan DEPRECIATION

YOUR TURN

On 1 January Year 1, Rendo Spa buys a machine for €96,000. The expected useful life is 5 years and the estimated residual value is €6,000. Prepare the full depreciation plan under the straight-line method, showing annual expense, accumulated depreciation and book value.

✎ ANNUAL CHARGE

✎ PLAN

Year	Annual depreciation	Accumulated depreciation	Book value
1 Jan Year 1			
Year 1			
Year 2			
Year 3			
Year 4			
Year 5			

B2 Straight-line plan and disposal at the end of life DISPOSAL

YOUR TURN

Use the machine of exercise B1. At the end of Year 5 Rendo sells it, in scenario (a) for €9,500 and in scenario (b) for €3,800. Compute the gain or loss on disposal in each scenario.

✎ WORKING

✎ COMMENT

TYPE B.2

Double-declining-balance and units-of-activity plans

Accelerated and usage-based depreciation, and the residual value as the floor.

① Worked example → ② B3 → ③ B4

WORKED EXAMPLE · SOLVED

Double-declining-balance and units-of-activity

(a) Sala Ltd buys equipment for €40,000, useful life 4 years, residual value €4,000. Prepare the double-declining-balance plan. **(b)** A truck costs €62,000, residual value €2,000, expected total use 300,000 km. Actual use: Year 1 80,000 km, Year 2 100,000 km, Year 3 70,000 km, Year 4 50,000 km. Prepare the units-of-activity plan.

Step 1 — (a) DDB: double the straight-line rate, applied to the opening book value

Rate = 2 × (1 ÷ 4) = 50% · the residual value is **ignored** in the rate but is the floor for the book value

Year	Opening book value	Rate	Depreciation	Accumulated	Closing book value
Year 1	40,000	50%	20,000	20,000	20,000
Year 2	20,000	50%	10,000	30,000	10,000
Year 3	10,000	50%	5,000	35,000	5,000
Year 4	**5,000**	**plug**	**1,000**	**36,000**	**4,000**

Year 4 at 50% would give 2,500 and a book value of 2,500, below the residual value: the last charge is the plug 5,000 − 4,000 = **1,000**.

Step 2 — (b) Units of activity: a rate per kilometre

Rate = (62,000 − 2,000) ÷ 300,000 km = **€0.20 per km**

Year	Kilometres	× Rate	Depreciation	Accumulated	Book value
Year 1	80,000	0.20	16,000	16,000	46,000
Year 2	100,000	0.20	20,000	36,000	26,000
Year 3	70,000	0.20	14,000	50,000	12,000
Year 4	**50,000**	**0.20**	**10,000**	**60,000**	**2,000**

Comment

DDB front-loads the expense; units of activity follows use. Both end at the residual value: check the last book value.

✎ EXTRA SPACE

B3 Double-declining-balance plan DEPRECIATION

YOUR TURN

Vandel Ltd buys equipment for €80,000 with a 5-year useful life and an estimated residual value of €5,000. Prepare the depreciation plan under the double-declining-balance method.

✎ RATE

✎ PLAN

Year	Opening book value	Rate	Depreciation	Accumulated	Closing book value
Year 1					
Year 2					
Year 3					
Year 4					
Year 5					

B4 Units-of-activity method DEPRECIATION

YOUR TURN

A delivery fleet costs €150,000, has an estimated residual value of €10,000 and an expected total output of 400,000 kilometres. Actual use is Year 1: 95,000 km, Year 2: 110,000 km, Year 3: 80,000 km, Year 4: 70,000 km, Year 5: 45,000 km. Prepare the depreciation plan under the units-of-activity method.

✎ RATE PER KM

✎ PLAN

Year	Kilometres	× Rate	Depreciation	Accumulated	Book value
Year 1					
Year 2					
Year 3					
Year 4					
Year 5					

✎ EXTRA SPACE

TYPE B.3

Gain or loss on disposal

Book value from cost and accumulated depreciation, then compare with the proceeds.

① Worked example → ② B5 → ③ B6

WORKED EXAMPLE · SOLVED

Gain or loss from cost and accumulated depreciation

(a) Equipment that cost €30,000, with accumulated depreciation of €17,500, is sold for €10,000. **(b)** Equipment that cost €45,000, with accumulated depreciation of €41,000, is sold for €6,200. Compute the gain or loss.

Step 1 — Book value first, then compare with the proceeds

Book value = Cost − Accumulated depreciation · Gain (loss) = Proceeds − Book value

(a) Book value	30,000 − 17,500	**12,500**
(a) Result	10,000 − 12,500	**Loss €2,500**
(b) Book value	45,000 − 41,000	**4,000**
(b) Result	6,200 − 4,000	**Gain €2,200**

Comment

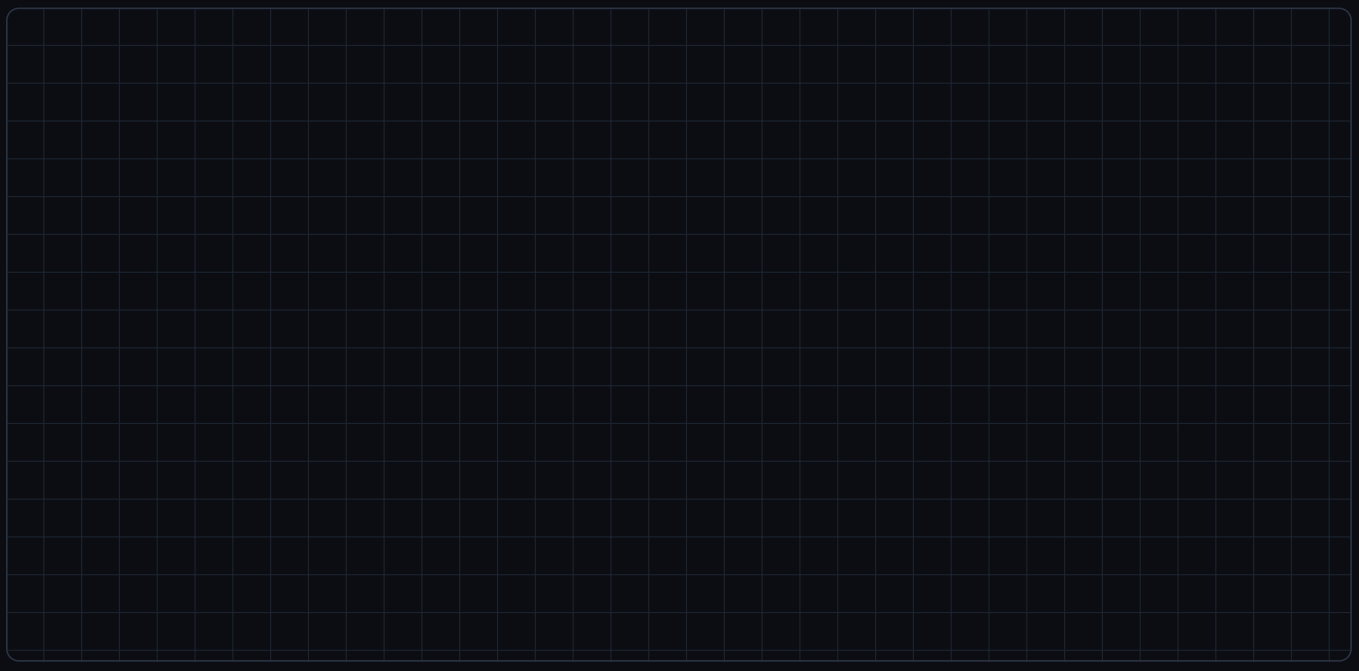
A nearly fully depreciated asset (b) easily produces a gain: the gain is not a sign of good management, only that past depreciation was heavier than the actual loss of value.

B5 Gain or loss from cost and accumulated depreciation DISPOSAL

YOUR TURN

During the year equipment was sold for €11,400 cash. The cost of the equipment was €26,000 originally and it had accumulated depreciation of €13,100 at the time of sale. Compute the gain or loss on the sale.

✎ WORKING



B6 Gain on a nearly fully depreciated asset DISPOSAL

YOUR TURN

Old equipment with an original cost of €64,000 and accumulated depreciation of €58,500 was sold for €14,200 cash. Compute the gain or loss.

✎ WORKING

TYPE B.4

Working backwards from the disposal

Recovering book value or accumulated depreciation from the gain or loss.

① Worked example → ② B7 → ③ B8

WORKED EXAMPLE · SOLVED

Working backwards through the disposal identity

(a) Equipment costing €36,000 with a book value of €14,000 is sold for €15,500. Compute the accumulated depreciation and the gain or loss. **(b)** Equipment costing €60,000 is sold for €21,000, producing a loss of €3,000. Compute the book value at the date of sale and the accumulated depreciation.

Step 1 — Two identities, three unknowns at most

Book value = Cost − Accumulated depreciation · Proceeds − Book value = Gain (+) or Loss (−)

Step 2 — Solve

(a) Accumulated depreciation	36,000 − 14,000	**22,000**
(a) Gain or loss	15,500 − 14,000	**Gain €1,500**
(b) Book value	Proceeds + loss = 21,000 + 3,000	**24,000**
(b) Accumulated depreciation	60,000 − 24,000	**36,000**

Comment

With a **gain** the book value is proceeds − gain; with a **loss** it is proceeds + loss. Write the identity first, then substitute — it avoids the sign error.

✎ EXTRA SPACE

B7 Work backwards to accumulated depreciation DISPOSAL

YOUR TURN

Old equipment costing €47,500 with a book value of €18,000 was sold for €12,750 cash. Compute the accumulated depreciation at the date of sale and the gain or loss.

✎ WORKING

B8 Work backwards from the gain DISPOSAL

YOUR TURN

Old equipment costing €52,000 was sold for €16,900, producing a gain of €6,400. Compute the book value at the date of sale and the accumulated depreciation.

✎ WORKING

TYPE B.5

Mid-year disposals and plant asset age

Depreciation by months to the date of sale; total life, age and remaining life.

① Worked example → ② B9 → ③ B10

WORKED EXAMPLE · SOLVED

A mid-year disposal, and the plant asset age measures

(a) Corti Srl bought a machine on 1 January Year 1 for €90,000, straight-line over 6 years, residual value €6,000. On 31 March Year 3 it is sold for €58,000. Compute the Year 3 depreciation to the date of sale, the accumulated depreciation at that date and the gain or loss.

(b) Fenni Group reports gross plant and equipment of €600,000, accumulated depreciation of €240,000 and depreciation expense for the year of €50,000. Compute the average total life span, average age and average remaining life.

Step 1 — (a) Depreciate up to the date of sale, by months

Annual charge	(90,000 − 6,000) ÷ 6	**14,000**
Year 3, 1 Jan – 31 Mar	14,000 × 3/12	**3,500**
Accumulated at date of sale	14,000 × 2 + 3,500	**31,500**
Book value	90,000 − 31,500	**58,500**
Gain or loss	58,000 − 58,500	**Loss €500**

Step 2 — (b) Every age measure divides by the depreciation expense

Average total life span	Gross PP&E ÷ Dep. expense = 600,000 ÷ 50,000	**12.0 years**
Average age	Accumulated dep. ÷ Dep. expense = 240,000 ÷ 50,000	**4.8 years**
Average remaining life	Net PP&E ÷ Dep. expense = 360,000 ÷ 50,000	**7.2 years**
Consistency check	4.8 + 7.2	**12.0 ✓**

Comment

The plant is 40 per cent through its life (4.8 ÷ 12): heavy replacement spending is about seven years away. The measures assume straight-line depreciation and exclude land, which is not depreciated.

✎ EXTRA SPACE

B9 Partial-year depreciation and a mid-year disposal DISPOSAL

YOUR TURN

Sarnico Srl owns a machine bought on 1 January Year 1 for €120,000, depreciated straight-line over 8 years with a residual value of €8,000. On 30 September Year 4 the machine is sold for €74,000. Compute (a) the depreciation to be recorded in Year 4 up to the date of disposal, (b) the accumulated depreciation at the date of disposal and (c) the gain or loss on the sale.

✎ WORKING

B10 Plant asset age measures ANALYSIS

YOUR TURN

Teoro Group reports gross plant and equipment of €860,000, accumulated depreciation of €344,000 and a current year depreciation expense of €86,000. Compute the average total life span, the average age and the average remaining life, check that they are consistent, and comment in two lines.

Measure	Formula	Computation	Result
Average total life span			
Average age			
Average remaining life			
Check			

✎ COMMENT

SECTION C

Inventory valuation

FIFO, LIFO, weighted average, the LIFO reserve and the liquidation effect. Chapter 4 covers all of this and the lecturer never set an exercise on it — which is a reason to be ready, not a reason to skip it.

TYPE C.1

FIFO, LIFO and weighted average

Goods available for sale, split between cost of goods sold and ending inventory.

① Worked example → ② C1 → ③ C2

WORKED EXAMPLE · SOLVED

FIFO, LIFO and weighted average on three cost layers

Merli Trading, Year 3 (periodic system). Opening inventory 40 units at €10; purchase in May of 60 units at €12; purchase in October of 20 units at €15. During the year 90 units were sold at €20 each.

> **Required**
>
> 1. Compute cost of goods sold and ending inventory under FIFO, LIFO and weighted average.
> 2. Compute gross profit under each method and comment.

Step 1 — Goods available for sale: the total every method must split

Layer	Units	Unit cost	Cost
Opening inventory	40	10	400
May purchase	60	12	720
October purchase	20	15	300
Available for sale	**120**		**1,420**

Units sold 90, so units in ending inventory = 120 − 90 = 30.

Step 2 — Assign the 90 units sold

FIFO — oldest first	COGS = 40 × 10 + 50 × 12 = 400 + 600	**1,000**
	Ending inventory = 10 × 12 + 20 × 15 = 120 + 300	**420**
LIFO — newest first	COGS = 20 × 15 + 60 × 12 + 10 × 10 = 300 + 720 + 100	**1,120**
	Ending inventory = 30 × 10	**300**
Weighted average	Average cost = 1,420 ÷ 120 = €11.833	
	COGS = 90 × 11.833	**1,065**
	Ending inventory = 30 × 11.833	**355**

Step 3 — Gross profit, and the check

Method	Revenue	COGS	Gross profit	Ending inventory	COGS + EI
FIFO	1,800	1,000	800	420	1,420
LIFO	1,800	1,120	680	300	1,420
Weighted average	1,800	1,065	735	355	1,420

Comment

With rising prices FIFO gives the highest gross profit and the most current inventory on the balance sheet; LIFO gives the most current cost of goods sold in the income statement and the lowest profit; weighted average sits between.

C1 FIFO, LIFO and weighted average INVENTORY

YOUR TURN

Vasco Trading reports the following for Year 2. Opening inventory 50 units at €20; purchase on 12 April of 30 units at €24; purchase on 3 October of 20 units at €30. During the year 70 units were sold at €38 each.

Required

1. Compute cost of goods sold and ending inventory under FIFO, LIFO and the weighted average method.
2. Compute gross profit under each method.
3. Comment in two lines.

✎ WORKING

✎ SUMMARY

Method	Revenue	COGS	Ending inventory	Gross profit
FIFO				
LIFO				
Weighted average				

✎ COMMENT

C2 Four cost layers INVENTORY

YOUR TURN

Pelmar Srl, Year 4. Opening inventory 100 units at €12; purchases of 150 units at €14 in March, 200 units at €15 in July and 50 units at €18 in November. 380 units were sold during the year.

> **Required**
>
> 1. Compute cost of goods sold and ending inventory under FIFO, LIFO and weighted average.
> 2. State which method gives the most current balance sheet value for inventory, and which gives the most current income statement.

✎ WORKING

✎ SUMMARY

Method	Cost of goods sold	Ending inventory	Check: COGS + EI
FIFO			
LIFO			
Weighted average			

✎ COMMENT

TYPE C.2

The method and the ratios; restating LIFO to FIFO

Turnover and current ratio under each method, and the LIFO reserve adjustment.

① Worked example → ② C3 → ③ C5

WORKED EXAMPLE · SOLVED

What the method does to the ratios, and restating LIFO to FIFO

(a) Use the Merli data of the previous example. Opening inventory was €400 under either method; other current assets are €1,000 and current liabilities €900. Compute inventory turnover and the current ratio under FIFO and LIFO.

(b) Sirmione Spa reports inventories of €1,500 thousand on a LIFO basis and a LIFO reserve of €400 thousand; tax rate 30 per cent. Restate inventory to FIFO and show the balancing adjustments.

Step 1 — (a) Ratios under each method

Inventory turnover, FIFO	COGS ÷ avg inventory = 1,000 ÷ [(400 + 420) ÷ 2 = 410]	**2.44x**
Inventory turnover, LIFO	1,120 ÷ [(400 + 300) ÷ 2 = 350]	**3.20x**
Current ratio, FIFO	(1,000 + 420) ÷ 900	**1.58**
Current ratio, LIFO	(1,000 + 300) ÷ 900	**1.44**

LIFO makes the company look **more efficient** (higher turnover) and **less liquid** (lower current ratio) — without anything real having changed. A creditor should put both companies on the same basis before comparing.

Step 2 — (b) Restate with the LIFO reserve

Adjustment	€ 000
Inventory, LIFO	1,500
Add LIFO reserve	+400
Inventory, FIFO	**1,900**
The two balancing adjustments	
Deferred tax liability (30% × 400)	+120
Retained earnings (70% × 400)	+280

Comment

Assets rise by 400; the financing side rises by 120 of tax and 280 of equity. The restated balance sheet still balances.

✎ EXTRA SPACE

C3 Which method, and what it does to the ratios INVENTORY

YOUR TURN

Use the Pelmar data of exercise C2. Opening inventory was €1,200 under either method. Other current assets are €3,200 and current liabilities are €2,900.

> **Required**
>
> 1. Compute the inventory turnover and the current ratio under FIFO and under LIFO.
> 2. Comment on what the choice of method does to a creditor's view of the company.

✎ WORKING

Ratio	FIFO	LIFO
Inventory turnover		
Current ratio		

✎ COMMENT

C5 Restating LIFO inventory to FIFO INVENTORY

YOUR TURN

Oristan Group reports inventories of €2,400 thousand on a LIFO basis and discloses a LIFO reserve of €600 thousand. The tax rate is 30 per cent. Restate the inventory to a FIFO basis and show the two balancing adjustments.

✎ RESTATEMENT

Adjustment	€ 000
Inventory, LIFO	
LIFO reserve	
Inventory, FIFO	
Balancing adjustment 1:	
Balancing adjustment 2:	

✎ WORKING

TYPE C.3

Holding gains and LIFO liquidations

The part of profit that comes from old costs rather than from trading.

① Worked example → ② C4 → ③ C6

WORKED EXAMPLE · SOLVED

Holding gains, and a LIFO liquidation

(a) Bruni Spa began the year with 20 units at €100 and bought 30 units at €130. It sold 25 units at €200. Split FIFO gross profit into economic profit and holding gain.

(b) Gaeta Srl carries LIFO inventory in two layers: 1,000 units at €4 (old) and 2,000 units at €7. This year it buys 3,000 units at €10 and sells 4,500 units at €16. Compute COGS and gross profit, and the effect of the liquidation.

Step 1 — (a) FIFO profit contains a holding gain

Revenue	25 × 200	**5,000**
FIFO COGS	20 × 100 + 5 × 130	**2,650**
FIFO gross profit	5,000 − 2,650	**2,350**
Economic profit	Revenue − current cost = 5,000 − 25 × 130	**1,750**
Holding gain	20 old units × (130 − 100)	**600**
Check	FIFO GP − holding gain = 2,350 − 600 = LIFO GP (25 × 130 at current cost)	**1,750 ✓**

Step 2 — (b) Selling more than you buy digs into old layers

Units	3,000 opening + 3,000 bought − 4,500 sold	**1,500 left**
LIFO COGS	3,000 × 10 + 1,500 × 7	**40,500**
Gross profit	4,500 × 16 − 40,500 = 72,000 − 40,500	**31,500**
COGS had it bought 4,500	4,500 × 10	**45,000**
Liquidation profit	1,500 × (10 − 7) = 45,000 − 40,500	**+4,500**

Comment

Both effects are one-off and come from old, cheap costs, not from selling better. The analyst removes them when judging sustainable margins; the liquidation also brings forward tax on the old layers.

✎ EXTRA SPACE

C4 Economic profit and holding gain INVENTORY

YOUR TURN

Tellaro Spa began the year with 50 units at €400 and purchased 70 units at €480 during the year. It sold 40 units at €650 each. Compute the gross profit under FIFO, split it into economic profit and holding gain, and show that the FIFO gross profit less the holding gain equals the LIFO gross profit.

✎ WORKING

C6 LIFO liquidation INVENTORY

YOUR TURN

Brienno Spa carries LIFO inventory in two layers: 2,000 units at €6, acquired years ago, and 3,000 units at €9. During the current year it purchases 4,000 units at €12 and sells 5,000 units at €20.

> **Required**
>
> 1. Compute cost of goods sold and gross profit for the year.
> 2. Compute what cost of goods sold would have been had the company purchased 5,000 units instead of 4,000, and quantify the effect of the liquidation on pretax income.
> 3. Comment.

✎ WORKING

✎ COMMENT

TYPE C.4

The cost of goods sold equation; product and period costs

Solving the inventory identity, and what an error does to profit.

① Worked example → ② C7 → ③ C8

WORKED EXAMPLE · SOLVED

The cost of goods sold equation, and product versus period costs

(a) Ferla Ltd shows opening inventory of €60,000, closing inventory of €75,000 and cost of goods sold of €395,000. Compute net purchases; then show the effect of closing inventory overstated by €5,000 (tax rate 30 per cent).

(b) Classify as product or period cost: factory rent; depreciation of delivery vans; import duties on goods bought; chief executive's salary; packaging of the product; interest on a working-capital loan.

Step 1 — (a) One equation, solved for the unknown

Opening inventory + Net purchases − Closing inventory = Cost of goods sold

Net purchases	395,000 + 75,000 − 60,000	**410,000**
COGS if closing is overstated by 5,000	60,000 + 410,000 − 80,000	**390,000**
Effect	COGS −5,000 → pretax income +5,000 → net income +3,500	**overstated**

Next year the error reverses: opening inventory is overstated, so next year's COGS is overstated by the same 5,000.

Step 2 — (b) Is the cost needed to bring the goods to saleable condition?

Cost	Classification
Factory rent	**Product** — manufacturing overhead
Depreciation of delivery vans	**Period** — selling expense
Import duties on goods bought	**Product** — part of purchase cost
Chief executive's salary	**Period** — administrative
Packaging of the product	**Product**
Interest on a working-capital loan	**Period** — financing

Comment

If a product cost is wrongly expensed as a period cost, it leaves cost of goods sold, so **gross profit is overstated**; but the part that belonged to closing inventory is charged now, so **operating profit and inventory are understated**.

✎ EXTRA SPACE

C7 The cost of goods sold equation INVENTORY

YOUR TURN

Filled in from a damaged file, Canova Ltd shows opening inventory of €85,000, closing inventory of €97,000 and cost of goods sold of €640,000. Compute net purchases for the year. Then show what cost of goods sold would have been if closing inventory had been overstated by €9,000, and what that would do to reported net income.

✎ WORKING

✎ COMMENT

C8 Expensing or capitalising inventory costs INVENTORY

YOUR TURN

For each cost below, say whether it is treated as a product cost (capitalised into inventory) or a period cost (expensed as incurred), and state the effect on the current year's gross profit of getting it wrong in the direction of expensing everything.

Cost	Product or period
Direct labour in the factory	
Office supplies at head office	
Freight-in on raw materials	
Advertising campaign	
Factory overhead allocated to units produced	
Sales commissions	

✎ EFFECT ON GROSS PROFIT OF EXPENSING EVERYTHING

SECTION D

Horizontal and year-to-year change analysis

Amount and percentage changes, the cases where the percentage cannot be computed, and the comment that goes under the table.

TYPE D.1

Horizontal analysis of a statement

Change in euro and in per cent against the base year, then the comment.

① Worked example → ② D1 → ③ D2

WORKED EXAMPLE · SOLVED

Horizontal analysis of an income statement

Prepare a horizontal analysis of the condensed income statement of Corvo Srl for 2025, base year 2024, and comment.

	2025	2024
Net sales	600,000	500,000
Cost of goods sold	390,000	310,000
Gross profit	210,000	190,000
Operating expenses	132,000	120,000
Net income	78,000	70,000

Step 1 — One formula for every line

Change € = Current year − Base year · Change % = Change € ÷ Base year × 100

Step 2 — The schedule

	2025	2024	Change €	Change %
Net sales	600,000	500,000	+100,000	+20.0%
Cost of goods sold	390,000	310,000	+80,000	+25.8%
Gross profit	210,000	190,000	+20,000	+10.5%
Operating expenses	132,000	120,000	+12,000	+10.0%
Net income	78,000	70,000	+8,000	+11.4%

Comment

Sales grew 20.0 per cent but cost of goods sold grew faster (25.8 per cent), so gross profit rose only 10.5 per cent: the gross margin is being squeezed. Operating expenses were held to +10.0 per cent, which lets net income grow 11.4 per cent. Favourable on volume and cost control, unfavourable on pricing or purchasing.

✎ EXTRA SPACE

D1 Horizontal analysis of a balance sheet HORIZONTAL

YOUR TURN

Prepare a schedule showing a horizontal analysis of the condensed statement of financial position of Trevi Group for 2025, using 2024 as the base year. Then comment in three lines on how the growth has been financed.

	2025	2024	Change €	Change %
Plant assets (net)	528,000	440,000		
Current assets	196,000	160,000		
Total assets	**724,000**	**600,000**		
Share capital	250,000	200,000		
Retained earnings	174,000	185,000		
Total equity	**424,000**	**385,000**		
Non-current liabilities	186,000	130,000		
Current liabilities	114,000	85,000		
Total liabilities	**300,000**	**215,000**		
Total equity and liabilities	**724,000**	**600,000**		

✎ HOW THE GROWTH HAS BEEN FINANCED

D2 Horizontal analysis of an income statement HORIZONTAL

YOUR TURN

Prepare a horizontal analysis of the condensed income statement of Hale Corporation for 2025, using 2024 as the base year, and say whether the picture is favourable.

	2025	2024	Change €	Change %
Net sales	840,000	700,000		
Cost of goods sold	571,200	490,000		
Gross profit	268,800	210,000		
Operating expenses	151,200	126,000		
Net income	117,600	84,000		

✎ WORKING

✎ COMMENT

TYPE D.2

When the percentage cannot be computed

Zero and negative bases, and the movements a creditor asks about.

① Worked example → ② D3 → ③ D4

WORKED EXAMPLE · SOLVED

When the percentage change cannot be computed

Compute the amount and percentage change for each account of Asola Spa, and flag the movements a creditor would ask about.

	Year 2	Year 1
Marketable securities	30,000	0
Inventories	84,000	70,000
Accounts payable	45,000	60,000
Bank loans	0	50,000
Net income (loss)	12,000	(8,000)

Step 1 — The amount can always be computed; the percentage needs a positive base

	Year 2	Year 1	Change €	Change %
Marketable securities	30,000	0	+30,000	n.m.
Inventories	84,000	70,000	+14,000	+20.0%
Accounts payable	45,000	60,000	−15,000	−25.0%
Bank loans	0	50,000	−50,000	−100.0%
Net income (loss)	12,000	(8,000)	+20,000	n.m.

Step 2 — Why "n.m." (not meaningful)

- **Base of zero** (marketable securities): division by zero — a percentage cannot be computed.
- **Negative base** (loss to profit): the formula gives −250 per cent for an improvement, so the sign is misleading.
- A positive base falling to zero (bank loans) is fine: −100 per cent.

Comment

A creditor would ask why inventories rose 20 per cent while payables fell 25 per cent — together they absorbed €29,000 of cash — and how the €50,000 loan was repaid while €30,000 was parked in securities.

✎ EXTRA SPACE

D3 When the percentage change cannot be computed HORIZONTAL

YOUR TURN

Compute the amount and the percentage of increase or decrease for each account below. Where a percentage cannot be computed, say so and explain why in one line.

	Year 2	Year 1	Change €	Change %
Short-term investments	225,000	180,000		
Accounts receivable	84,000	96,000		
Notes payable	45,000	0		
Deficit on retained earnings	15,000	(40,000)		

✎ WHERE THE PERCENTAGE CANNOT BE COMPUTED, AND WHY

D4 Year-to-year change over a set of accounts HORIZONTAL

YOUR TURN

Compute the amount and percentage of increase or decrease for each account of Orvieto Spa, and identify which movements a creditor would want explained.

	Year 2	Year 1	Change €	Change %
Cash	52,000	65,000		
Inventories	148,000	120,000		
Prepaid expenses	9,000	12,000		
Goodwill	90,000	90,000		
Bank loans	70,000	0		
Accrued liabilities	34,500	30,000		

✎ MOVEMENTS A CREDITOR WOULD WANT EXPLAINED

TYPE D.3

Recovering and choosing the base year

Working backwards from a percentage, and why the base changes the story.

① Worked example → ② D5 → ③ D6

WORKED EXAMPLE · SOLVED

Recovering the base year, and choosing it

(a) Sales rose 15.0 per cent in Year 2 to €690,000. Cost of goods sold rose 20.0 per cent from a Year 1 figure of €360,000. Compute Year 1 sales, Year 2 cost of goods sold, gross profit for both years and its change.

(b) Danae Ltd reports net sales of €400,000 in 2023, €320,000 in 2024 and €360,000 in 2025. Analyse 2025 against 2024 and against 2023.

Step 1 — (a) Undo the percentage by dividing, apply it by multiplying

Year 1 sales	690,000 ÷ 1.15	**600,000**
Year 2 cost of goods sold	360,000 × 1.20	**432,000**
Gross profit Year 1	600,000 − 360,000	**240,000**
Gross profit Year 2	690,000 − 432,000	**258,000**
Change	+18,000 ÷ 240,000	**+7.5%**

Never compute Year 1 as 690,000 × 0.85 = 586,500: the 15 per cent is a percentage of Year 1, not of Year 2.

Step 2 — (b) Same 2025 figure, two bases

2025 vs 2024	(360,000 − 320,000) ÷ 320,000	**+12.5%**
2025 vs 2023	(360,000 − 400,000) ÷ 400,000	**−10.0%**

Comment

(a) Sales +15.0 per cent but gross profit only +7.5 per cent: costs outran prices. (b) Both answers are correct — one tells a story of recovery, the other of a business still below its 2023 level. Always state the base year, and check the base chosen by whoever prepared an analysis you are reading.

✎ EXTRA SPACE

D5 Recovering the base year HORIZONTAL

YOUR TURN

Sales increased 18.0 per cent during Year 2 and reached €1,062,000. Cost of goods sold increased 24.0 per cent from a Year 1 figure of €600,000.

> **Required**
>
> 1. Compute Year 1 sales and Year 2 cost of goods sold.
> 2. Compute gross profit for both years and its percentage change.
> 3. Comment.

✎ WORKING

✎ COMMENT

D6 Choosing the base year changes the story HORIZONTAL

YOUR TURN

Kerrel Ltd reports net sales of €500,000 in 2023, €750,000 in 2024 and €690,000 in 2025.

> **Required**
>
> 1. Prepare a horizontal analysis of 2025 using 2024 as the base.
> 2. Prepare a horizontal analysis of 2025 using 2023 as the base.
> 3. Explain in two lines why both are correct and what that implies for reading someone else's analysis.

Comparison	Base year	Change €	Change %
2025 vs 2024			
2025 vs 2023			

✎ WORKING

✎ COMMENT

TYPE D.4

Sign changes and the written comment

Expense lines, and commenting on a table without over-reaching.

① Worked example → ② D7 → ③ D8

WORKED EXAMPLE · SOLVED

Horizontal analysis with a sign change, and writing the comment

(a) Complete the horizontal analysis of this extract from the income statement of Lerma NV and write the comment.

	2025	2024
Operating profit	50,000	40,000
Interest expense	(30,000)	(10,000)
Income before tax	20,000	30,000
Income tax expense	(6,000)	(9,000)
Net income	14,000	21,000

(b) A completed analysis of Tosca Spa shows: sales +10.0%, receivables +25.0%, inventories +5.0%, net income +12.0%. Write the comment (six lines at most) and state what you cannot conclude.

Step 1 — (a) Compute on the size of each line; read the sign as more or less expense

	2025	2024	Change €	Change %
Operating profit	50,000	40,000	+10,000	+25.0%
Interest expense	(30,000)	(10,000)	+20,000	+200.0%
Income before tax	20,000	30,000	−10,000	−33.3%
Income tax expense	(6,000)	(9,000)	−3,000	−33.3%
Net income	14,000	21,000	−7,000	−33.3%

"Interest expense +200%" means the expense tripled. State the convention once, then apply it to every expense line.

Step 2 — (a) The comment

Operating profit improved by 25.0 per cent, but interest expense tripled and wiped out the gain: net income fell 33.3 per cent. The deterioration comes from financing, not from operations.

Step 3 — (b) Comment on a table you did not prepare

Receivables grew 25 per cent against sales growth of 10 per cent: collection is slowing or credit terms were loosened, and part of the 12 per cent rise in net income is not yet cash. Inventories grew less than sales, which is favourable. **What we cannot conclude:** whether the receivables reflect one large customer, a year-end timing effect or bad debts — we need the days' sales in receivables, an ageing schedule and the cash flow statement.

D7 Horizontal analysis with a sign change HORIZONTAL

YOUR TURN

Below is an extract from the income statement of Sendra NV. Complete the horizontal analysis and write the two-line comment.

	2025	2024	Change €	Change %
Operating profit	96,000	80,000		
Interest expense	(54,000)	(20,000)		
Income before tax	42,000	60,000		
Income tax expense	(12,600)	(18,000)		
Net income	29,400	42,000		

✎ COMMENT

D8 Read the table, write the comment HORIZONTAL

YOUR TURN

You are handed the following completed horizontal analysis of Alcide Group. No further information is available. Write the comment, no more than six lines, and state clearly what you cannot conclude.

	Change	Percent
Net sales	+420,000	+14.0%
Cost of goods sold	+357,000	+17.0%
Gross profit	+63,000	+7.0%
Selling expenses	+96,000	+24.0%
Administrative expenses	−18,000	−6.0%
Operating profit	−15,000	−7.5%
Accounts receivable	+198,000	+33.0%
Inventories	+240,000	+30.0%

✎ COMMENT (SIX LINES) AND WHAT YOU CANNOT CONCLUDE

SECTION E

Index-number trend analysis

Trend percents, rebasing, and what a series of indices can and cannot support.

TYPE E.1

Trend percents

Every year divided by the base year, then the reading.

① Worked example → ② E1 → ③ E2

WORKED EXAMPLE · SOLVED

Trend percents for two accounts

Compute index-number trend percents for Gelso Spa, Year 1 = 100, and say whether the trend is favourable.

	Year 1	Year 2	Year 3	Year 4
Sales	200,000	220,000	250,000	270,000
Cost of goods sold	120,000	135,600	157,500	175,500

Step 1 — Divide every year by the base year

$\text{Index}_t = \text{Amount}_t \div \text{Amount}_{base} \times 100$ — e.g. COGS Year 3 = 157,500 ÷ 120,000 × 100 = 131.3

Trend percents	Year 1	Year 2	Year 3	Year 4
Sales	100.0	110.0	125.0	135.0
Cost of goods sold	100.0	113.0	131.3	146.3

Comment

Cost of goods sold grows faster than sales in every year (146 against 135 by Year 4), so the gross margin is shrinking: unfavourable, even though both lines are growing.

✎ EXTRA SPACE

E1 Trend percents for three accounts TREND

YOUR TURN

Compute index-number trend percents for the following accounts of Donati Spa, using Year 1 as the base year. State whether the situation revealed by the trends appears favourable or unfavourable.

	Year 1	Year 2	Year 3	Year 4	Year 5
Sales	420,000	462,000	508,200	541,800	571,200
Cost of goods sold	252,000	282,240	317,520	347,760	378,000
Accounts receivable	60,000	68,400	78,000	85,800	94,800
Trend percents (Year 1 = 100)					
Sales					
Cost of goods sold					
Accounts receivable					

✎ WORKING

✎ COMMENT

E2 Trend in assets against trend in equity TREND

YOUR TURN

Compute trend percents with Year 1 as the base, then say what the divergence implies for the capital structure.

	Year 1	Year 2	Year 3	Year 4	Year 5
Total assets	1,200,000	1,260,000	1,392,000	1,560,000	1,776,000
Total equity	700,000	721,000	756,000	798,000	840,000
Trend percents (Year 1 = 100)					
Total assets					
Total equity					

✎ WORKING

✎ WHAT THE DIVERGENCE IMPLIES

TYPE E.2

Rebasing and reading an index table

Changing the base, and inferring ratios from indices alone.

① Worked example → ② E3 → ③ E4

WORKED EXAMPLE · SOLVED

Rebasing a series, and reading an index table

(a) A sales index with Year 1 = 100 reads 100, 120, 150, 165. Rebase it to Year 2 = 100.

(b) Only these indices are available (Year 1 = 100). Did return on sales and asset turnover rise or fall? What can't you say?

	Year 1	Year 2	Year 3
Sales	100	115	130
Net income	100	120	140
Total assets	100	110	125

Step 1 — (a) Divide every index by the new base index

	Year 1	Year 2	Year 3	Year 4
Year 1 = 100	100.0	120.0	150.0	165.0
Year 2 = 100 (÷ 1.20)	83.3	100.0	125.0	137.5

What changes: the level of every number. What does not: the relative movement between any two years (Year 4 ÷ Year 3 = 1.10 in both series).

Step 2 — (b) A ratio rises when its numerator index outgrows its denominator index

Return on sales (NI ÷ Sales)	Year 3: 140 ÷ 130 = 1.077 → about 7.7% higher than in Year 1	**rose**
Asset turnover (Sales ÷ TA)	Year 3: 130 ÷ 125 = 1.04 → about 4% higher than in Year 1	**rose**

What we cannot say: the level of either ratio (is a margin of 3% or 30%?), and anything about a competitor — indices only give the direction relative to the company's own base.

✎ EXTRA SPACE

E3 Rebasing a trend series TREND

YOUR TURN

The sales trend index of Donati Spa with Year 1 as base is 100, 110, 121, 129, 136. The company now wants the series rebased to Year 3 = 100. Produce the rebased series and explain what changes and what does not.

	Year 1	Year 2	Year 3	Year 4	Year 5
Year 1 = 100	100	110	121	129	136
Year 3 = 100					

✎ WORKING

✎ WHAT CHANGES AND WHAT DOES NOT

E4 Reading margins out of a trend table TREND

YOUR TURN

You are given only the trend indices below, with Year 1 as base. Without any absolute figures, state whether (a) the return on sales rose or fell, (b) asset turnover rose or fell, and (c) what you cannot say.

	Year 1	Year 2	Year 3
Sales	100	118	136
Net income	100	112	128
Total assets	100	126	148

✎ WORKING

✎ (A), (B) AND (C)

TYPE E.3

One trend against another

Inventory or receivables against sales; debt against its cost.

① Worked example → ② E5 → ③ E6

WORKED EXAMPLE · SOLVED

One trend against another

(a) The sales index of Arno Srl is 100, 108, 117. Receivables were €50,000, €60,000 and €75,000. Compare the two trends. **(b)** Non-current liabilities were €100,000, €130,000 and €160,000; interest expense €6,000, €8,450 and €12,000. Compute both trends and the implied interest rate.

Step 1 — (a) Receivables against sales

	Year 1	Year 2	Year 3
Sales index	100	108	117
Receivables	50,000	60,000	75,000
Receivables index	100	120	150

Receivables grow almost three times as fast as sales: collection is slowing, so check the days' sales in receivables and the ageing.

Step 2 — (b) Debt, its cost, and the implied rate

	Year 1	Year 2	Year 3
Non-current liabilities index	100	130	160
Interest expense index	100	140.8	200
Implied rate = Interest ÷ Debt	6.0%	6.5%	7.5%

Comment

Interest doubled while debt rose 60 per cent: the company is paying more per euro borrowed — lenders see it as riskier, or rates have risen. The trend in the cost of debt is as important as the trend in the debt.

✎ EXTRA SPACE

E5 Inventory outpacing sales TREND

YOUR TURN

Using Year 1 as the base, compute the trend percents for inventories of Lanova Srl and compare them with the sales series of exercise E1 (100, 110, 121, 129, 136). Comment.

	Year 1	Year 2	Year 3	Year 4	Year 5
Inventories	180,000	212,400	250,200	297,000	352,800
Inventory index					
Sales index (E1)	100	110	121	129	136

✎ WORKING

✎ COMMENT

E6 Trend in debt and in its cost TREND

YOUR TURN

Compute trend percents with Year 1 as base for the non-current liabilities and the interest expense of Verona Group, then compute the implied average interest rate in Year 1 and Year 5 and comment.

	Year 1	Year 2	Year 3	Year 4	Year 5
Non-current liabilities	200,000	230,000	290,000	360,000	450,000
Interest expense	16,000	18,400	24,650	32,320	42,750
Trend percents (Year 1 = 100)					
Non-current liabilities					
Interest expense					
Implied rate					

✎ WORKING

✎ COMMENT

TYPE E.4

Trend with common-size, and the limits of trends

Back to euros, and the conclusions a trend cannot carry.

① Worked example → ② E7 → ③ E8

WORKED EXAMPLE · SOLVED

Trend and common-size together; what a trend cannot tell you

(a) Sales of Lago Ltd were €400,000 in Year 1 and the sales index is 100, 115, 125. Cost of goods sold as a percentage of sales was 55.0, 58.0 and 61.0 per cent. Compute gross profit for each year and reconcile the two stories.

(b) A colleague notes that Company R's net income index went from 100 to 180 while Company S's went from 100 to 130, and concludes that R is the more profitable company. Comment.

Step 1 — (a) Turn indices and percentages back into euros

	Year 1	Year 2	Year 3
Sales (400,000 × index)	400,000	460,000	500,000
COGS (% × sales)	220,000	266,800	305,000
Gross profit	180,000	193,200	195,000
Gross profit index	100	107.3	108.3

The trend says the company grew 25 per cent; the common-size says each euro of sales keeps 6 cents less. Together: gross profit grew only 8.3 per cent.

Step 2 — (b) Growth is not profitability

- Each index is on its **own base**: R may have started from a tiny profit, so +80 per cent may be few euros.
- Profitability is a **return** (net income ÷ sales, ROCE, RNOA), which indices do not show.
- A single year's base can be **abnormal** (a bad Year 1 flatters every later index).

✎ EXTRA SPACE

E7 Trend and common-size together TREND

YOUR TURN

Sales of Castano Ltd were €500,000 in Year 1 and the sales trend index is 100, 112, 126. Cost of goods sold, expressed as a common-size percentage of sales, was 60.0 per cent, 63.0 per cent and 66.0 per cent. Compute gross profit in absolute terms for each year and reconcile the two stories.

	Year 1	Year 2	Year 3
Sales			
Cost of goods sold			
Gross profit			
Gross profit index			

✎ RECONCILE THE TWO STORIES

E8 What a trend series cannot do TREND

YOUR TURN

Two competitors report the following sales trend indices, each on its own Year 1 base. A colleague concludes that Company Q is the better investment. List the three reasons the conclusion is unsupported.

	Year 1	Year 2	Year 3	Year 4
Company P	100	104	109	113
Company Q	100	128	161	204

✎ THREE REASONS

SECTION F

Common-size (vertical) analysis

Income statement and balance sheet in percentages, two-company comparison, composition analysis, and the two traps.

TYPE F.1

Common-size income statement

Every line as a percentage of net sales, two years side by side.

① Worked example → ② F1 → ③ F4

WORKED EXAMPLE · SOLVED

Common-size income statement, two years

Express the income statement of Pino Srl in common-size percents for both years and assess the situation.

	2025	2024
Net sales	800,000	640,000
Cost of goods sold	488,000	371,200
Gross profit	312,000	268,800
Operating expenses	208,000	179,200
Net income	104,000	89,600

Step 1 — Every line divided by net sales of the same year

Common-size % = Line ÷ Net sales × 100 — e.g. COGS 2025 = 488,000 ÷ 800,000 = 61.0%

	2025	2024	% 2025	% 2024
Net sales	800,000	640,000	100.0%	100.0%
Cost of goods sold	488,000	371,200	61.0%	58.0%
Gross profit	312,000	268,800	39.0%	42.0%
Operating expenses	208,000	179,200	26.0%	28.0%
Net income	104,000	89,600	13.0%	14.0%

Comment

Net income grew in euros, but each euro of sales now earns 13.0 cents against 14.0: cost of goods sold took 3 points more of sales (pricing pressure or input costs), only partly offset by 2 points saved on operating expenses. Mildly unfavourable.

✎ EXTRA SPACE

F1 Common-size income statement, two years COMMON-SIZE

YOUR TURN

Express the income statement of Carro Spa in common-size percents for both years and assess whether the company's situation is favourable or unfavourable.

	2025	2024	% 2025	% 2024
Net sales	1,500,000	1,200,000		
Cost of goods sold	915,000	708,000		
Gross profit	585,000	492,000		
Selling expenses	240,000	180,000		
Administrative expenses	135,000	108,000		
Total operating expenses	375,000	288,000		
Operating profit	210,000	204,000		
Interest expense	30,000	24,000		
Income before tax	180,000	180,000		
Income tax expense	54,000	54,000		
Net income	126,000	126,000		

✎ COMMENT

F4 Favourable or unfavourable COMMON-SIZE

YOUR TURN

Express the income statement information of Harman Corporation in common-size percents and assess whether the company's situation is favourable or unfavourable.

	2025	2024	% 2025	% 2024
Sales	960,000	750,000		
Cost of goods sold	604,800	435,000		
Gross profit	355,200	315,000		
Operating expenses	201,600	150,000		
Net income	153,600	165,000		

✎ WORKING

✎ COMMENT

TYPE F.2

Common-size balance sheet and capital structure

Every line as a percentage of total assets.

① Worked example → ② F2 → ③ F7

WORKED EXAMPLE · SOLVED

Common-size balance sheet

Express the balance sheet of Alto Spa in common-size percents (one decimal) and comment on the change in structure.

	2025	2024
Cash	30,000	40,000
Accounts receivable	90,000	60,000
Inventories	120,000	100,000
Plant assets, net	360,000	300,000
Accounts payable	90,000	50,000
Long-term debt	150,000	150,000
Share capital	200,000	200,000
Retained earnings	160,000	100,000
Total (assets = equity and liabilities)	**600,000**	**500,000**

Step 1 — Divide by total assets (= total equity and liabilities)

	2025	2024	% 2025	% 2024
Cash	30,000	40,000	5.0%	8.0%
Accounts receivable	90,000	60,000	15.0%	12.0%
Inventories	120,000	100,000	20.0%	20.0%
Plant assets, net	360,000	300,000	60.0%	60.0%
Total assets	**600,000**	**500,000**	**100.0%**	**100.0%**
Accounts payable	90,000	50,000	15.0%	10.0%
Long-term debt	150,000	150,000	25.0%	30.0%
Share capital	200,000	200,000	33.3%	40.0%
Retained earnings	160,000	100,000	26.7%	20.0%
Total equity and liabilities	**600,000**	**500,000**	**100.0%**	**100.0%**

Comment

The asset mix is stable except that cash (8.0 → 5.0 per cent) has turned into receivables (12.0 → 15.0 per cent): less liquid. On the financing side equity stays at 60 per cent, but suppliers (10 → 15 per cent) have replaced long-term debt (30 → 25 per cent): cheaper, but shorter-term financing.

F2 Common-size balance sheet COMMON-SIZE

YOUR TURN

Express the balance sheet of Nordia Ltd in common-size percents, rounding to one decimal place, and comment on the change in the structure of the company.

	2025	2024	% 2025	% 2024
Cash	40,000	44,800		
Accounts receivable	128,000	83,200		
Inventories	176,000	102,400		
Prepaid expenses	16,000	12,800		
Plant assets, net	440,000	396,800		
Accounts payable	152,000	76,800		
Accrued liabilities	48,000	44,800		
Long-term debt	200,000	166,400		
Share capital	280,000	280,000		
Retained earnings	120,000	72,000		
Total	**800,000**	**640,000**		

✎ COMMENT

F7 Common-size capital structure COMMON-SIZE

YOUR TURN

Construct a common-size statement of the liabilities and equity side of the balance sheet of Marenco Spa and comment on the composition. Total liabilities and equity are €3,000,000.

	Amount	%
Current liabilities	540,000	
Long-term debt	810,000	
Preferred stock	300,000	
Common stock	960,000	
Paid-in capital	90,000	
Retained earnings	300,000	
Total liabilities and equity	**3,000,000**	

✎ COMMENT

TYPE F.3

Comparing companies; composition of current assets

Removing size to compare cost structures, and the quality of a total.

① Worked example → ② F3 → ③ F6

WORKED EXAMPLE · SOLVED

Two companies of different size; the composition of current assets

(a) Prepare a vertical analysis of Rossa (large) and Blu (small) and compare their profitability.

	Rossa	Blu
Net sales	5,000,000	800,000
Cost of goods sold	3,250,000	480,000
Operating expenses	1,000,000	200,000
Operating income	750,000	120,000
Net income	500,000	72,000

(b) Current assets of Neva Srl were Year 1: cash 20,000, receivables 30,000, inventories 30,000; Year 2: cash 8,000, receivables 28,000, inventories 44,000. Analyse the change in liquidity.

Step 1 — (a) Percentages remove size

	Rossa	Blu	Rossa %	Blu %
Net sales	5,000,000	800,000	100.0%	100.0%
Cost of goods sold	3,250,000	480,000	65.0%	60.0%
Operating expenses	1,000,000	200,000	20.0%	25.0%
Operating income	750,000	120,000	15.0%	15.0%
Net income	500,000	72,000	10.0%	9.0%

Same operating margin, two cost structures: Blu buys or produces more cheaply (60 against 65 per cent) but carries heavier overheads (25 against 20 per cent), typical of a smaller scale. Below operating income Blu loses 6 points to interest and tax against Rossa's 5, so Rossa ends slightly ahead.

Step 2 — (b) Same total, different quality

	Year 1	%	Year 2	%
Cash	20,000	25.0%	8,000	10.0%
Receivables	30,000	37.5%	28,000	35.0%
Inventories	30,000	37.5%	44,000	55.0%
Total current assets	**80,000**	**100.0%**	**80,000**	**100.0%**

The total is unchanged, but liquidity worsened: cash fell from a quarter to a tenth of current assets, while inventory — the least liquid item — rose to more than half. The current ratio would not show this.

F3 Common-size comparison of two competitors COMMON-SIZE

YOUR TURN

Prepare a vertical analysis of the 2025 income statement data for Alpha Company and Beta Company in columnar form, and comment on their relative profitability.

	Alpha	Beta	Alpha %	Beta %
Net sales	2,400,000	480,000		
Cost of goods sold	1,632,000	345,600		
Gross profit	768,000	134,400		
Operating expenses	456,000	115,200		
Income from operations	312,000	19,200		
Interest expense	14,400	3,360		
Income before income taxes	297,600	15,840		
Income tax expense	96,000	5,760		
Net income	201,600	10,080		

✎ COMMENT

F6 Current asset composition COMMON-SIZE

YOUR TURN

The current assets of Lerici Energia are reproduced below for Year 1 and Year 2, with their common-size percentages. Complete the percentages and analyse the change in current asset liquidity.

	Year 1	%	Year 2	%
Cash	36,000		18,000	
Accounts receivable	54,000		45,000	
Inventories	60,000		87,000	
Total current assets	**150,000**		**150,000**	

✎ COMMENT

TYPE F.4

Back from percentages; the traps

Reconstructing euros from percentages, and the wrong bases.

① Worked example → ② F5 → ③ F8

WORKED EXAMPLE · SOLVED

Back from percentages to euros; two traps

(a) A common-size income statement shows cost of goods sold 70.0%, operating expenses 18.0% and operating profit 12.0%. Gross profit is €270,000. Reconstruct the statement down to operating profit.

(b) Explain what is wrong: (1) a student divides each liability by *total liabilities* in a common-size balance sheet; (2) a student concludes that the company with the lower cost-of-goods-sold percentage is the more efficient one.

Step 1 — (a) Find the anchor: the line whose euro amount and percentage are both known

Gross profit %	100% − 70.0%	**30.0%**
Net sales	270,000 ÷ 0.30	**900,000**
Cost of goods sold	70.0% × 900,000	**630,000**
Operating expenses	18.0% × 900,000	**162,000**
Operating profit	12.0% × 900,000 = 270,000 − 162,000	**108,000**

Step 2 — (b) The traps

- (1) The base of a common-size balance sheet is **total assets** (= total equity and liabilities). Dividing by total liabilities makes the financing side incomparable with the asset side and hides the weight of debt against equity.
- (2) Percentages remove size, not **business model** or **classification**: one company may book depreciation or distribution in cost of goods sold, the other in operating expenses. Compare operating margin, not one line.

✎ EXTRA SPACE

F5 Recovering absolute figures from percentages COMMON-SIZE

YOUR TURN

An extract of a common-size income statement shows cost of goods sold at 64.0 per cent, operating expenses at 22.0 per cent and operating profit at 14.0 per cent. Gross profit in absolute terms is €468,000. Reconstruct the income statement down to operating profit.

✎ WORKING

Line	%	€
Net sales		
Cost of goods sold		
Gross profit		
Operating expenses		
Operating profit		

F8 Two traps in a common-size statement COMMON-SIZE

YOUR TURN

Answer both in three lines each.

1. A student's common-size balance sheet column adds to 100.2 per cent. The arithmetic has been checked and is right. What happened, and what should the student write?
2. A second student prepares a common-size income statement by dividing every line by net income rather than by net sales, reasoning that net income is the bottom line. Explain why this is wrong and what it would do to a comparison between two companies.

✎ 1.

✎ 2.

SECTION G

Liquidity and working capital

Current, acid test and cash ratios, the turnover and days measures, and what they do and do not reveal. The shape of Exercises A and B of the third review session.

TYPE G.1

Current, acid test and cash ratios

From the widest to the strictest measure, and the quality behind the total.

① Worked example → ② G1 → ③ G9

WORKED EXAMPLE · SOLVED

Current, acid test and cash ratios — and what is behind them

The current section of the balance sheet of Delta Srl at 31 December is given below. Compute the three liquidity ratios and the composition of current assets, and say whether a supplier should grant 60-day credit.

Current assets	€	Current liabilities	€
Cash	40,000	Accounts payable	160,000
Marketable securities	20,000	Accrued liabilities	40,000
Accounts receivable	140,000		
Inventory	180,000		
Prepaid expenses	20,000		
Total	**400,000**	**Total**	**200,000**

Step 1 — The three ratios, from the widest to the strictest

Current ratio	Current assets ÷ Current liabilities = 400,000 ÷ 200,000	**2.00**
Acid test ratio	(Cash + Mkt sec. + Receivables) ÷ CL = (40 + 20 + 140) ÷ 200	**1.00**
Cash ratio	(Cash + Mkt sec.) ÷ CL = (40 + 20) ÷ 200	**0.30**
Working capital	400,000 − 200,000	**200,000**

Step 2 — Composition of current assets

	€	% of current assets
Cash and marketable securities	60,000	15.0%
Accounts receivable	140,000	35.0%
Inventory	180,000	45.0%
Prepaid expenses	20,000	5.0%

Comment

The current ratio meets the 2:1 rule of thumb, but half of current assets are inventory and prepaid expenses. Without them the company covers its current liabilities exactly once, and only if receivables are collected; cash covers 30 per cent. A supplier can grant 60-day credit, but should first check the age of the receivables.

G1 Liquidity ratios for two companies LIQUIDITY

YOUR TURN

The statements of Orso Spa and Lince Srl at 31 December 2025 are reproduced below. Calculate the current ratio, the acid test ratio and the cash ratio for both companies, and say which one a supplier would rather extend 60-day credit to.

Income statement 2025	Orso	Lince
Sales	1,200,000	900,000
Cost of goods sold	780,000	495,000
Selling and administrative expense	210,000	165,000
Operating income	210,000	240,000
Interest expense	36,000	60,000
Income before taxes	174,000	180,000
Income tax expense	52,200	54,000
Net income	**121,800**	**126,000**

Balance sheet, 31 December 2025	Orso	Lince
Assets		
Fixed assets, net	760,000	640,000
Inventory	240,000	120,000
Accounts receivable, net	180,000	105,000
Marketable securities	30,000	10,000
Cash	50,000	45,000
Total current assets	500,000	280,000
Total assets	**1,260,000**	**920,000**
Equity and liabilities		
Share capital and retained earnings	660,000	470,000
Bonds payable	240,000	210,000
Accounts payable	300,000	180,000
Income taxes payable	60,000	60,000
Total current liabilities	360,000	240,000
Total liabilities	600,000	450,000
Total equity and liabilities	**1,260,000**	**920,000**

Ratio	Formula	Orso	Lince
Current ratio			
Acid test ratio			
Cash ratio			

✎ WORKING

✎ WHICH ONE A SUPPLIER WOULD PREFER

G9 Quality of current assets LIQUIDITY

YOUR TURN

Two companies report identical current ratios of 2.0 and identical current liabilities of €500,000. Their current assets are composed as follows.

	Company X	Company Y
Cash	250,000	40,000
Accounts receivable, net	400,000	260,000
Inventories	300,000	620,000
Prepaid expenses	50,000	80,000
Total current assets	**1,000,000**	**1,000,000**

Required

1. Compute the acid test and cash ratios, and the common-size composition of current assets, for each company.
2. State which company is more liquid and why the current ratio did not show it.

Measure	Company X	Company Y
Acid test ratio		
Cash ratio		
Cash %		
Receivables %		
Inventories %		
Prepaid %		

✎ COMMENT

TYPE G.2

Receivables and inventory turnover

Turnover on averages, days on closing balances, and why they differ.

① Worked example → ② G2 → ③ G3

WORKED EXAMPLE · SOLVED

Receivables and inventory: turnover and days

Eco Spa: sales (all on credit) €1,080,000; cost of goods sold €720,000; accounts receivable €100,000 at the start of the year and €140,000 at the end; inventory €110,000 at the start and €130,000 at the end. Compute the turnover and days measures.

Step 1 — Receivables

Accounts receivable turnover	Sales ÷ Average AR = 1,080,000 ÷ [(100,000 + 140,000) ÷ 2 = 120,000]	**9.0x**
Collection period	360 ÷ AR turnover = 360 ÷ 9.0	**40.0 days**
Days' sales in receivables	Closing AR ÷ (Sales ÷ 360) = 140,000 ÷ 3,000	**46.7 days**

Step 2 — Inventory

Inventory turnover	COGS ÷ Average inventory = 720,000 ÷ [(110,000 + 130,000) ÷ 2 = 120,000]	**6.0x**
Days to sell inventory	360 ÷ inventory turnover = 360 ÷ 6.0	**60.0 days**
Days' sales in inventory	Closing inventory ÷ (COGS ÷ 360) = 130,000 ÷ 2,000	**65.0 days**

Step 3 — Why the two days measures differ

The closing-balance measures (46.7 and 65.0 days) are longer than the average-based ones (40 and 60 days) because both balances **rose during the year**: the year-end position is worse than the year's average. If the two were equal, balances were flat.

NOTE Switching from FIFO to LIFO with rising prices raises cost of goods sold and lowers inventory, so inventory turnover rises and both days measures fall — the company looks faster without selling faster.

✎ EXTRA SPACE

G2 Receivables liquidity LIQUIDITY

YOUR TURN

Using the data of Orso Spa in exercise G1, and assuming that accounts receivable at the start of the year were €156,000 and that all sales are on credit, compute the accounts receivable turnover, the days' sales in receivables on the closing balance, and the receivables collection period. Comment on the difference between the last two.

Measure	Computation	Result
Accounts receivable turnover		
Days' sales in receivables		
Collection period		

✎ WHY THE LAST TWO DIFFER

G3 Inventory liquidity LIQUIDITY

YOUR TURN

Still with Orso Spa. Opening inventory was €200,000. Compute the inventory turnover, the days' sales in inventory on the closing balance, and the days-to-sell-inventory ratio. Then state what would happen to each if the company switched from FIFO to LIFO in a period of rising prices.

Measure	Computation	Result	If FIFO → LIFO
Inventory turnover			
Days' sales in inventory			
Days to sell inventory			

✎ COMMENT

TYPE G.3

Payables and the net trade cycle

How many days of operations the company must finance itself.

① Worked example → ② G4 → ③ G5

WORKED EXAMPLE · SOLVED

Payables and the net trade cycle

Continue with Eco Spa: accounts payable at year end are €72,000. Assume all cost of sales items are purchased. Compute the days' purchases in accounts payable and the net trade cycle, and comment.

Step 1 — Days' purchases in accounts payable

Purchases (assumed = COGS)	720,000 ÷ 360 = 2,000 per day	
Days' purchases in AP	Closing AP ÷ (Purchases ÷ 360) = 72,000 ÷ 2,000	**36.0 days**

Step 2 — The net trade cycle

Net trade cycle = Days' sales in receivables + Days' sales in inventory − Days' purchases in AP

Net trade cycle	46.7 + 65.0 − 36.0	**75.7 days**

Step 3 — Two companies or two years: one column each

Days	Eco Spa	How to read it
Receivables	46.7	money owed by customers
+ Inventory	65.0	goods waiting to be sold
− Payables	36.0	credit received from suppliers
= Net trade cycle	**75.7**	**days of sales the company must finance itself**

Comment

Eco pays its suppliers in 36 days but recovers cash from sales only after about 112 days: almost 76 days of operations must be financed with equity or debt. Shortening the cycle (collecting faster, holding less stock) frees cash.

✎ EXTRA SPACE

G4 Payables and the trade cycle LIQUIDITY

YOUR TURN

Compute the days' purchases in accounts payable for Orso and for Lince, assume all cost of sales items are purchased, and then build the net trade cycle for each company using the figures already computed in exercises G1 to G3 (Lince: days' sales in receivables 42, days' sales in inventory 87). Comment.

Days	Orso	Lince
Days' sales in receivables		
+ Days' sales in inventory		
− Days' purchases in accounts payable		
= Net trade cycle		

✎ WORKING

✎ COMMENT

G5 Full liquidity set, two years LIQUIDITY

YOUR TURN

The statements of Vela Group for 2025 and 2024 are given below, in thousands of euro. Compute the current ratio, acid test ratio, cash ratio, days' sales in receivables, days' sales in inventory and days' purchases in accounts payable for both years, then comment on the change in liquidity.

Income statement	2025	2024
Sales	14,400	12,000
Cost of goods sold	9,000	7,200
Selling and administrative expense	2,160	1,800
Operating income	3,240	3,000
Interest expense	600	720
Income before taxes	2,640	2,280
Income tax expense	1,056	912
Net income	**1,584**	**1,368**

✎ EXTRA SPACE

Balance sheet	2025	2024
Assets		
Fixed assets, net	5,400	4,200
Inventory	1,500	1,200
Accounts receivable, net	1,680	1,200
Marketable securities	0	0
Cash	420	600
Total current assets	3,600	3,000
Total assets	**9,000**	**7,200**
Equity and liabilities		
Share capital and retained earnings	5,400	3,900
Bonds payable	1,200	1,500
Accounts payable	2,100	1,500
Income taxes payable	300	300
Total current liabilities	2,400	1,800
Total liabilities	3,600	3,300
Total equity and liabilities	**9,000**	**7,200**

Ratio	2025	2024
Current ratio		
Acid test ratio		
Cash ratio		
Days' sales in receivables		
Days' sales in inventory		
Days' purchases in accounts payable		

✎ WORKING

✎ COMMENT

TYPE G.4

Transactions, ratios and window dressing

Tracking the totals, and why equal changes move a ratio.

① Worked example → ② G6 → ③ G7

WORKED EXAMPLE · SOLVED

The effect of transactions on the ratios; window dressing

(a) At 1 March Sorgente Srl has current assets of €300,000 (including inventory €60,000 and prepaid expenses €10,000) and current liabilities of €150,000. Trace the current ratio and the acid test ratio through: (1) €20,000 of receivables collected; (2) inventory of €30,000 bought on credit; (3) €40,000 of payables paid; (4) a cash dividend of €15,000 declared; (5) equipment bought for €25,000 cash.

(b) Show the effect of paying €100,000 of payables early, for a company with CA 500 / CL 250 and one with CA 200 / CL 250 (thousands).

Step 1 — (a) Track the three totals, then divide

	Current assets	Quick assets	Current liabilities	Current ratio	Acid test
1 March	300,000	230,000	150,000	2.00	1.53
(1) Collect receivables	300,000	230,000	150,000	2.00	1.53
(2) Inventory on credit	330,000	230,000	180,000	1.83	1.28
(3) Pay payables	290,000	190,000	140,000	2.07	1.36
(4) Dividend declared	290,000	190,000	155,000	1.87	1.23
(5) Equipment for cash	265,000	165,000	155,000	1.71	1.06

Quick assets = current assets − inventory − prepaid expenses = 300,000 − 60,000 − 10,000 = 230,000 at the start. (1) only swaps one current asset for another; (4) creates a liability without touching assets.

Step 2 — (b) Window dressing

€ 000	CR before	WC before	CR after	WC after
CA 500, CL 250, pay 100	2.00	250	400 ÷ 150 = 2.67	250
CA 200, CL 250, pay 100	0.80	−50	100 ÷ 150 = 0.67	−50

Comment

Subtracting the same amount from both terms pushes a ratio away from 1: up if it was above 1, down if below. Working capital does not move. A year-end ratio can therefore be dressed up; the analyst looks at working capital, at the ratio during the year and at the cash flow statement.

G6 Effect of transactions on the ratios LIQUIDITY

YOUR TURN

Leach SpA had the following transactions involving current assets and current liabilities during February.

Date	Transaction
Feb 3	Accounts receivable of €30,000 are collected
Feb 7	Equipment is purchased for €56,000 cash
Feb 11	Paid €6,000 for a three-year insurance policy
Feb 14	Accounts payable of €24,000 are paid
Feb 18	Cash dividends of €10,000 are declared

At 1 February current assets were €260,000 and current liabilities €100,000. Current assets included €30,000 of inventory and €4,000 of prepaid expenses.

> **Required**
>
> 1. Compute the current ratio at the beginning of the month and after each transaction.
> 2. Compute the acid test ratio at the beginning of the month and after each transaction.

	Current assets	Quick assets	Current liabilities	Current ratio	Acid test
1 February					
Feb 3					
Feb 7					
Feb 11					
Feb 14					
Feb 18					

✎ WORKING

G7 Window dressing LIQUIDITY

YOUR TURN

Towards the close of the year a finance director instructs that €200,000 of accounts payable be settled early out of cash. Before the payment, current assets are €600,000 and current liabilities €400,000.

> **Required**
>
> 1. Compute the current ratio and working capital before and after.
> 2. Repeat the computation for a second company whose current assets are €300,000 and current liabilities €400,000, paying €100,000.
> 3. Explain what this tells an analyst.

	CR before	WC before	CR after	WC after
First company				
Second company				

✎ WORKING

✎ WHAT THIS TELLS AN ANALYST

TYPE G.5

Back from the ratios; the written credit paragraph

Recovering balances, then the eight-line memorandum.

① Worked example → ② G8 → ③ G10

WORKED EXAMPLE · SOLVED

Working back from the ratios; writing the liquidity paragraph

(a) Rivo Ltd: receivables turnover 8.0 times on net credit sales of €960,000, opening receivables €100,000; inventory turnover 5.0 times on cost of goods sold of €600,000, opening inventory €110,000. Current liabilities €300,000, cash €50,000, no other current assets. Compute the closing balances, the collection period, days' sales in inventory, current and acid test ratios.

(b) Write the liquidity section of a credit memorandum on Rivo (eight lines at most).

Step 1 — (a) Turnover → average → closing balance

Average receivables	960,000 ÷ 8.0	**120,000**
Closing receivables	2 × 120,000 − 100,000	**140,000**
Average inventory	600,000 ÷ 5.0	**120,000**
Closing inventory	2 × 120,000 − 110,000	**130,000**
Collection period	360 ÷ 8.0	**45 days**
Days' sales in inventory	130,000 ÷ (600,000 ÷ 360)	**78 days**
Current ratio	(50,000 + 140,000 + 130,000) ÷ 300,000	**1.07**
Acid test ratio	(50,000 + 140,000) ÷ 300,000	**0.63**

Step 2 — (b) A model paragraph

Rivo's liquidity is weak. The current ratio of 1.07 is well below the 2:1 benchmark and the acid test of 0.63 means that, without selling inventory, the company can meet only 63 per cent of its current liabilities. Inventory sits for 78 days and customers pay in 45, so cash comes back slowly. ***Recommendation:*** *grant only short-term credit, secured and with a limit. Before lending we would request (1) an ageing schedule of receivables and (2) a monthly cash flow forecast for the next twelve months.*

Comment

Structure of every written answer: the figures with their benchmark → what they mean → the recommendation → the information you would ask for.

✎ EXTRA SPACE

G8 Backwards from the ratios LIQUIDITY

YOUR TURN

Monrio Ltd discloses that its accounts receivable turnover for the year was 7.2 times and its inventory turnover 6.0 times. Net credit sales were €2,160,000 and cost of goods sold €1,440,000. Opening accounts receivable were €280,000 and opening inventory €210,000.

> **Required**
>
> 1. Compute closing accounts receivable and closing inventory.
> 2. Compute the collection period and days' sales in inventory.
> 3. The company reports current liabilities of €520,000 and cash of €60,000, with no marketable securities and no prepaid expenses. Compute the current ratio and the acid test ratio.

✎ WORKING

G10 Write the liquidity section of a credit report LIQUIDITY

YOUR TURN

You are the lending officer. Using the ratios computed for Orso and Lince in exercises G1 to G4, write the liquidity section of a credit memorandum: no more than eight lines, with a recommendation and the two pieces of information you would request before lending.

✎ CREDIT MEMORANDUM — LIQUIDITY

SECTION H

Solvency and capital structure

The four structure ratios, earnings coverage, and the cases where the solvency answer contradicts the liquidity answer.

TYPE H.1

The capital structure ratios

From the balance sheet or straight from a common-size statement.

① Worked example → ② H1 → ③ H7

WORKED EXAMPLE · SOLVED

The four capital structure ratios — from the balance sheet and from percentages

(a) Ferro Spa: total assets €2,000,000; current liabilities €400,000; bonds payable (long-term) €600,000; equity €1,000,000. Compute the four capital structure ratios. **(b)** Show the same ratios from a common-size financing side: current liabilities 20%, long-term debt 30%, equity 50%.

Step 1 — (a) Four ratios, two families

Total debt ratio	Total liabilities ÷ Total assets = 1,000,000 ÷ 2,000,000	**50.0%**
Total debt to equity	Total liabilities ÷ Equity = 1,000,000 ÷ 1,000,000	**1.00**
Long-term debt to equity	Non-current liabilities ÷ Equity = 600,000 ÷ 1,000,000	**0.60**
Short-term debt to total debt	Current liabilities ÷ Total liabilities = 400,000 ÷ 1,000,000	**40.0%**

Step 2 — (b) Size cancels: the percentages give the ratios directly

Total debt ratio	(20 + 30) ÷ 100	**50.0%**
Total debt to equity	50 ÷ 50	**1.00**
Long-term debt to equity	30 ÷ 50	**0.60**
Short-term debt to total debt	20 ÷ 50	**40.0%**

NOTE Preferred stock is normally counted as equity; if it is redeemable or has a fixed maturity, treat it as debt. Either way, state the choice.

Comment

Half of Ferro's assets are financed by creditors, one euro of debt per euro of equity; 40 per cent of the debt falls due within a year.

✎ EXTRA SPACE

H1 The four capital structure ratios SOLVENCY

YOUR TURN

Using the statements of Orso Spa and Lince Srl from exercise G1, compute the total debt ratio, the total debt to equity ratio, the long-term debt to equity ratio and the short-term debt to total debt ratio for both companies.

Ratio	Formula	Orso	Lince
Total debt ratio			
Total debt to equity			
Long-term debt to equity			
Short-term debt to total debt			

✎ WORKING

H7 Solvency from a common-size statement SOLVENCY

YOUR TURN

Only the common-size financing side of Marenco Spa is available. Total liabilities and equity are €3,000,000. Compute the four capital structure ratios directly from the percentages.

	%
Current liabilities	18.0%
Long-term debt	27.0%
Preferred stock	10.0%
Common stock	32.0%
Paid-in capital	3.0%
Retained earnings	10.0%

Ratio	From the percentages	Result
Total debt ratio		
Total debt to equity		
Long-term debt to equity		
Short-term debt to total debt		

✎ HOW YOU TREATED PREFERRED STOCK

TYPE H.2

Times interest earned

Both routes, why they agree, and when they do not.

① Worked example → ② H2 → ③ H8

WORKED EXAMPLE · SOLVED

Times interest earned by both routes, and the trap

Gemma Spa reports operating income of €300,000, interest expense of €60,000, income before tax of €240,000, tax of €72,000 and net income of €168,000. **(a)** Compute times interest earned by both routes. **(b)** Repeat if the year also included a €90,000 gain on the sale of a subsidiary (tax 30 per cent on the higher income).

Step 1 — (a) Two routes to the same numerator, earnings before interest and taxes

Route 1	(Net income + Tax + Interest) ÷ Interest = (168 + 72 + 60) ÷ 60	**5.0x**
Route 2	Operating income ÷ Interest = 300 ÷ 60	**5.0x**

Limitation: the ratio treats interest as the **only fixed charge** (it ignores lease payments and principal repayments) and uses accrual earnings, not cash.

Step 2 — (b) A one-off gain separates the two routes

Income before tax	300 + 90 − 60	**330**
Tax 30% / Net income	99 / 231	
Route 1	(231 + 99 + 60) ÷ 60	**6.5x**
Route 2	300 ÷ 60	**5.0x**

Comment

Put **5.0x** in a credit report: interest is paid year after year out of recurring operating income, and the gain on the subsidiary will not come back. Route 1 flatters coverage whenever net income contains non-operating items.

✎ EXTRA SPACE

H2 Times interest earned, by both routes SOLVENCY

YOUR TURN

Compute the times interest earned ratio for Orso and Lince by both of the routes given in the formula, and show that they agree. Then state the limitation of the ratio named on the chapter 10 slide.

Route	Orso	Lince
(Net income + Tax + Interest) ÷ Interest		
Operating income ÷ Interest		

✎ WORKING

✎ LIMITATION

H8 A trap in times interest earned SOLVENCY

YOUR TURN

Fenice Srl reports operating income of €480,000, a gain on the disposal of a subsidiary of €220,000, interest expense of €120,000, tax of €174,000 and net income of €406,000. Compute times interest earned by both routes, explain the discrepancy and say which figure you would put in a credit report.

✎ WORKING

✎ COMMENT

TYPE H.3

The full solvency set

Five ratios, two columns, one direction of travel.

① Worked example → ② H3 → ③ H4

WORKED EXAMPLE · SOLVED

The full solvency set over two years

Lupo Spa, in € thousands. 2025: total assets 5,000; current liabilities 1,000; long-term debt 1,500; equity 2,500; operating income 600; interest 150. 2024: total assets 4,000; current liabilities 900; long-term debt 1,100; equity 2,000; operating income 560; interest 110. Compute the five solvency ratios and comment on the direction of travel.

Step 1 — One column per year

Ratio	Working 2025	2025	2024
Total debt ratio	2,500 ÷ 5,000	50.0%	50.0%
Total debt to equity	2,500 ÷ 2,500	1.00	1.00
Long-term debt to equity	1,500 ÷ 2,500	0.60	0.55
Short-term debt to total debt	1,000 ÷ 2,500	40.0%	45.0%
Times interest earned	600 ÷ 150	4.0x	5.1x

2024 workings: 2,000 ÷ 4,000; 2,000 ÷ 2,000; 1,100 ÷ 2,000; 900 ÷ 2,000; 560 ÷ 110 = 5.09.

Comment

The overall weight of debt is unchanged at half of assets, and the mix has moved towards long-term debt — less refinancing risk. But coverage fell from 5.1 to 4.0 times because interest grew 36 per cent against 7 per cent for operating income: the new debt is expensive. Stable structure, weaker capacity to service it.

✎ EXTRA SPACE

H3 Full solvency set, two companies SOLVENCY

YOUR TURN

Calculate the five solvency ratios for Orso and Lince and say which company a bondholder should prefer.

Ratio	Orso	Lince
Total debt ratio		
Total debt to equity		
Long-term debt to equity		
Short-term debt to total debt		
Times interest earned		

✎ WORKING

✎ WHICH ONE A BONDHOLDER SHOULD PREFER

H4 Solvency over two years SOLVENCY

YOUR TURN

Using the Vela Group statements of exercise G5, compute the five solvency ratios for 2025 and 2024 and comment on the direction of travel.

Ratio	2025	2024
Total debt ratio		
Total debt to equity		
Long-term debt to equity		
Short-term debt to total debt		
Times interest earned		

✎ WORKING

✎ COMMENT

TYPE H.4

Financing decisions and the tax shield

Before-and-after ratios, and why interest costs less than it seems.

① Worked example → ② H5 → ③ H6

WORKED EXAMPLE · SOLVED

A financing decision through the ratios; the tax shield of interest

(a) Ponte Srl (€ thousands): current assets 600, current liabilities 400, long-term debt 200, total assets 1,500, equity 900, operating income 180, interest 20. It issues 150 of bonds at 8 per cent and uses the cash to pay 150 of accounts payable. Show the effect on the ratios.

(b) Two identical companies earn income before interest and taxes of €400; Leva Spa has €2,000 of 8 per cent debt, Piana Spa none. Tax rate 30 per cent. Compare the total return to all security holders.

Step 1 — (a) Before and after

Ratio	Before	After	Why
Current ratio	600 ÷ 400 = 1.50	600 ÷ 250 = 2.40	cash in and out; CL −150
Total debt ratio	600 ÷ 1,500 = 40.0%	600 ÷ 1,500 = 40.0%	total debt unchanged
Long-term debt to equity	200 ÷ 900 = 0.22	350 ÷ 900 = 0.39	+150 bonds
Short-term to total debt	400 ÷ 600 = 66.7%	250 ÷ 600 = 41.7%	mix lengthens
Times interest earned	180 ÷ 20 = 9.0x	180 ÷ 32 = 5.6x	+12 interest (8% × 150)

Step 2 — (b) Interest is deductible; dividends are not

	Leva (debt)	Piana (no debt)
Income before interest and taxes	400	400
Interest (8% × 2,000)	(160)	—
Income before tax	240	400
Tax 30%	(72)	(120)
Net income to shareholders	168	280
Interest to lenders	160	—
Total return to all security holders	**328**	**280**

Comment

(a) Liquidity improves and total leverage is unchanged, but the company swaps free supplier credit for interest-bearing debt, so coverage falls. (b) Leva's holders receive 48 more: the tax saved on interest, 30% × 160. Debt is cheaper than it looks after tax.

✎ EXTRA SPACE

H5 Effect of a new bond issue SOLVENCY

YOUR TURN

Orso Spa is considering issuing €300,000 of ten-year bonds at 7 per cent and using the proceeds to repay €300,000 of accounts payable. Using the data of exercise G1, and assuming the operating income and the tax rate are unchanged, compute the effect on the current ratio, the total debt ratio, the long-term debt to equity ratio, the short-term debt to total debt ratio and times interest earned.

Ratio	Before	After
Current ratio		
Total debt ratio		
Long-term debt to equity		
Short-term debt to total debt		
Times interest earned		

✎ WORKING

✎ COMMENT

H6 Tax deductibility of interest SOLVENCY

YOUR TURN

Two identical companies each earn income before interest and taxes of €250. Risky Spa is financed with €1,000 of 10 per cent debt; Safety Spa has no debt. The tax rate is 40 per cent. Prepare the comparison that shows the total return to all security holders, and state the conclusion.

	Risky Spa	Safety Spa
Income before interest and taxes		
Interest		
Income before tax		
Tax (40%)		
Net income		
Interest to lenders		
Total return to all security holders		

✎ CONCLUSION

TYPE H.5

Back from the ratios; liquidity against solvency

Recovering the blocks, and the written conclusion.

① Worked example → ② H9 → ③ H10

WORKED EXAMPLE · SOLVED

Working back from the ratios; when liquidity and solvency disagree

(a) Total assets are €5,000,000, the total debt ratio is 40 per cent, short-term debt to total debt 30 per cent, times interest earned 6.0, and the average interest rate on all debt 5 per cent. Recover the balance sheet blocks and the income figures.

(b) Over two years a company's current ratio went from 1.4 to 1.8 and its acid test from 0.8 to 1.0, while its total debt ratio rose from 45 to 58 per cent and times interest earned fell from 6.0 to 3.2. Write the conclusion.

Step 1 — (a) Peel the ratios one at a time

Total liabilities	40% × 5,000,000	**2,000,000**
Current liabilities	30% × 2,000,000	**600,000**
Non-current liabilities	2,000,000 − 600,000	**1,400,000**
Total equity	5,000,000 − 2,000,000	**3,000,000**
Interest expense	5% × 2,000,000	**100,000**
Operating income	6.0 × 100,000	**600,000**
Total debt to equity	2,000,000 ÷ 3,000,000	**0.67**
Long-term debt to equity	1,400,000 ÷ 3,000,000	**0.47**

Step 2 — (b) A model conclusion

Liquidity has improved: the current ratio rose from 1.4 to 1.8 and the acid test reached 1.0. Solvency has deteriorated: debt now finances 58 per cent of assets and coverage has almost halved to 3.2 times. The two diverge because the company has raised long-term debt and kept part of it in current assets — it is liquid today because it has borrowed for tomorrow. We would monitor times interest earned, the maturity schedule of the new debt and whether operating income grows enough to service it.

✎ EXTRA SPACE

H9 Working backwards from the ratios SOLVENCY

YOUR TURN

Dorelli NV discloses a total debt ratio of 60 per cent, a short-term debt to total debt ratio of 35 per cent and times interest earned of 4.5. Total assets are €8,000,000 and the average interest rate on all debt is 6 per cent.

> **Required**
>
> 1. Compute total liabilities, current liabilities, non-current liabilities and total equity.
> 2. Compute interest expense and operating income.
> 3. Compute the total debt to equity and the long-term debt to equity ratios.

✎ WORKING



H10 Liquidity and solvency can point opposite ways SOLVENCY

YOUR TURN

Using everything computed for Vela Group in exercises G5 and H4, write the eight-line conclusion of a credit analysis: state the direction of liquidity, the direction of solvency, the reason they diverge, and what you would monitor.

✎ CONCLUSION OF THE CREDIT ANALYSIS

SECTION I

Profitability, NOPAT and RNOA

The substitutions the lecturer makes, the operating / financial cut of the balance sheet, and the two-way decomposition of the return on operating capital.

TYPE I.1

NOPAT and the operating profit margin

Name the substitution, compute, then break the margin into its lines.

① Worked example → ② I1 → ③ I5

WORKED EXAMPLE · SOLVED

NOPAT, the operating profit margin and its components

NOVA SPA

Statements for 2025 — used in every worked example of Sections I and J. Tax rate 30 per cent.

Income statement 2025	€ 000	Balance sheet 31 Dec 2025	€ 000
Sales	2,000	Fixed assets, net	900
Cost of goods sold	1,200	Inventory	250
Selling and administrative expense	500	Accounts receivable	200
Operating income	**300**	Marketable securities	50
Interest expense	40	Cash	100
Income before tax	**260**	**Total assets**	**1,500**
Income tax (30%)	78	Equity	800
Net income	**182**	Bonds payable	400
		Accounts payable	220
		Income taxes payable	80
		Total equity and liabilities	**1,500**

Compute the profit margin of Nova Spa and disaggregate it.

Step 1 — Name the substitution, then the formula

"Operating profit margin (instead of Profit margin)" — the course measures the margin on operations, after tax, before financing: NOPAT ÷ Sales. Tax rate assumed: 30 per cent.

NOPAT	Operating income × (1 − t) = 300 × 0.70	**210**
Operating profit margin	NOPAT ÷ Sales = 210 ÷ 2,000	**10.5%**

Step 2 — Disaggregate the margin line by line

Gross profit margin	(2,000 − 1,200) ÷ 2,000	**40.0%**
− S&A as % of sales	500 ÷ 2,000	**25.0%**
= Operating income margin, pre-tax	40.0% − 25.0% = 300 ÷ 2,000	**15.0%**
× (1 − tax rate)	15.0% × 0.70	**10.5%** ✓

Comment

Nova keeps 40 cents of gross profit per euro of sales and spends 25 on selling and administration; after tax, 10.5 cents of each euro of sales is operating profit. To compare two companies, compare the components: that is where the difference arises.

I1 NOPAT and the operating profit margin PROFITABILITY

YOUR TURN

Using the statements of Orso Spa and Lince Srl given in exercise G1, compute the profit margin for both companies. Assume a tax rate of 30 per cent.

✎ SUBSTITUTION AND FORMULA

	Orso	Lince
Operating income		
NOPAT = Operating income × (1 − 0.30)		
Sales		
Operating profit margin		

✎ COMMENT

I5 Disaggregating the margin PROFITABILITY

YOUR TURN

For Orso Spa, disaggregate the operating profit margin into its components: gross profit margin, and selling and administrative expense as a percentage of sales. Do the same for Lince and comment on where the difference arises.

Component	Orso	Lince
Gross profit margin		
− S&A expense as % of sales		
= Operating income margin (pre-tax)		
× (1 − 0.30) = Operating profit margin		

✎ WORKING

✎ WHERE THE DIFFERENCE ARISES

TYPE I.2

Net operating assets and the turnover

The operating / financial cut, the identity, and the turnover.

① Worked example → ② I2 → ③ I3

WORKED EXAMPLE · SOLVED

Net operating assets, net financial obligations and the turnover

Use the Nova Spa statements. Treat marketable securities as a financial asset and all current liabilities as operating. Compute NOA and NFO, check the identity, and compute the asset turnover.

Step 1 — Re-cut the balance sheet

Operating side	€ 000	Financial side	€ 000
Total assets	1,500	Bonds payable	400
− Marketable securities	(50)	− Marketable securities	(50)
= Operating assets	1,450	**= Net financial obligations (NFO)**	**350**
− Operating liabilities (all CL: 220 + 80)	(300)	Equity	800
= Net operating assets (NOA)	**1,150**	**NFO + Equity**	**1,150 ✓**

Step 2 — The turnover, on what is available

"Operating asset turnover (instead of Asset turnover)". Can we calculate the average? No, we can't — only one balance sheet date is given, so we use year-end NOA.

Operating asset turnover	Sales ÷ NOA = 2,000 ÷ 1,150	**1.74x**

Comment

Every euro of capital invested in operations generates €1.74 of sales. NOA = NFO + Equity is the check that nothing was lost or counted twice in the re-cut.

✎ EXTRA SPACE

I2 Net operating assets and net financial obligations PROFITABILITY

YOUR TURN

For both Orso and Lince, re-cut the balance sheet into operating and financial components. Treat marketable securities as a financial asset and all current liabilities as operating. Show that NOA equals NFO plus equity.

	Orso	Lince
Total assets		
− Financial assets (marketable securities)		
= Operating assets		
− Operating liabilities		
= NOA		
Financial obligations (bonds)		
− Financial assets		
= NFO		
Equity		
Check: NFO + Equity		

I3 Operating asset turnover PROFITABILITY

YOUR TURN

Compute the asset turnover for Orso and Lince. Only one year of balance sheet data is available.

✎ SUBSTITUTION, AND THE AVERAGE

	Orso	Lince
Sales		
NOA		
Operating asset turnover		

✎ COMMENT

TYPE I.3

RNOA = margin × turnover

The decomposition, and why margin and turnover are substitutes.

① Worked example → ② I4 → ③ I7

WORKED EXAMPLE · SOLVED

RNOA as margin times turnover

(a) Compute the return on assets of Nova Spa and verify that it decomposes into margin × turnover. **(b)** Two companies each earn NOPAT of 300 on NOA of 2,000; X has sales of 1,000, Y of 6,000. Complete the decomposition.

Step 1 — (a) Substitution, formula, check

"Return on net operating assets, RNOA (instead of Return on assets)".

RNOA	NOPAT ÷ NOA = 210 ÷ 1,150	**18.3%**
Check	Operating profit margin × Operating asset turnover = 10.5% × 1.739	**18.3%** ✓

Step 2 — (b) Same return, two business models

	X	Y
NOPAT	300	300
Sales	1,000	6,000
Net operating assets	2,000	2,000
NOPAT margin (NOPAT ÷ Sales)	30.0%	5.0%
NOA turnover (Sales ÷ NOA)	0.5x	3.0x
RNOA (margin × turnover)	**15.0%**	**15.0%**

Comment

X earns a lot on each sale but turns its capital slowly (a luxury or capital-intensive model); Y earns little per sale but turns its capital six times faster (a discount retailer). The return on capital is identical: margin and turnover are substitutes.

✎ EXTRA SPACE

I4 RNOA, and the check that it decomposes PROFITABILITY

YOUR TURN

Compute the return on assets for Orso and Lince, then verify that it equals the operating profit margin multiplied by the operating asset turnover.

✎ SUBSTITUTION

	Orso	Lince
NOPAT		
NOA		
RNOA = NOPAT ÷ NOA		
Check: margin × turnover		

✎ COMMENT

I7 Margin and turnover are interchangeable PROFITABILITY

YOUR TURN

Three companies report the following. Complete the table and explain in four lines what it says about business models.

	A	B	C
NOPAT	500,000	500,000	100,000
Sales	5,000,000	10,000,000	10,000,000
Net operating assets	5,000,000	5,000,000	1,000,000
NOPAT margin			
NOA turnover			
RNOA			

✎ COMMENT

TYPE I.4

Averages and operating liabilities leverage

How much the denominator matters, and what OLLEV adds.

① Worked example → ② I6 → ③ I8

WORKED EXAMPLE · SOLVED

Year-end or average NOA; operating liabilities leverage

(a) Nova's NOA at 31 December 2024 was €850 thousand. Compute RNOA on year-end and on average NOA. **(b)** Compute OLLEV at year end and verify RNOA = (NOPAT ÷ Sales) × (Sales ÷ OA) × (1 + OLLEV).

Step 1 — (a) Two dates → use the average

RNOA on year-end NOA	210 ÷ 1,150	**18.3%**
Average NOA	(850 + 1,150) ÷ 2	**1,000**
RNOA on average NOA	210 ÷ 1,000	**21.0%**

NOA grew 35 per cent during the year. NOPAT was earned on the capital employed through the year, not on the year-end figure, which understates the return by 2.7 points. With two dates, present the average.

Step 2 — (b) Operating liabilities lift the return on the capital the firm provides

OLLEV	Operating liabilities ÷ NOA = 300 ÷ 1,150	**0.261**
NOPAT ÷ Sales	210 ÷ 2,000	**10.5%**
Sales ÷ Operating assets	2,000 ÷ 1,450	**1.379x**
RNOA	10.5% × 1.379 × (1 + 0.261)	**18.3%** ✓

Comment

In words: suppliers and the tax authority finance 300 of Nova's 1,450 of operating assets free of interest. The return on operating assets (14.5 per cent) is multiplied by 1.26 to give the return on the net capital Nova itself invests.

✎ EXTRA SPACE

I6 Year-end or average, and how much it matters PROFITABILITY

YOUR TURN

Using the Vela Group statements of exercise G5, compute RNOA for 2025 twice: once on the year-end NOA, once on the average of the opening and closing NOA. The tax rate is 40 per cent. Comment on the difference and say which you would present.

✎ WORKING

✎ COMMENT

I8 Operating liabilities leverage PROFITABILITY

YOUR TURN

For Orso Spa, compute the operating liabilities leverage ratio OLLEV and verify the identity

RNOA = (NOPAT ÷ Sales) × (Sales ÷ Average OA) × (1 + OLLEV)

Then state what the identity means in words.

✎ WORKING

✎ WHAT THE IDENTITY MEANS

TYPE I.5

Disaggregating the turnover; the written comparison

Days for every piece of NOA, then the paragraph.

① Worked example → ② I9 → ③ I10

WORKED EXAMPLE · SOLVED

Disaggregating the turnover; writing the profitability comparison

(a) At the start of 2025 Nova had receivables of €160 thousand, inventory 230, accounts payable 180 and long-term operating assets 820. Compute the component turnovers and days on average balances. **(b)** A competitor, Stella, has an operating profit margin of 8.0 per cent, an operating asset turnover of 2.50 and an RNOA of 20.0 per cent. Write the comparison.

Step 1 — (a) One turnover for each piece of NOA

Component	Average balance	Turnover	Days (360 ÷ turnover)
Receivables (Sales ÷ avg AR)	(160 + 200) ÷ 2 = 180	2,000 ÷ 180 = 11.1x	32.4
Inventory (COGS ÷ avg inv.)	(230 + 250) ÷ 2 = 240	1,200 ÷ 240 = 5.0x	72.0
Accounts payable (COGS ÷ avg AP)	(180 + 220) ÷ 2 = 200	1,200 ÷ 200 = 6.0x	60.0
Long-term operating assets (Sales ÷ avg)	(820 + 900) ÷ 2 = 860	2,000 ÷ 860 = 2.33x	—

The weakest component is **inventory**: 72 days of stock against 32 days to collect. Payables at 60 days finance most of it.

Step 2 — (b) A model comparison, in the comparative register

Nova earns a higher operating profit margin than Stella (10.5 against 8.0 per cent), thanks to a 40 per cent gross margin, but it turns its net operating assets more slowly (1.74 against 2.50 times). The slower turnover more than offsets the higher margin: Nova's RNOA is 18.3 per cent against Stella's 20.0. The gap lies in asset use, chiefly the 72 days of inventory; reducing stock would raise Nova's RNOA without touching its margin.

✎ EXTRA SPACE

I9 Disaggregating the turnover PROFITABILITY

YOUR TURN

Orso Spa opened the year with receivables of €156,000, inventory of €200,000, accounts payable of €264,000 and long-term operating assets of €700,000. Compute the receivables turnover, the inventory turnover, the accounts payable turnover and the long-term operating asset turnover, with the accompanying days ratios, and say which component of the operating asset turnover is weakest.

Component	Average balance	Turnover	Days
Receivables			
Inventory			
Accounts payable			
Long-term operating assets			

✎ WORKING

✎ WEAKEST COMPONENT

I10 Write the profitability comparison PROFITABILITY

YOUR TURN

Using everything computed in exercises I1 to I5, write the profitability paragraph of a report comparing Orso and Lince. Eight lines at most, in the comparative register the published solutions use.

✎ PROFITABILITY — ORSO AND LINCE

SECTION J

Return on common equity and its disaggregation

ROCE, the six-line chain that produces LEV and Spread, and the cases where leverage works against the shareholder.

TYPE J.1

Return on common equity

Net income to ordinary shareholders over their equity — and the preferred trap.

① Worked example → ② J1 → ③ J6

WORKED EXAMPLE · SOLVED

Return on common equity — with and without preferred shares

(a) Compute ROCE for Nova Spa (net income €182 thousand, equity €800 thousand, no preferred shares; one balance sheet date). **(b)** Sole Spa reports net income of €300,000; its equity of €2,000,000 includes €400,000 of 8 per cent preferred stock, whose dividend was paid in full. Compute ROCE.

Step 1 — (a) The common shareholders' return

ROCE = (Net income − Preferred dividends) ÷ Average common equity

Can we calculate the average? No, we can't — year-end equity is used.

ROCE, Nova	182 ÷ 800	**22.75%**

Step 2 — (b) Take the preferred out of both terms

Preferred dividend	8% × 400,000	**32,000**
Earnings to common	300,000 − 32,000	**268,000**
Common equity	2,000,000 − 400,000	**1,600,000**
ROCE, Sole	268,000 ÷ 1,600,000	**16.75%**
The careless answer	300,000 ÷ 2,000,000	**15.0% ×**

Comment

ROCE measures the return to the **ordinary** shareholders: preferred dividends leave the numerator and preferred stock leaves the denominator. Forgetting either understates (or overstates) the return.

✎ EXTRA SPACE

J1 Return on common equity ROCE

YOUR TURN

Compute the return on ordinary shareholders' equity for Orso and Lince, using the data of exercise G1. Neither company has preferred shares.

	Orso	Lince
Net income		
Equity (year-end — average not available)		
ROCE		

✎ COMMENT

J6 ROCE with preferred shares ROCE

YOUR TURN

Aurelia NV reports net income of €420,000. Its shareholders' equity of €2,900,000 includes €500,000 of 9 per cent preferred stock, on which the dividend was declared and paid in full. Compute the return on common equity, and state what a careless answer would have been.

✎ WORKING

✎ COMMENT

TYPE J.2

ROCE = RNOA + LEV × Spread

The full chain from NOPAT to ROCE, with the check.

① Worked example → ② J2 → ③ J3

WORKED EXAMPLE · SOLVED

The full disaggregation: ROCE = RNOA + LEV × Spread

Disaggregate Nova Spa's ROCE (statements in Section I). All assets except marketable securities, and all current liabilities, are operating; tax rate 30 per cent; no interest income on the securities; year-end figures.

Step 1 — The chain, one line at a time

NOPAT	300 × (1 − 0.30)	**210**
NOA	(1,500 − 50) − 300	**1,150**
RNOA	210 ÷ 1,150	**18.26%**
NFE	40 × (1 − 0.30)	**28**
NFO	400 − 50	**350**
NFR	28 ÷ 350	**8.00%**
LEV	NFO ÷ Equity = 350 ÷ 800	**0.4375**
Spread	RNOA − NFR = 18.26% − 8.00%	**10.26%**
ROCE	18.26% + 0.4375 × 10.26% = 18.26% + 4.49%	**22.75%**
Check	Net income ÷ Equity = 182 ÷ 800 (and NOPAT − NFE = 210 − 28 = 182)	**22.75%** ✓

Comment

Operations earn 18.3 per cent on net operating assets; debt costs 8.0 per cent after tax. The positive spread of 10.3 points, applied to 0.44 euros of net debt per euro of equity, adds 4.5 points: financial leverage is working for the shareholders.

✎ EXTRA SPACE

J2 Full disaggregation — Orso ROCE

YOUR TURN

Disaggregate Orso's ROCE into its operating and non-operating components, using ROCE = RNOA + (LEV × Spread). Assume all assets except marketable securities, and all current liabilities, are operating, and a 30 per cent tax rate. Comment on the use of financial leverage.

Step	Working	Result
NOPAT = Operating income × (1 − t)		
NOA		
RNOA = NOPAT ÷ NOA		
NFE = Interest × (1 − t)		
NFO		
NFR = NFE ÷ NFO		
LEV = NFO ÷ Equity		
Spread = RNOA − NFR		
ROCE = RNOA + LEV × Spread		
Check: Net income ÷ Equity		

✎ COMMENT

J3 Full disaggregation — Lince ROCE

YOUR TURN

Repeat the disaggregation for Lince Srl and compare the two leverage stories.

Step	Working	Result
NOPAT = Operating income × (1 − t)		
NOA		
RNOA = NOPAT ÷ NOA		
NFE = Interest × (1 − t)		
NFO		
NFR = NFE ÷ NFO		
LEV = NFO ÷ Equity		
Spread = RNOA − NFR		
ROCE = RNOA + LEV × Spread		
Check: Net income ÷ Equity		

✎ THE TWO LEVERAGE STORIES

TYPE J.3

Negative spread

When leverage subtracts from the shareholders' return.

① Worked example → ② J4 → ③ J8

WORKED EXAMPLE · SOLVED

When the spread turns negative

Mira Spa, € thousands, year-end figures. Year A: NOPAT 120, NOA 1,000, NFE 24, NFO 400, equity 600. Year B: NOPAT 60, NOA 1,000, NFE 32, NFO 400, equity 600. Disaggregate ROCE for both years; then state the rule.

Step 1 — Two columns of the same chain

	Year A	Year B
RNOA = NOPAT ÷ NOA	120 ÷ 1,000 = 12.0%	60 ÷ 1,000 = 6.0%
NFR = NFE ÷ NFO	24 ÷ 400 = 6.0%	32 ÷ 400 = 8.0%
LEV = NFO ÷ Equity	400 ÷ 600 = 0.667	0.667
Spread = RNOA − NFR	+6.0%	−2.0%
LEV × Spread	+4.0%	−1.33%
ROCE	**16.0%**	**4.67%**
Check: (NOPAT − NFE) ÷ Equity	96 ÷ 600 = 16.0%	28 ÷ 600 = 4.67%

Step 2 — What happened

Operations halved their return (12 → 6 per cent) while the cost of debt rose (6 → 8 per cent). The spread turned negative, so the same leverage that added 4.0 points in Year A now **subtracts** 1.3 points: an unlevered twin would earn 6.0 per cent, more than Mira.

Comment

THE RULE IN ONE SENTENCE Leverage magnifies: it raises ROCE when RNOA exceeds the net financial rate and lowers it when RNOA falls below it — ROCE = RNOA + LEV × (RNOA − NFR).

✎ EXTRA SPACE

J4 When the spread turns negative ROCE

YOUR TURN

Using the Vela Group statements of exercise G5 and a tax rate of 40 per cent, disaggregate ROCE for 2025 and for 2024 on year-end figures. Explain what happened between the two years.

Step	2025	2024
NOPAT = Operating income × (1 − t)		
NOA		
RNOA = NOPAT ÷ NOA		
NFE = Interest × (1 − t)		
NFO		
NFR = NFE ÷ NFO		
LEV = NFO ÷ Equity		
Spread = RNOA − NFR		
ROCE = RNOA + LEV × Spread		
Check: Net income ÷ Equity		

✎ WHAT HAPPENED

J8 Leverage that destroys value ROCE

YOUR TURN

Two companies have identical operating businesses, each earning an RNOA of 7.0 per cent. Company U is financed entirely by equity. Company L has net financial obligations equal to 1.5 times its equity, at an after-tax net financial rate of 9.0 per cent.

> **Required**
>
> 1. Compute ROCE for both.
> 2. Compute what Company L's ROCE would be if its RNOA rose to 12.0 per cent, leverage and NFR unchanged.
> 3. State the rule in one sentence.

✎ WORKING

✎ THE RULE

TYPE J.4

Disaggregation from summary data

Long-term assets plus working capital as NOA.

① Worked example → ② J5 → ③ J9

WORKED EXAMPLE · SOLVED

Disaggregation from long-term assets and working capital

Fiume Srl: revenue 3,000; operating income 360; interest expense 50; pre-tax income 310; income taxes (40%) 124; net income 186. Balance sheet: long-term assets 1,400; working capital 600; total liabilities 700; total shareholders' equity 1,300. Long-term assets and working capital are operating; tax rate 40 per cent.

Step 1 — Read the condensed balance sheet

Working capital is already **net** of current liabilities, so long-term assets + working capital = NOA = 2,000, and the liabilities shown (700) are the financing ones = NFO. Check: NOA 2,000 = NFO 700 + Equity 1,300 ✓.

Step 2 — The chain

NOPAT	360 × (1 − 0.40)	**216**
RNOA	216 ÷ 2,000	**10.80%**
NFE	50 × (1 − 0.40)	**30**
NFR	30 ÷ 700	**4.29%**
LEV	700 ÷ 1,300	**0.538**
Spread	10.80% − 4.29%	**6.51%**
ROCE	10.80% + 0.538 × 6.51% = 10.80% + 3.51%	**14.31%**
Check	186 ÷ 1,300	**14.31%** ✓

Comment

ROCE of 14.3 per cent: 10.8 points come from operations and 3.5 from leverage, which works because debt costs 4.3 per cent after tax against an operating return of 10.8.

✎ EXTRA SPACE

J5 Disaggregation from summary data ROCE

YOUR TURN

Financial information for Prisma Spa follows. Calculate the ROCE and then its disaggregation ROCE = RNOA + Leverage × Spread. Assume long-term assets and working capital are operating and a 40 per cent tax rate.

Income statement data	2025	Balance sheet data	2025
Revenue	2,500	Long-term assets	700
Operating income	300	Working capital	500
Interest expense	40	Total liabilities	320
Pre-tax income	260	Total shareholders' equity	880
Income taxes	104		
Net income	156		

Step	Working	Result
ROCE = Net income ÷ Equity		
NOPAT = Operating income × (1 − t)		
NOA		
RNOA = NOPAT ÷ NOA		
NFE = Interest × (1 − t)		
NFO		
NFR = NFE ÷ NFO		
LEV = NFO ÷ Equity		
Spread = RNOA − NFR		
ROCE = RNOA + LEV × Spread		

✎ EXTRA SPACE

J9 Disaggregation from long-term assets and working capital ROCE

YOUR TURN

Montaldo Srl reports revenue of €5,000, operating income of €450, interest expense of €60, income taxes of €156 and net income of €234. Its long-term assets are €2,000, its working capital €600, its total liabilities €900 and its equity €1,700. Assume long-term assets and working capital are operating. Compute RNOA, ROCE and the full disaggregation.

✎ TAX RATE IMPLIED

Step	Working	Result
NOPAT = Operating income × (1 − t)		
NOA		
RNOA = NOPAT ÷ NOA		
NFE = Interest × (1 − t)		
NFO		
NFR = NFE ÷ NFO		
LEV = NFO ÷ Equity		
Spread = RNOA − NFR		
ROCE = RNOA + LEV × Spread		
Check: Net income ÷ Equity		

✎ EXTRA SPACE

TYPE J.5

The alternate view; the written return analysis

Margin × turnover × leverage, and the research-note paragraph.

① Worked example → ② J7 → ③ J10

WORKED EXAMPLE · SOLVED

The alternate view of ROCE; writing the return analysis

(a) Sole Spa (example J.1) reports sales of €4,000,000, average total assets of €3,200,000 and common equity of €1,600,000. Verify ROCE = Adjusted profit margin × Asset turnover × Leverage. **(b)** Write the return section of a note on Nova Spa.

Step 1 — (a) Three ratios that multiply back to ROCE

Adjusted profit margin	(Net income − Pref. div.) ÷ Sales = 268,000 ÷ 4,000,000	**6.70%**
Asset turnover	Sales ÷ Average total assets = 4,000,000 ÷ 3,200,000	**1.25x**
Leverage	Average total assets ÷ Average common equity = 3,200,000 ÷ 1,600,000	**2.00**
ROCE	6.70% × 1.25 × 2.00	**16.75%** ✓

Difference in purpose. The alternate view mixes operating and financing items (net income after interest, total assets including financial assets): it tells you *how much* margin, turnover and leverage there is. RNOA + LEV × Spread separates the two activities and tells you *whether* leverage creates value.

Step 2 — (b) A model return section (ten lines at most)

Nova earned a ROCE of 22.8 per cent. Of this, 18.3 points come from operations (RNOA), driven by a 10.5 per cent operating margin and a turnover of 1.74 times; 4.5 points come from financial leverage. Net debt is 0.44 times equity and costs 8.0 per cent after tax, so the spread of 10.3 points is comfortably positive and leverage is creating value. The operating return is the engine and the turnover its weak spot. ***Recommendation to management:*** *reduce the 72 days of inventory to lift turnover; additional debt remains value-creating only while RNOA stays above the 8 per cent cost of borrowing.*

✎ EXTRA SPACE

J7 The alternate view of the disaggregation ROCE

YOUR TURN

Continuing from exercise J6, Aurelia reports sales of €6,000,000 and average total assets of €4,000,000. Verify the alternate view of the ROCE disaggregation

> ROCE = Adjusted profit margin × Asset turnover × Leverage

and say how it differs in purpose from the RNOA + LEV × Spread version.

Component	Working	Result
Adjusted profit margin		
Asset turnover		
Leverage		
ROCE (product)		

✎ DIFFERENCE IN PURPOSE

J10 Write the return analysis ROCE

YOUR TURN

Using the Vela Group disaggregation of exercise J4, write the return section of an equity research note: ten lines at most, ending with a recommendation to management.

✎ RETURN ANALYSIS — VELA GROUP

SECTION K

Statement of cash flows — indirect method

The change-and-classification table, then the statement. This is the single most heavily examined topic in the course: Exercise Session 3 and Review Session 1 are nothing else.

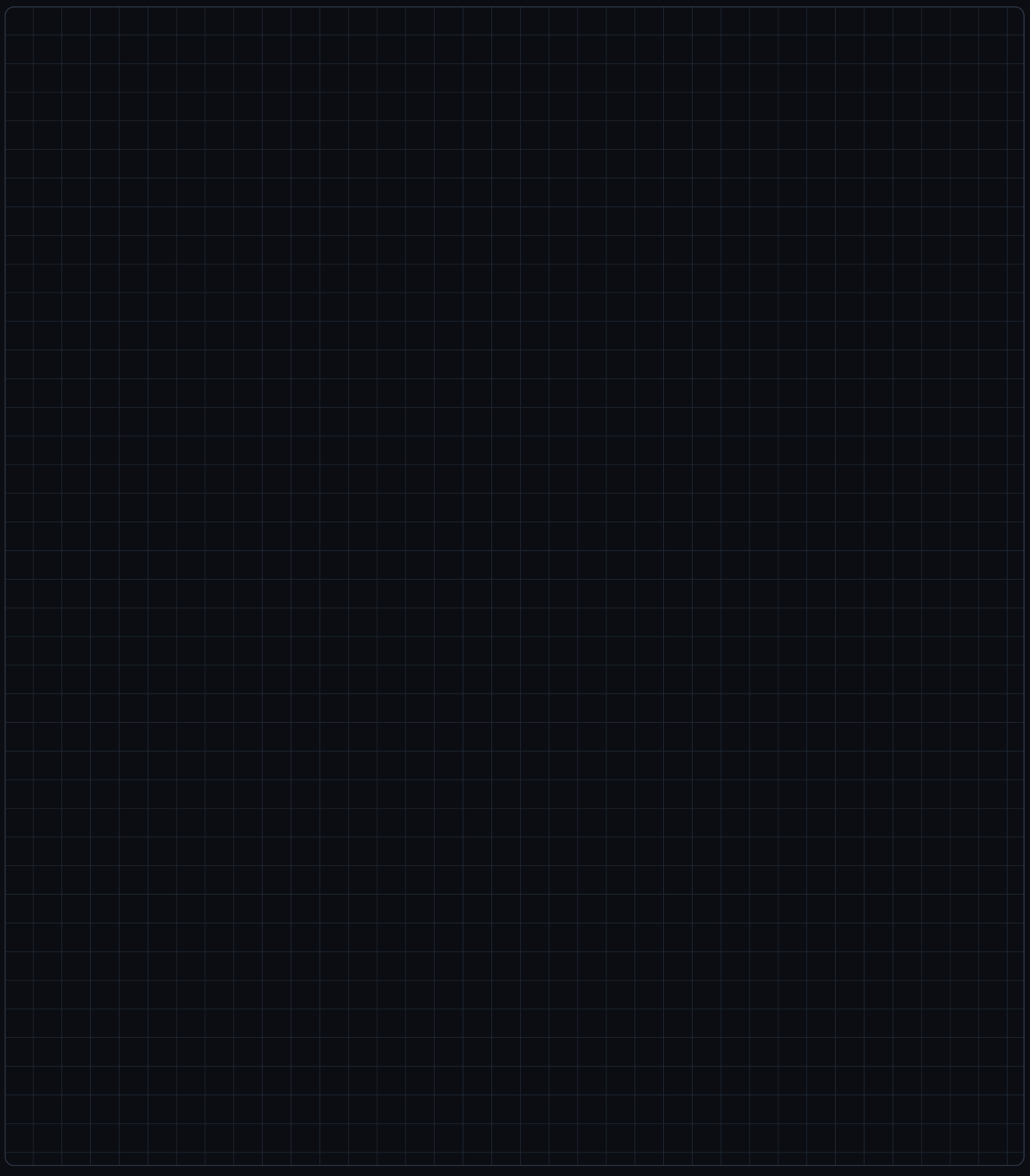
TYPE K.1

The indirect method, from the comparative balance sheet

Change and classification of every line, then the three sections and the cash check.

① Worked example → ② K1 → ③ K5

✎ EXTRA SPACE



WORKED EXAMPLE · SOLVED

The change-and-classification table, then the statement

PAVONE SRL

Comparative Statements of Financial Position — 31 December

	2025	2024
Assets		
Land	€50,000	€70,000
Equipment	210,000	160,000
Accumulated depreciation	(64,000)	(40,000)
Inventories	90,000	100,000
Accounts receivable	70,000	55,000
Cash	49,000	20,000
Total	**€405,000**	**€365,000**
Equity and liabilities		
Share capital — ordinary	€180,000	€150,000
Retained earnings	125,000	85,000
Bonds payable	60,000	80,000
Accounts payable	40,000	50,000
Total	**€405,000**	**€365,000**

Additional information. Net income for 2025 was €70,000. Cash dividends of €30,000 were declared and paid. Bonds of €20,000 were redeemed for cash. Ordinary shares were issued for €30,000 cash. No equipment was sold; land was sold at cost.

Step 1 — Change and classification of every line

	2025	2024	Change	Classification
Land	50,000	70,000	−20,000	Investing: sale at cost, inflow 20,000
Equipment	210,000	160,000	+50,000	Investing: purchase, outflow 50,000
Accumulated depreciation	(64,000)	(40,000)	+24,000	Operating: depreciation, add back
Inventories	90,000	100,000	−10,000	Operating: decrease, +10,000
Accounts receivable	70,000	55,000	+15,000	Operating: increase, −15,000
Cash	49,000	20,000	+29,000	the result to explain
Share capital	180,000	150,000	+30,000	Financing: issue, +30,000
Retained earnings	125,000	85,000	+40,000	Operating NI +70,000; Financing dividends −30,000
Bonds payable	60,000	80,000	−20,000	Financing: redemption, −20,000
Accounts payable	40,000	50,000	−10,000	Operating: decrease, −10,000

SIGNS Asset up = cash down; asset down = cash up. Liability or equity up = cash up; down = cash down. Depreciation is the change in accumulated depreciation when nothing was sold.

Step 2 — The statement

Pavone Srl — Statement of cash flows 2025	€
Cash flows from operating activities	
Net income	70,000
Depreciation expense	24,000
Decrease in inventories	10,000
Increase in accounts receivable	(15,000)
Decrease in accounts payable	(10,000)
Net cash provided by operating activities	**79,000**
Cash flows from investing activities	
Sale of land	20,000
Purchase of equipment	(50,000)
Net cash used in investing activities	**(30,000)**
Cash flows from financing activities	
Issue of ordinary shares	30,000
Redemption of bonds	(20,000)
Dividends paid	(30,000)
Net cash used in financing activities	**(20,000)**
Net increase in cash	**29,000**
Cash at 1 January	20,000
Cash at 31 December	**49,000**

Comment

Close with the check: 79,000 − 30,000 − 20,000 = 29,000, exactly the change in the cash line of the balance sheet. If it does not reconcile, a line of the change table has not been used, or has been used twice.

✎ EXTRA SPACE

K1 Indirect statement of cash flows CASH FLOW

YOUR TURN

ARDEA LTD

Comparative Statements of Financial Position — 31 December

	2025	2024	Change	Classification
Assets				
Land	€60,000	€90,000		
Equipment	320,000	250,000		
Accumulated depreciation	(78,000)	(46,000)		
Inventories	140,000	165,000		
Accounts receivable	95,000	82,000		
Cash	88,000	35,000		
Total	**€625,000**	**€576,000**		
Equity and liabilities				
Share capital — ordinary	€260,000	€215,000		
Retained earnings	215,000	150,000		
Bonds payable	110,000	150,000		
Accounts payable	40,000	61,000		
Total	**€625,000**	**€576,000**		

✎ EXTRA SPACE

Additional information. Net income for 2025 was €125,000. Cash dividends of €60,000 were declared and paid. Bonds payable of €40,000 were redeemed for cash. Ordinary shares were issued for €45,000 cash. No equipment was sold during 2025, but land was sold at cost.

> **Required**
>
> 1. Prepare the comparative statement with the change and the classification of each difference (fill in the columns above).
> 2. Prepare a statement of cash flows for 2025 using the indirect method.

Statement of cash flows — indirect method	€
Cash flows from operating activities	
Net cash from operating activities	
Cash flows from investing activities	
Net cash from investing activities	
Cash flows from financing activities	
Net cash from financing activities	
Net increase (decrease) in cash	
Cash at 1 January	
Cash at 31 December	

K5 A fuller set of current accounts CASH FLOW

YOUR TURN

DORIA NV

Comparative Statements of Financial Position — 31 December

	2025	2024	Change	Classification
Assets				
Land	€120,000	€120,000		
Buildings	400,000	400,000		
Accumulated depreciation	(90,000)	(72,000)		
Long-term investments	65,000	95,000		
Inventory	112,000	96,000		
Accounts receivable	84,000	91,000		
Prepaid expenses	11,000	7,000		
Cash	53,000	38,000		
Total	**€755,000**	**€775,000**		
Equity and liabilities				
Share capital — ordinary	€300,000	€300,000		
Retained earnings	182,000	150,000		
Bonds payable	150,000	200,000		
Accounts payable	94,000	101,000		
Accrued liabilities	29,000	24,000		
Total	**€755,000**	**€775,000**		

✎ EXTRA SPACE

Additional information. Net income was €72,000 and dividends of €40,000 were declared and paid. Long-term investments were sold at cost. No buildings were bought or sold. Bonds of €50,000 were redeemed.

> **Required**
>
> 1. Prepare the statement of cash flows for 2025, indirect method.
> 2. Comment in three lines on how the company funded its dividend.

Statement of cash flows — indirect method	€
Cash flows from operating activities	
Net cash from operating activities	
Cash flows from investing activities	
Net cash from investing activities	
Cash flows from financing activities	
Net cash from financing activities	
Net increase (decrease) in cash	
Cash at 1 January	
Cash at 31 December	

✎ EXTRA SPACE

✎ HOW THE DIVIDEND WAS FUNDED

TYPE K.2

Disposals at a gain or a loss

T-accounts for depreciation and purchases; where the gain and the proceeds go.

① Worked example → ② K2 → ③ K3

WORKED EXAMPLE · SOLVED

Disposals: the gain or loss, the depreciation and the purchases

Rondine Spa reports net income of €50,000. Equipment was €260,000 at the start of 2025 and €300,000 at the end; accumulated depreciation €90,000 and €110,000. During the year equipment that cost €40,000, with accumulated depreciation of €28,000, was sold for €16,000. Find the gain, the depreciation expense and the purchases, and show how they enter the statement.

Step 1 — Gain or loss on the sale

Book value sold	40,000 − 28,000	**12,000**
Gain	16,000 − 12,000	**4,000**

Step 2 — Two T-accounts: opening + increases − decreases = closing

Accumulated depreciation	90,000 + Depreciation − 28,000 (removed on sale) = 110,000	**Dep. 48,000**
Equipment	260,000 + Purchases − 40,000 (cost sold) = 300,000	**Purch. 80,000**

Step 3 — Where each figure goes

Extract of the statement	€
Operating activities	
Net income	50,000
Depreciation expense	48,000
Gain on sale of equipment	(4,000)
Investing activities	
Proceeds from sale of equipment	16,000
Purchase of equipment	(80,000)
Net cash used in investing activities	**(64,000)**

Comment

The gain appears twice with opposite signs: it is inside net income (+4,000), and it is deducted in operating (−4,000) because the cash it relates to — the whole 16,000 of proceeds — belongs to investing. Had the equipment been sold for 9,000, the loss of 3,000 would be **added back** in operating and 9,000 shown in investing.

✎ EXTRA SPACE

K2 Disposal at a gain CASH FLOW

YOUR TURN

BOREA SPA

Comparative Statements of Financial Position — 31 December

	2025	2024	Change	Classification
Assets				
Equipment	€180,000	€200,000		
Accumulated depreciation	(52,000)	(44,000)		
Investments	50,000	38,000		
Accounts receivable	72,000	64,000		
Cash	44,000	51,000		
Total	**€294,000**	**€309,000**		
Equity and liabilities				
Share capital — ordinary	€130,000	€118,000		
Retained earnings	96,000	84,000		
Bonds payable	28,000	70,000		
Accounts payable	40,000	37,000		
Total	**€294,000**	**€309,000**		

Additional information. Net income was €46,000 and dividends declared and paid were €34,000. Equipment that cost €20,000 and had accumulated depreciation of €14,000 was sold for €9,000; no equipment was purchased. All other changes in non-current account balances had a direct effect on cash flows, except the change in accumulated depreciation.

> **Required**
>
> 1. Prepare a statement of cash flows for 2025 using the indirect method.
> 2. Explain in two lines why the gain appears twice in the statement, with opposite signs.

✎ GAIN AND DEPRECIATION

Statement of cash flows — indirect method	€
Cash flows from operating activities	
Net cash from operating activities	
Cash flows from investing activities	
Net cash from investing activities	
Cash flows from financing activities	
Net cash from financing activities	
Net increase (decrease) in cash	
Cash at 1 January	
Cash at 31 December	

✎ WHY THE GAIN APPEARS TWICE

K3 Disposal at a loss, and a derived purchase CASH FLOW

YOUR TURN

CALIPSO SRL

Comparative Statements of Financial Position — 31 December

	2025	2024	Change	Classification
Assets				
Equipment	€240,000	€210,000		
Accumulated depreciation	(62,000)	(55,000)		
Inventory	88,000	70,000		
Accounts receivable	64,000	78,000		
Prepaid expenses	6,000	9,000		
Cash	29,000	41,000		
Total	**€365,000**	**€353,000**		
Equity and liabilities				
Share capital — ordinary	€150,000	€150,000		
Retained earnings	92,000	71,000		
Long-term notes payable	60,000	85,000		
Accounts payable	48,000	35,000		
Income taxes payable	15,000	12,000		
Total	**€365,000**	**€353,000**		

Additional information. Net income was €54,000 and dividends of €33,000 were declared and paid. Equipment costing €25,000 with accumulated depreciation of €17,000 was sold for €5,000. Long-term notes of €25,000 were repaid.

> **Required**
>
> 1. Prepare the statement of cash flows for 2025, indirect method.
> 2. State the amount of equipment purchased during the year and how you derived it.

✎ T-ACCOUNTS: LOSS, DEPRECIATION, PURCHASES

Statement of cash flows — indirect method	€
Cash flows from operating activities	
Net cash from operating activities	
Cash flows from investing activities	
Net cash from investing activities	
Cash flows from financing activities	
Net cash from financing activities	
Net increase (decrease) in cash	
Cash at 1 January	
Cash at 31 December	

✎ EQUIPMENT PURCHASED, AND HOW

TYPE K.3

Income statement data and non-cash activities

Deriving proceeds, purchases and dividends; the schedule of non-cash activities.

① Worked example → ② K6 → ③ K7

WORKED EXAMPLE · SOLVED

Starting from the income statement; a non-cash acquisition

IBIS SRL

Income statement 2025 and comparative statements of financial position

Income statement 2025	€
Net sales	500,000
Cost of goods sold	(300,000)
Gross profit	**200,000**
Operating expenses excluding depreciation	(100,000)
Depreciation expense	(30,000)
Operating profit	**70,000**
Gain on sale of equipment	6,000
Interest expense	(10,000)
Income before tax	**66,000**
Income tax expense	(19,800)
Net income	**46,200**

	2025	2024		2025	2024
Cash	45,200	30,000	Accounts payable	45,000	40,000
Accounts receivable	60,000	50,000	Interest payable	2,000	3,000
Inventory	70,000	75,000	Long-term note payable	80,000	0
Equipment	400,000	300,000	Bank loan	50,000	70,000
Accumulated depreciation	(100,000)	(90,000)	Share capital	200,000	180,000
			Retained earnings	98,200	72,000
Total assets	**475,200**	**365,000**	**Total**	**475,200**	**365,000**

Additional information. Equipment costing €50,000 with accumulated depreciation of €20,000 was sold. A machine costing €80,000 was acquired by issuing a long-term note to the supplier; all other equipment was bought for cash. Shares were issued for cash, part of the bank loan was repaid and dividends were paid.

✎ EXTRA SPACE

Step 1 — Derive the missing cash figures

Proceeds from the sale	Book value + gain = (50,000 − 20,000) + 6,000	**36,000**
Check on depreciation	90,000 + 30,000 − 20,000 = closing 100,000	✓
Total equipment acquired	400,000 − 300,000 + 50,000	**150,000**
of which for cash	150,000 − 80,000 (note to supplier)	**70,000**
Dividends paid	72,000 + 46,200 − 98,200	**20,000**
Shares issued / loan repaid	200,000 − 180,000 / 70,000 − 50,000	**20,000 / 20,000**

Step 2 — The statement and the non-cash schedule

Ibis Srl — Statement of cash flows 2025	**€**
Net income	46,200
Depreciation expense	30,000
Gain on sale of equipment	(6,000)
Increase in accounts receivable	(10,000)
Decrease in inventory	5,000
Increase in accounts payable	5,000
Decrease in interest payable	(1,000)
Net cash provided by operating activities	**69,200**
Proceeds from sale of equipment	36,000
Purchase of equipment	(70,000)
Net cash used in investing activities	**(34,000)**
Issue of shares	20,000
Repayment of bank loan	(20,000)
Dividends paid	(20,000)
Net cash used in financing activities	**(20,000)**
Net increase in cash (30,000 → 45,200)	**15,200**
Non-cash investing and financing activities	
Machine acquired by issuing a long-term note to the supplier	80,000

Comment

The €80,000 machine moved no cash: showing it as an investing outflow and a financing inflow would inflate both sections. It is disclosed in the schedule of non-cash activities instead.

✎ EXTRA SPACE

K6 Non-cash investing and financing CASH FLOW

YOUR TURN

EOLO SPA

Comparative Statements of Financial Position — 31 December

	2025	2024	Change	Classification
Assets				
Cash	€62,000	€47,000		
Accounts receivable	58,000	49,000		
Inventory	71,000	80,000		
Prepaid expenses	8,000	11,000		
Plant assets	520,000	410,000		
Accumulated depreciation	(168,000)	(145,000)		
Total assets	**€551,000**	**€452,000**		
Equity and liabilities				
Accounts payable	€54,000	€61,000		
Accrued expenses	21,000	16,000		
Long-term note payable	90,000	0		
Mortgage payable	0	120,000		
Common stock	240,000	180,000		
Retained earnings	146,000	75,000		
Total liabilities and equity	**€551,000**	**€452,000**		

Additional information.

1. A machine was purchased during the year at a cost of €90,000, financed in full by a long-term note issued to the manufacturer.
2. Other equipment was purchased for €45,000 cash.
3. A machine with a cost of €25,000 and a net book value of €4,000 was sold for €11,000.
4. Net income was €111,000 and dividends paid were €40,000.
5. Common stock was issued for €60,000 cash and the mortgage was repaid in full.

> **Required**
>
> 1. Prepare the statement of cash flows for 2025, indirect method, including the schedule of non-cash activities.
> 2. Explain why the €90,000 machine does not appear in the investing section.

✎ GAIN AND DEPRECIATION

Statement of cash flows — indirect method	€
Cash flows from operating activities	
Net cash from operating activities	
Cash flows from investing activities	
Net cash from investing activities	
Cash flows from financing activities	
Net cash from financing activities	
Net increase (decrease) in cash	
Cash at 1 January	
Cash at 31 December	
Non-cash investing and financing activities	

✎ WHY THE MACHINE IS NOT IN INVESTING

K7 Starting from the income statement CASH FLOW

YOUR TURN

Faro Srl reports the following income statement for 2025.

Net sales	640,000
Cost of goods sold	(384,000)
Gross profit	**256,000**
Operating expenses excluding depreciation	(142,000)
Depreciation expense	(38,000)
Operating profit	**76,000**
Loss on sale of equipment	(6,000)
Interest expense	(12,000)
Income before tax	**58,000**
Income tax expense	(17,400)
Net income	**40,600**

✎ EXTRA SPACE

FARO SRL

Comparative Statements of Financial Position — 31 December

	2025	2024	Change	Classification
Assets				
Cash	€51,000	€33,000		
Accounts receivable	76,000	68,000		
Inventory	92,000	104,000		
Equipment	310,000	280,000		
Accumulated depreciation	(126,000)	(102,000)		
Total assets	**€403,000**	**€383,000**		
Equity and liabilities				
Accounts payable	€58,000	€66,000		
Interest payable	4,000	3,000		
Bank loan	100,000	120,000		
Share capital	180,000	150,000		
Retained earnings	61,000	44,000		
Total liabilities and equity	**€403,000**	**€383,000**		

Additional information. Equipment costing €34,000 with accumulated depreciation of €14,000 was sold during the year. Shares were issued for cash and part of the bank loan was repaid. Dividends were declared and paid.

Required

1. Compute the proceeds from the equipment sale, the equipment purchased and the dividends paid.
2. Prepare the statement of cash flows for 2025, indirect method.

Figure	Working	Result
Proceeds from equipment sale		
Equipment purchased		
Dividends paid		

✎ WORKING

Statement of cash flows — indirect method	€
Cash flows from operating activities	
Net cash from operating activities	
Cash flows from investing activities	
Net cash from investing activities	
Cash flows from financing activities	
Net cash from financing activities	
Net increase (decrease) in cash	
Cash at 1 January	
Cash at 31 December	

✎ EXTRA SPACE

TYPE K.4

Accumulated depreciation, dividends and treasury shares

The puzzles that unlock most cash flow questions.

① Worked example → ② K8 → ③ K9

WORKED EXAMPLE · SOLVED

Accumulated depreciation puzzles; dividends and treasury shares

(a) Compute depreciation expense: case 1 — opening accumulated depreciation 50,000, closing 64,000, an asset with accumulated depreciation of 9,000 disposed of; case 2 — opening 120,000, closing 110,000, an asset that cost 30,000 and was 80 per cent depreciated disposed of.

(b) Retained earnings went from €140,000 to €152,000 with net income of €45,000. Common stock rose from €200,000 to €230,000, bonds fell from €60,000 to €40,000 and treasury stock went from 0 to (€12,000). Prepare the financing section.

Step 1 — (a) One formula

Depreciation expense = Closing accumulated depreciation − Opening + Accumulated depreciation on disposals

Case 1	64,000 − 50,000 + 9,000	**23,000**
Case 2	110,000 − 120,000 + (80% × 30,000 = 24,000)	**14,000**

A **fall** in accumulated depreciation does not mean negative depreciation: the disposal removed more than the year added.

Step 2 — (b) Dividends from retained earnings, then the section

Dividends paid	140,000 + 45,000 − 152,000	**33,000**

Cash flows from financing activities	€
Issue of common stock	30,000
Repayment of bonds	(20,000)
Purchase of treasury stock	(12,000)
Dividends paid	(33,000)
Net cash used in financing activities	**(35,000)**

Comment

Treasury stock is a contra-equity account: its increase is cash paid to buy back shares, a financing outflow. Net, the company returned 35,000 to its capital providers — 45,000 to shareholders through dividends and buyback, less 30,000 of new shares, plus 20,000 to bondholders.

✎ EXTRA SPACE

K8 Four accumulated depreciation puzzles CASH FLOW

YOUR TURN

For each independent case, compute the depreciation expense for the year. This is the single step that unlocks most cash flow questions.

Case	Opening acc. dep.	Closing	Disposals during the year	Depreciation expense
1	92,000	118,000	none	
2	92,000	118,000	an asset with accumulated depreciation of 15,000	
3	140,000	126,000	an asset with accumulated depreciation of 48,000	
4	86,000	86,000	an asset that cost 40,000 and was two-thirds depreciated	

✎ WORKING

K9 Treasury stock in the financing section CASH FLOW

YOUR TURN

NEMO SPA

Comparative Statements of Financial Position — 31 December

	2025	2024	Change	Classification
Assets				
Cash	€40,500	€35,000		
Accounts receivable	62,000	55,000		
Inventory	88,000	92,000		
Fixed assets	300,000	260,000		
Accumulated depreciation	(95,000)	(80,000)		
Total assets	**€395,500**	**€362,000**		
Equity and liabilities				
Accounts payable	€47,000	€52,000		
Bonds payable	70,000	40,000		
Common stock	200,000	180,000		
Retained earnings	96,500	90,000		
Treasury stock	(18,000)	0		
Total liabilities and equity	**€395,500**	**€362,000**		

Additional information. Net income was €38,500. No fixed assets were sold. Bonds and common stock were issued for cash, treasury stock was purchased for cash, and dividends were declared and paid.

> **Required**
>
> 1. Compute the dividends paid.
> 2. Prepare the statement of cash flows for 2025, indirect method.
> 3. Comment on the financing section.

✎ DIVIDENDS PAID

Statement of cash flows — indirect method	€
Cash flows from operating activities	
Net cash from operating activities	
Cash flows from investing activities	
Net cash from investing activities	
Cash flows from financing activities	
Net cash from financing activities	
Net increase (decrease) in cash	
Cash at 1 January	
Cash at 31 December	

✎ COMMENT ON FINANCING

TYPE K.5

Free cash flow and the net income reconciliation

What is left after investing and dividends, and why profit is not cash.

① Worked example → ② K4 → ③ K10

WORKED EXAMPLE · SOLVED

Free cash flow; why net income and operating cash flow differ

Use the Pavone Srl statement of example K.1 (net income 70,000; net cash from operating activities 79,000; equipment purchased 50,000; dividends 30,000). Compute free cash flow, then reconcile net income to operating cash flow and classify each reconciling item.

Step 1 — Free cash flow (course definition)

FCF = Cash from operations − Capital expenditures − Cash dividends

FCF	79,000 − 50,000 − 30,000	**(1,000)**
FCF without the dividend	79,000 − 50,000	**29,000**

Step 2 — The reconciliation

Reconciling item	€	Type
Net income	70,000	
Depreciation expense	+24,000	non-cash expense
Increase in accounts receivable	(15,000)	working capital movement
Decrease in inventories	+10,000	working capital movement
Decrease in accounts payable	(10,000)	working capital movement
Net cash provided by operating activities	**79,000**	**difference +9,000**

A **reclassification** item would be a gain or loss on a disposal: removed from operating because its cash sits in investing.

Comment

Operations covered capital spending but not capital spending plus the dividend: the 1,000 gap, and more, was filled by selling land and issuing shares — not repeatable every year. Net income measures profitability on an accrual basis; cash from operations measures the cash the business actually generated, i.e. liquidity.

✎ EXTRA SPACE

K4 Free cash flow CASH FLOW

YOUR TURN

Using the statement of cash flows prepared for Calipso Srl in exercise K3, compute free cash flow and interpret it in three lines. Then state what free cash flow would have been if the company had not paid a dividend.

✎ WORKING

✎ INTERPRETATION

✎ WITHOUT THE DIVIDEND

K10 Why net income and operating cash flow differ CASH FLOW

YOUR TURN

Eolo Spa (exercise K6) reports net income of €111,000 and net cash provided by operating activities of €149,000. Prepare a reconciliation that explains the €38,000 difference, classify each reconciling item as a non-cash expense, a reclassification or a working capital movement, and write a four-line comment on what the two numbers each measure.

Reconciling item	€	Type
Net income		
Net cash provided by operating activities		

✎ COMMENT

SECTION L

Cash flow analysis beyond the statement

Classification, the accrual-versus-cash comparison, the direct-method conversions, free cash flow, and the reading of a company whose profit and cash are moving in opposite directions.

TYPE L.1

Classifying flows and disclosures

Operating, investing, financing or non-cash — and where each item is reported.

① Worked example → ② L1 → ③ L5

WORKED EXAMPLE · SOLVED

Classifying cash flows, and where each disclosure goes

(a) Classify each transaction as operating, investing, financing or non-cash, inflow or outflow. **(b)** Say where each item is reported: body of the statement, schedule of non-cash activities, supplemental disclosures, or nowhere.

Step 1 — (a) Ask what the cash was for

Transaction	Activity	Direction
Cash paid to employees	**Operating**	outflow
Interest received on a loan to a customer	**Operating**	inflow
Dividends received from an associate	**Operating**	inflow
Purchase of a patent for cash	**Investing**	outflow
Sale of shares held as an investment	**Investing**	inflow
Proceeds from a bank loan	**Financing**	inflow
Repurchase of own shares	**Financing**	outflow
Repayment of lease principal	**Financing**	outflow
Land acquired by issuing a mortgage to the seller	**Non-cash**	—

COURSE CONVENTION Operating = running the business, including interest paid and received and dividends received. Investing = buying and selling long-term assets and investments, lending. Financing = raising and repaying capital, including dividends paid. No cash = schedule.

Step 2 — (b) Where it is reported

Item	Where
Depreciation expense	**Body** — add-back in operating (indirect method)
Cash paid for interest and for taxes	**Supplemental disclosures** (indirect method)
Land acquired by issuing shares	**Schedule of non-cash activities**
A stock split	**Nowhere** — no resources move, only the number of shares

✎ EXTRA SPACE

L1 Classify eighteen transactions CASH FLOW

YOUR TURN

Classify each transaction as operating, investing, financing or non-cash, and state whether it is an inflow or an outflow.

Transaction	Activity	In / out
Cash collected from customers		
Cash paid to suppliers for inventory		
Interest received on a bank deposit		
Interest paid to lenders		
Income taxes paid		
Dividends received from an investment		
Purchase of a delivery van for cash		
Proceeds from selling a warehouse		
Loan made to an associated company		
Collection of the principal of a note receivable		
Purchase of shares in another company		
Proceeds from issuing ordinary shares		
Dividends paid to shareholders		
Repayment of the principal of a bond		
Purchase of treasury shares		
Issue of shares to acquire a building		
Conversion of bonds into ordinary shares		
Exchange of one plant asset for another		

✎ NOTES

L5 Which disclosures are required CASH FLOW

YOUR TURN

For each item, state whether it appears in the body of the statement of cash flows, in the schedule of non-cash activities, in the supplemental disclosures, or nowhere.

Item	Where it appears
1. Cash paid during the year for interest	
2. Cash paid during the year for income taxes	
3. A capital lease of €1,000 incurred on the lease of equipment	
4. Acquisition of fixed assets by issuing bonds of €30,000	
5. Conversion of bonds into common stock of €10,000	
6. A stock dividend declared and distributed	
7. A three-for-one stock split	
8. Depreciation expense for the year	

✎ NOTES

TYPE L.2

Accrual against cash

Line-by-line comparison, the reconciliation, and single-transaction effects.

① Worked example → ② L2 → ③ L6

WORKED EXAMPLE · SOLVED

Accrual reporting against cash reporting; the effect on cash from operations

(a) Mora Srl reports the statement below. Receivables rose by €10,000 and payables rose by €6,000. Contrast net income with cash flow from operations side by side, then reconcile.

Sales	150,000
Gain on sale of land	4,000
Purchases	(85,000)
Increase in inventory (credited to cost of sales)	5,000
Depreciation	(12,000)
Expenses paid in cash	(25,000)
Net income	**37,000**

(b) Effect on cash from operations of: (1) collecting €20,000 on account; (2) selling equipment with a book value of €30,000 for €35,000; (3) recording €8,000 of amortisation; (4) paying a previously declared €10,000 dividend; (5) paying €12,000 of salaries; (6) buying €7,000 of inventory for cash.

Step 1 — (a) Side by side: for each line, how much cash moved?

Line	Accrual basis	Cash basis	Why
Sales	150,000	140,000	receivables +10,000 not collected
Gain on sale of land	4,000	—	proceeds are investing
Purchases	(85,000)	(79,000)	payables +6,000 not paid
Increase in inventory	5,000	—	no cash
Depreciation	(12,000)	—	non-cash
Expenses paid in cash	(25,000)	(25,000)	already cash
Total	**37,000**	**36,000**	**net income / CFO**

Step 2 — (a) The indirect reconciliation

Net income		**37,000**
+ Depreciation	non-cash expense	**12,000**
− Gain on sale of land	reclassified to investing	**(4,000)**
− Increase in receivables		**(10,000)**
− Increase in inventory		**(5,000)**
+ Increase in payables		**6,000**
= Cash from operations		**36,000**

Net income reflects **profitability**; cash from operations reflects **liquidity**.

Step 3 — (b) Only cash that comes from or goes to operations counts

Transaction	Effect on cash from operations
(1) Collect €20,000 on account	**Increase 20,000**
(2) Sell equipment, book value 30,000, for 35,000	**No effect** — 35,000 in investing; the gain is deducted
(3) Amortisation 8,000	**No effect** — non-cash
(4) Pay a declared dividend 10,000	**No effect** — financing
(5) Pay salaries 12,000	**Decrease 12,000**
(6) Buy inventory for cash 7,000	**Decrease 7,000**

✎ EXTRA SPACE

L2 Accrual reporting against cash reporting CASH FLOW

YOUR TURN

Pirra Srl reports the following for the year. Accounts receivable increased by €12,000, inventory decreased by €6,000 and accounts payable decreased by €9,000.

Sales	220,000
Gain on sale of investments	8,000
	228,000
Purchases	(130,000)
Decrease in inventory charged to cost of sales	(6,000)
Depreciation	(14,000)
Expenses paid in cash	(42,000)
Loss on sale of fixed assets	(3,000)
Net income	**33,000**

Required

1. Prepare a side-by-side comparative statement contrasting the two bases of reporting: net income and cash flows from operations.
2. Reconcile the two figures by the indirect method.
3. Which of the two better reflects profitability, and which liquidity?

✎ 1. SIDE BY SIDE

Line	Accrual basis	Cash basis
Sales		
Gain on sale of investments		
Purchases		
Decrease in inventory		
Depreciation		
Expenses paid in cash		
Loss on sale of fixed assets		
Total		

✎ 2. INDIRECT RECONCILIATION

L6 Effect on cash from operations CASH FLOW

YOUR TURN

State the effect of each independent transaction on cash from operations: increase, decrease or no effect, and by how much.

Transaction	Effect on cash from operations
1. A customer pays €40,000 owed on account.	
2. Inventory is purchased on credit for €25,000.	
3. A machine is sold for €18,000; its book value was €22,000.	
4. Depreciation of €60,000 is recorded.	
5. A dividend of €30,000 is declared but not yet paid.	
6. A supplier is paid €15,000.	
7. €9,000 of rent is prepaid for next year.	
8. Equipment is bought for €80,000 cash.	

✎ NOTES

TYPE L.3

Direct-method conversions

Cash collected from customers and cash paid to suppliers.

① Worked example → ② L3 → ③ L4

WORKED EXAMPLE · SOLVED

From accrual figures to cash: customers and suppliers

(a) Tirso Spa reports net sales of €600,000. Accounts receivable, net, were €80,000 at the start of the year and €95,000 at the end; the allowance for doubtful accounts rose from €4,000 to €6,000 and bad debt expense was €5,000. Compute cash collected from customers.

(b) Cost of goods sold was €400,000; inventory rose from €50,000 to €60,000 and accounts payable from €30,000 to €36,000. Compute purchases and cash paid to suppliers.

Step 1 — (a) Collections, first ignoring the allowance

Collections (naive)	Sales − increase in net AR = 600,000 − 15,000	**585,000**

Step 2 — (a) Now with the allowance: bad debts reduce receivables without cash

Write-offs	Opening allowance + bad debt expense − closing = 4,000 + 5,000 − 6,000	**3,000**
Gross receivables	84,000 opening → 101,000 closing: increase 17,000	
Collections	Sales − increase in gross AR − write-offs = 600,000 − 17,000 − 3,000	**580,000**
Same, on net AR	Sales − increase in net AR − bad debt expense = 600,000 − 15,000 − 5,000	**580,000** ✓

The naive figure overstates collections by exactly the bad debt expense (5,000): that part of sales will never be collected.

Step 3 — (b) Two adjustments: inventory, then payables

Purchases	COGS + increase in inventory = 400,000 + 10,000	**410,000**
Cash paid to suppliers	Purchases − increase in payables = 410,000 − 6,000	**404,000**

✎ EXTRA SPACE

L3 From sales to cash collected CASH FLOW

YOUR TURN

Nerva Spa reports net sales of €960,000. Accounts receivable, net, were €118,000 at the start of the year and €142,000 at the end. The allowance for doubtful accounts rose from €6,000 to €9,000 and the bad debt expense for the year was €11,000.

> **Required**
>
> 1. Compute cash collected from customers ignoring the allowance.
> 2. Compute cash collected from customers taking the allowance into account, and reconcile the two answers.

✎ WORKING

✎ RECONCILIATION OF THE TWO ANSWERS

L4 From cost of goods sold to cash paid CASH FLOW

YOUR TURN

Ostia Srl reports cost of goods sold of €540,000. Inventory moved from €96,000 to €128,000 and accounts payable from €74,000 to €61,000. Compute purchases for the year and cash paid to suppliers.

✎ WORKING

TYPE L.4

Free cash flow and the profit–cash gap

Ranking flexibility, and reading growth that consumes cash.

① Worked example → ② L7 → ③ L8

WORKED EXAMPLE · SOLVED

Free cash flow across companies; profit rising, cash falling

(a) Rank by financial flexibility: Pietra (CFO 200, capex 120, dividends 30, net income 140) and Quarzo (CFO 150, capex 30, dividends 60, net income 110), € thousands.

(b) Lumo Group, € thousands:

	Year 1	Year 2
Net sales	1,000	1,300
Net income	60	80
Net cash provided by operating activities	70	10
Accounts receivable	150	300
Inventory	200	320
Accounts payable	120	140

Step 1 — (a) FCF = CFO − capex − dividends

Pietra	200 − 120 − 30	**50**
Quarzo	150 − 30 − 60	**60**

Quarzo has more free cash despite lower operating cash flow, so more flexibility today; Pietra is investing heavily (capex 60 per cent of CFO), which may mean growth tomorrow. Neither ranking is complete without knowing what the capex buys.

Step 2 — (b) Read the gap between profit and cash

Sales growth	1,300 ÷ 1,000 − 1	**+30%**
Receivables growth	300 ÷ 150 − 1	**+100%**
Days' sales in receivables	150 ÷ (1,000/360) → 300 ÷ (1,300/360)	**54 → 83**
CFO ÷ Net income	70 ÷ 60 → 10 ÷ 80	**1.17 → 0.13**

Profit grows but cash does not: receivables doubled and inventory rose 60 per cent against 30 per cent sales growth, while payables barely moved. Growth is being financed with working capital, and earnings quality is falling. Next: days' sales in receivables and inventory, net trade cycle, CFO ÷ net income. Ask management about credit terms, the largest customers' balances, bad debts and slow-moving stock.

✎ EXTRA SPACE

L7 Free cash flow across three companies CASH FLOW

YOUR TURN

Compute free cash flow for each company and rank them by financial flexibility. Comment in four lines.

	Alfa	Beta	Gamma
Net cash provided by operating activities	180,000	180,000	95,000
Capital expenditures	40,000	150,000	20,000
Cash dividends paid	60,000	20,000	0
Net income	120,000	165,000	40,000
Free cash flow			
Rank			

✎ COMMENT

L8 Rising profit, falling cash CASH FLOW

YOUR TURN

Below are three years of summary data for Talea Group. Write the analysis: what is happening, which ratios you would compute next, and what you would ask management.

	Year 1	Year 2	Year 3
Net sales	4,000,000	5,200,000	6,760,000
Net income	240,000	338,000	473,000
Net cash provided by operating activities	310,000	190,000	(40,000)
Accounts receivable	500,000	866,000	1,420,000
Inventory	600,000	910,000	1,420,000
Accounts payable	400,000	480,000	520,000

✎ SUPPORTING FIGURES

✎ ANALYSIS

SECTION M

Bonds and present value

Exercises 1-13 and 1-14 of the Subramanyam chapter 1 problem set, which the lecturer listed for Exercise Session 2 and never solved in class.

TYPE M.1

Pricing a bond with annual coupons

Periods remaining, annuity factor, discount factor — and the sanity check.

① Worked example → ② M1 → ③ M2

WORKED EXAMPLE · SOLVED

Pricing a bond with annual coupons — at a discount and at a premium

(a) An 8-year bond, face value €1,000, coupon 6 per cent paid at the end of each year, is priced at the end of its third year, when the required rate is 8 per cent. **(b)** A 10-year bond, face €1,000, coupon 9 per cent annual, is priced at the beginning of its seventh year, when the required rate is 7 per cent. Compute both prices.

Step 1 — Count the periods that are left

Price = Coupon × $[1 - (1 + r)^{-n}] \div r$ + Face × $(1 + r)^{-n}$ — r = required rate per period, n = periods **remaining**

(a) End of year 3 of 8	8 − 3	**n = 5**
(b) Beginning of year 7 = end of year 6	10 − 6	**n = 4**

Step 2 — Discount the coupons and the face value

(a) Annuity factor 8%, 5	$(1 - 1.08^{-5}) \div 0.08$	**3.992710**
(a) PV factor 8%, 5	1.08^{-5}	**0.680583**
(a) Price	60 × 3.992710 + 1,000 × 0.680583 = 239.563 + 680.583	**€920.15**
(b) Annuity factor 7%, 4	$(1 - 1.07^{-4}) \div 0.07$	**3.387211**
(b) PV factor 7%, 4	1.07^{-4}	**0.762895**
(b) Price	90 × 3.387211 + 1,000 × 0.762895 = 304.849 + 762.895	**€1,067.74**

Comment

(a) Coupon 6 per cent below the 8 per cent required → the bond sells at a **discount**. (b) Coupon 9 per cent above the 7 per cent required → **premium**. The sign of the gap tells you the answer before you compute: use it as a check.

✎ EXTRA SPACE

M1 Priced at a discount BONDS

YOUR TURN

Compute the present value of a 10-year bond with a face value of $100 and a contract (coupon) rate of 10 per cent per annum, payable at the end of each year, priced at the end of its fifth year, when the effective (required) interest rate is 14 per cent per annum.

✎ WORKING

M2 Priced at a premium BONDS

YOUR TURN

Compute the present value of a 14-year bond with a face value of $1,000 and a contract rate of 8 per cent per annum, payable at the end of each year, priced at the beginning of its tenth year, when the required rate is 6 per cent per annum.

✎ WORKING

TYPE M.2

Semiannual coupons and the required rate

Halve the rate, double the periods; risk in the required rate.

① Worked example → ② M3 → ③ M4

WORKED EXAMPLE · SOLVED

Semiannual coupons; pricing a purchase at two required rates

(a) Reprice bond (b) of the previous example assuming the coupon is paid semiannually. **(b)** You consider buying $5,000 of 6 per cent bonds due in 5 years, interest paid semiannually. Price them at a required rate of 8 per cent and of 4 per cent, rounded to the nearest dollar.

Step 1 — Halve the rates, double the periods

(a) Per period	coupon 90 ÷ 2 = 45; rate 7% ÷ 2 = 3.5%; n = 4 × 2 = 8	
(a) Price	45 × 6.873956 + 1,000 × 0.759412 = 309.328 + 759.412	**€1,068.74**

Slightly higher than with annual coupons (€1,067.74): half of each coupon arrives six months earlier, and money received earlier is worth more.

Step 2 — (b) One bond, two required rates

Coupon per period	$5,000 × 6% ÷ 2	**$150**
At 8% → 4% × 10 periods	150 × 8.110896 + 5,000 × 0.675564 = 1,216.63 + 3,377.82	**$4,594**
At 4% → 2% × 10 periods	150 × 8.982585 + 5,000 × 0.820348 = 1,347.39 + 4,101.74	**$5,449**

Comment

Risk — the chance that interest or principal is not paid in full and on time — is priced through the required rate: the riskier you judge the issuer, the higher the rate you require and the lower the price you are willing to pay.

✎ EXTRA SPACE

M3 The same bond, semiannual interest BONDS

YOUR TURN

Recompute exercise M2 assuming the bond interest is payable in equal semiannual amounts. Explain why the answer differs.

✎ WORKING

✎ WHY THE ANSWER DIFFERS

M4 Pricing a purchase BONDS

YOUR TURN

On 1 January Year 1 you are considering the purchase of $10,000 of Colvin Company's 8 per cent bonds. The bonds are due in 10 years, with interest payable semiannually on 30 June and 31 December. Based on your analysis of Colvin you determine that a 6 per cent required interest rate is appropriate.

> **Required**
>
> 1. Compute the price you will pay for the bonds using the present value model, rounded to the nearest dollar.
> 2. Recompute the price if your required rate of return is 10 per cent.
> 3. Describe risk and explain how it is reflected in your required rate of return.

✎ 1.

✎ 2.

✎ 3.

TYPE M.3

The implied rate and bonds issued at a discount

Working back from a price; proceeds, discount and the ratios.

① Worked example → ② M5 → ③ M6

WORKED EXAMPLE · SOLVED

Working back to the required rate; a bond issued at a discount

(a) A 2-year bond, face €1,000, annual coupon 8 per cent, is quoted at €965.29. Find the required rate. **(b)** Lario Spa issues €200,000 of 5-year bonds with a 4 per cent annual coupon when the market requires 6 per cent. Before the issue: total assets €1,000,000, total liabilities €400,000, operating income €150,000, interest expense €20,000. Compute the proceeds, the discount, and the effect on the total debt ratio and times interest earned.

Step 1 — (a) Trial and error, guided by the price

Price below par → the required rate is above the 8 per cent coupon. Try 10 per cent:

At 10%	$80 \div 1.10 + 1{,}080 \div 1.10^2 = 72.73 + 892.56$	**965.29 ✓**

The market requires 10 per cent, 2 points above the coupon: it sees the issuer as riskier than when the bond was issued, or rates have risen.

Step 2 — (b) Proceeds and discount

Proceeds	8,000 × 4.212364 + 200,000 × 0.747258 = 33,699 + 149,452	**€183,151**
Discount	200,000 − 183,151	**€16,849**

The discount is extra interest: investors accept a 4 per cent coupon only if they pay less today; it is amortised over five years, so interest expense (6% × carrying amount) exceeds the coupon.

Step 3 — (b) Effect on the ratios

Total debt ratio before	400,000 ÷ 1,000,000	**40.0%**
Total debt ratio after	(400,000 + 183,151) ÷ (1,000,000 + 183,151)	**49.3%**
TIE before	150,000 ÷ 20,000	**7.5x**
TIE after (effective interest)	150,000 ÷ (20,000 + 6% × 183,151 = 10,989)	**4.84x**
TIE on the coupon only	150,000 ÷ (20,000 + 8,000) — overstated	**5.36x**

✎ EXTRA SPACE

M5 Working backwards to the required rate BONDS

YOUR TURN

A three-year bond with a face value of €1,000 and an annual coupon of 10 per cent is quoted at €951.96. The coupon is paid at the end of each year. Determine the required rate implied by that price, showing your reasoning, and say what it tells you about the market's view of the issuer.

✎ WORKING

✎ COMMENT

M6 A bond issued at a discount BONDS

YOUR TURN

Nera Spa issues €500,000 of ten-year bonds carrying a 5 per cent annual coupon at a time when the market requires 7 per cent on debt of this quality.

> **Required**
>
> 1. Compute the cash proceeds of the issue.
> 2. State the discount and explain what it represents.
> 3. State the effect of the issue on the total debt ratio and on times interest earned, given that before the issue the company had total assets of €4,000,000, total liabilities of €1,600,000, operating income of €600,000 and interest expense of €96,000.

✎ WORKING

✎ COMMENT

TYPE M.4

Discount, premium, par — and bonds in the statements

Answers without computing, and where bonds sit in the statements.

① Worked example → ② M7 → ③ M8

WORKED EXAMPLE · SOLVED

Discount, premium or par without computing; bonds in the statements

(a) Without computing, say whether each bond prices at a discount, premium or par: coupon 5% / required 5%; 8% / 6%; 3% / 6%; zero coupon / 4%; 6% / 6% but the issuer is upgraded the next day. **(b)** Answer: (1) where does a bond issued at a premium appear and what happens to it over its life? (2) A bond is redeemed early at a loss — where do the cash paid and the loss go in the cash flow statement?

Step 1 — (a) Compare the coupon with the required rate

Coupon / required	Price	Reason
5% / 5%	**Par**	the coupon pays exactly what the market requires
8% / 6%	**Premium**	coupon above the required rate: investors pay more for it
3% / 6%	**Discount**	coupon below the required rate
Zero coupon / 4%	**Discount**	all the return comes from buying below face
6% / 6%, then upgraded	**Premium**	lower risk → lower required rate → price rises above par

Step 2 — (b) Bonds inside the statements

- (1) A **non-current liability** at its carrying amount = face value + unamortised premium. The premium is amortised each year, so the carrying amount falls to face value at maturity and interest expense is lower than the coupon paid.
- (2) The **cash paid** to redeem is a financing outflow; the **loss** is a non-cash item inside net income, added back in the operating section.

✎ EXTRA SPACE

M7 Discount, premium or par BONDS

YOUR TURN

For each bond, state without computing whether it prices at a discount, at a premium or at par, and give the one-line reason.

Bond	Coupon rate	Required rate	Discount / premium / par	Reason
A	6%	6%		
B	9%	5%		
C	4%	7%		
D	7%	7%, but the issuer is downgraded the next day		
E	0% (zero coupon)	5%		

✎ NOTES

M8 Bonds inside the financial statements BONDS

YOUR TURN

Answer in three or four lines each.

1. Where does a bond issued at a discount appear on the balance sheet, and what happens to that figure over the life of the bond?
2. Why does the cash flow statement show the proceeds of a bond issue in financing, while the interest paid on the same bond sits in operating?
3. Two companies have identical operating businesses. One issued its debt at par at 8 per cent; the other issued at a deep discount with a 2 per cent coupon and the same 8 per cent effective rate. Which reports the higher times interest earned if both use the coupon rate to measure interest expense, and what should the analyst do?

✎ 1.

✎ 2.

✎ 3.

SECTION N

Multiple choice and true or false

Modelled on the chapter 1 question file on the course page, extended to every chapter covered. These are the cheap marks: answer them in two minutes and spend the time saved on the computational part.

TYPE N.1

Chapters 1 and 2

Overview of analysis and the reporting environment.

① Worked example → ② N1 → ③ N2

WORKED EXAMPLE · SOLVED

Chapters 1–2: overview and reporting environment

METHOD Read the stem, cross out the options that are clearly wrong, choose, then justify in one line — the justification is what makes the answer stick, and on a true-or-false with reasoning it is worth the marks.

Multiple choice

1. Which statement reports a company's financial position at a point in time?

A. Income statement
B. Balance sheet
C. Statement of cash flows
D. Statement of changes in equity

Answer: **B** — the other three cover a period

2. The purpose of an external audit is to:

A. detect all fraud
B. provide reasonable assurance that the statements are free of material misstatement
C. guarantee the company's solvency
D. prepare the financial statements

Answer: **B** — assurance on fair presentation, not a guarantee; management prepares the statements

True or false

1. Comparing a company's figures across years is called horizontal analysis.

Answer: **True** — year-to-year change and trend analysis compare the same company over time

2. Accrual accounting recognises revenue when cash is received.

Answer: **False** — revenue is recognised when earned; cash timing is irrelevant

N1 Chapter 1 — overview THEORY

YOUR TURN

ANSWER

1. Which of the following is likely to be the most informative source if you were interested in a company's business plan or strategy?

A. Auditor's letter
B. Management discussion and analysis
C. Proxy statement
D. Footnotes

2. Which of the following would not be considered a source of financing?

ANSWER

A. Notes receivable
B. Common stockholders' equity
C. Retained earnings
D. Debentures

3. The major reason for analysing a company's financial statements, as stated in class, is to:

ANSWER

A. forecast the share price
B. determine credit value
C. satisfy the regulator
D. compute the tax charge

4. A common-size income statement is prepared by dividing:

ANSWER

A. all items in year t by the corresponding value in year t−1
B. all items in year t by the corresponding balance sheet accounts
C. all items in year t by net income in year t
D. all items in year t by sales in year t

5. Liquidity is generally defined as a measure of:

ANSWER

A. the ability to pay employees on time
B. the ability to pay interest and principal on all debt
C. the ability to pay dividends
D. the ability to pay current liabilities

6. While determining the most profitable among a number of companies, which would be the best indicator of relative profitability?

ANSWER

A. Highest net income
B. Highest retained earnings
C. Highest return on equity
D. Highest sales

True or false

1. Financial statement analysis is an exact science. T F

2. The explanatory notes accompanying the financial statements are generally of little value to an analyst. T F

3. Two popular techniques of comparative analysis are year-to-year change analysis and index-number trend analysis. T F

4. In a common-size balance sheet, total assets are expressed as 100 per cent. T F

5. In a common-size income statement, net income is expressed as 100 per cent. T F

6. The income statement is the only one of the four basic statements that does not contain balances at a specific point in time. T F

✎ EXTRA SPACE

N2 Chapter 2 — reporting environment THEORY

YOUR TURN

1. If a company receives an unqualified audit opinion it means the auditors: ANSWER

A. did not complete a full audit

B. are providing assurance that the company will remain financially viable

C. are providing assurance that the statements fairly present the company's financial performance and position

D. are guaranteeing that the statements are free from fraud

2. The Management Discussion and Analysis section of an annual report: ANSWER

A. is required by the SEC

B. is optional but normally included

C. is required only if the company has suffered unfavourable trends

D. is required only if the audit opinion is qualified

3. Which body sets IFRS? ANSWER

A. The SEC

B. The FASB

C. The International Accounting Standards Board

D. The AICPA

4. Which is not one of the three audit opinion outcomes named on the slide? ANSWER

A. Clean

B. Qualified

C. Disclaimer

D. Provisional

5. Accruals are the cornerstone of accounting because: ANSWER

A. net income equals operating cash flow

B. net income equals operating cash flow plus accruals

C. accruals eliminate estimation error

D. accruals are required only under IFRS

6. Recording the least optimistic information about uncertain events is the principle of: ANSWER

A. materiality

B. conservatism

C. accrual accounting

D. historical cost

✎ EXTRA SPACE

True or false

1. Managers of companies are a major lobbyist on GAAP. T F

2. Fair value accounting measures assets and liabilities at their values on the measurement date, approximately the date of the financial statements. T F

3. Timeliness, frequency and forward-looking content are listed as desirable qualities of accounting information. T F

4. A big bath is a strategy in which managers record huge write-offs in one period to relieve other periods of expenses. T F

5. Income smoothing increases the volatility of reported income. T F

TYPE N.2

Chapters 3 and 4

Financing and investing activities.

① Worked example → ② N3 → ③ N4

WORKED EXAMPLE · SOLVED

Chapters 3–4: financing and investing activities

METHOD Read the stem, cross out the options that are clearly wrong, choose, then justify in one line — the justification is what makes the answer stick, and on a true-or-false with reasoning it is worth the marks.

Multiple choice

1. A provision for product warranties is:

A. an asset
B. an operating liability
C. a financing liability
D. part of equity

Answer: **B** — it arises from selling goods, carries no interest and is not borrowed money

2. Which method assigns the most recent costs to ending inventory?

A. FIFO
B. LIFO
C. Weighted average
D. None of them

Answer: **A** — first in, first out: the oldest costs go to cost of goods sold, the newest stay in inventory

True or false

1. Goodwill is amortised over a maximum of 40 years under IFRS.

Answer: **False** — goodwill is not amortised; it is tested for impairment at least annually

2. Retained earnings are part of earned capital.

Answer: **True** — contributed capital is paid in by shareholders; retained earnings are earned and kept

N3 Chapter 3 — financing activities THEORY

YOUR TURN

1. Which is an operating liability?

ANSWER

- **A.** Bonds payable
- **B.** Accounts payable
- **C.** Current portion of long-term debt
- **D.** Finance lease obligation

2. Treasury stock is:

ANSWER

- **A.** an asset
- **B.** a contra-equity account
- **C.** a current liability
- **D.** a non-current liability

3. A spin-off differs from a split-off in that in a spin-off:

ANSWER

- **A.** subsidiary stock is distributed to shareholders as a dividend
- **B.** subsidiary stock is exchanged for the company's own shares
- **C.** the subsidiary is sold for cash
- **D.** the subsidiary is liquidated

4. To record a contingent liability, the loss must be:

ANSWER

- **A.** possible and estimable
- **B.** probable and reasonably estimable
- **C.** remote but disclosed
- **D.** certain

5. A lease must be accounted for as a capital lease if, at inception, it meets:

ANSWER

- **A.** all four criteria
- **B.** any one of four criteria
- **C.** at least three of four criteria
- **D.** none of the criteria

6. Which is not listed as a transaction sometimes used for off-balance-sheet financing?

ANSWER

- **A.** Take-or-pay arrangements
- **B.** Through-put agreements
- **C.** Issuing ordinary shares at a premium
- **D.** Selling receivables with recourse and recording them as sales

True or false

1. A contingent asset is recorded as soon as realisation is probable. T F

2. Loss reserves alter a company's risk exposure and have cash flow consequences. T F

3. A lease whose term is 75 per cent or more of the asset's estimated economic life is a capital lease. T F

4. Paid-in capital in excess of par is part of contributed capital. T F

N4 Chapter 4 — investing activities THEORY

YOUR TURN

1. In a period of rising prices, which method produces the highest ending inventory? ANSWER

A. LIFO
B. FIFO
C. Weighted average
D. They are equal

2. The LIFO reserve is: ANSWER

A. the difference between FIFO and weighted average inventory
B. the amount by which current cost exceeds the reported cost of LIFO inventories
C. a liability for future inventory purchases
D. the gain on a LIFO liquidation

3. Receivables are reported at: ANSWER

A. gross invoice value
B. net realisable value
C. replacement cost
D. present value

4. A securitisation of receivables with recourse requires the seller to: ANSWER

A. remove the receivables from the balance sheet
B. record both an asset and a compensating liability for the amount factored
C. recognise a gain equal to the discount
D. do nothing until the receivables are collected

5. Impairment is recognised when: ANSWER

A. market value falls below cost
B. expected undiscounted cash flows are less than carrying value
C. the asset is fully depreciated
D. the useful life is revised

6. Average age of plant assets is measured as: ANSWER

A. gross plant ÷ depreciation expense
B. accumulated depreciation ÷ depreciation expense
C. net plant ÷ depreciation expense
D. accumulated depreciation ÷ gross plant

True or false

1. Capitalising a cost means reporting it in the period when it is incurred. T F

2. The double-declining-balance method ignores salvage value in computing the annual charge. T F

3. Acquisition cost of a plant asset includes financing charges. T F

4. In a period of rising prices, FIFO gross profit contains a holding gain as well as an economic profit. T F

TYPE N.3

Chapters 7 and 8

Cash flow, and return on invested capital.

① Worked example → ② N5 → ③ N6

WORKED EXAMPLE · SOLVED

Chapters 7–8: cash flow and return on invested capital

METHOD Read the stem, cross out the options that are clearly wrong, choose, then justify in one line — the justification is what makes the answer stick, and on a true-or-false with reasoning it is worth the marks.

Multiple choice

1. Dividends paid to shareholders are classified as:

A. operating
B. investing
C. financing
D. non-cash

Answer: **C** — a distribution to capital providers; dividends *received* are operating

2. If NOA is 800 and equity is 500, net financial obligations are:

A. 1,300
B. 300
C. −300
D. 500

Answer: **B** — NOA = NFO + Equity, so NFO = 800 − 500

True or false

1. Under the indirect method a decrease in accounts receivable is added to net income.

Answer: **True** — customers paid more than the period's sales: cash above income

2. RNOA changes when the company replaces equity with debt.

Answer: **False** — RNOA = NOPAT ÷ NOA is independent of financing; the effect of debt shows in LEV × Spread

✎ EXTRA SPACE

N5 Chapter 7 — cash flow THEORY

YOUR TURN

1. Interest paid is classified in the statement of cash flows as: ANSWER

- **A.** operating
- **B.** investing
- **C.** financing
- **D.** non-cash

2. Which section always appears first? ANSWER

- **A.** Investing
- **B.** Financing
- **C.** Operating
- **D.** Non-cash

3. Under the indirect method, a gain on the sale of equipment is: ANSWER

- **A.** added to net income
- **B.** deducted from net income
- **C.** ignored
- **D.** shown in financing

4. An increase in accounts payable is: ANSWER

- **A.** a cash outflow in operating
- **B.** a cash inflow in operating
- **C.** a financing inflow
- **D.** a non-cash item

5. The purchase of equipment financed in full by a note issued to the seller is: ANSWER

- **A.** an investing outflow and a financing inflow
- **B.** an investing outflow only
- **C.** disclosed as a non-cash activity
- **D.** ignored entirely

6. Free cash flow is: ANSWER

- **A.** cash from operations
- **B.** cash from operations less capital expenditures less cash dividends
- **C.** net income plus depreciation
- **D.** the net increase in cash

True or false

1. Both the direct and the indirect method give the same cash from operations. T F

2. Cash from operations is a measure of profitability. T F

3. An increase in inventory is added to net income in the operating section. T F

4. Depreciation is a source of cash. T F

N6 Chapter 8 — return on invested capital THEORY

YOUR TURN

1. NOPAT is:

ANSWER

A. net income plus interest
B. operating income × (1 − tax rate)
C. net income × (1 − tax rate)
D. pretax income less tax

2. Net operating assets equal:

ANSWER

A. total assets
B. total assets less total liabilities
C. operating assets less operating liabilities
D. operating assets less financial liabilities

3. RNOA can be decomposed as:

ANSWER

A. margin + turnover
B. margin × turnover
C. margin ÷ turnover
D. turnover − margin

4. In the expression ROCE = RNOA + (LEV × Spread), Spread is:

ANSWER

A. RNOA − NFR
B. NFR − RNOA
C. RNOA − LEV
D. NFE ÷ NFO

5. Net financial obligations are computed as:

ANSWER

A. total liabilities
B. NOA − equity
C. equity − NOA
D. current liabilities + bonds

6. If RNOA is below the net financial rate, additional leverage:

ANSWER

A. raises ROCE
B. lowers ROCE
C. leaves ROCE unchanged
D. raises RNOA

True or false

1. Invested capital is usually computed on average balances. T F

2. Operating liabilities leverage increases RNOA. T F

3. ROCE excludes preferred dividends from the numerator and preferred stock from the denominator. T F

4. Two companies with the same RNOA must have the same margin. T F

TYPE N.4

Chapter 10 and the marking conventions

Credit analysis, and the conventions the lecturer marks against.

① Worked example → ② N7 → ③ N8

WORKED EXAMPLE · SOLVED

Chapter 10 and the marking conventions

METHOD Read the stem, cross out the options that are clearly wrong, choose, then justify in one line — the justification is what makes the answer stick, and on a true-or-false with reasoning it is worth the marks.

Multiple choice

1. The cash ratio is:

A. cash ÷ current assets

B. (cash + marketable securities) ÷ current liabilities

C. cash ÷ total liabilities

D. (cash + receivables) ÷ current liabilities

Answer: **B** — the strictest liquidity measure: only cash and near-cash against current liabilities

2. The total debt ratio is:

A. total liabilities ÷ equity

B. total liabilities ÷ total assets

C. long-term debt ÷ equity

D. current liabilities ÷ total liabilities

Answer: **B** — A is debt to equity, C long-term debt to equity, D short-term to total debt

True or false

1. A company whose current ratio is below 1 has negative working capital.

Answer: **True** — current assets below current liabilities means CA − CL is negative

2. The collection period is computed on year-end receivables.

Answer: **False** — collection period = 360 ÷ turnover on average receivables; days' sales in receivables uses the closing balance

✎ EXTRA SPACE

N7 Chapter 10 — credit analysis THEORY

YOUR TURN

1. Working capital is: ANSWER

- **A.** current assets
- **B.** current assets ÷ current liabilities
- **C.** current assets − current liabilities
- **D.** total assets − total liabilities

2. The acid-test ratio excludes: ANSWER

- **A.** cash
- **B.** accounts receivable
- **C.** marketable securities
- **D.** inventory and prepaid expenses

3. Days' sales in inventory is: ANSWER

- **A.** inventory ÷ sales × 360
- **B.** inventory ÷ cost of goods sold × 360
- **C.** cost of goods sold ÷ inventory
- **D.** 360 ÷ inventory

4. The rule-of-thumb current ratio named on the slide is: ANSWER

- **A.** 1:1
- **B.** 1.5:1
- **C.** 2:1
- **D.** 3:1

5. Times interest earned is potentially misleading because: ANSWER

- **A.** it uses net income
- **B.** it ignores taxes
- **C.** it treats interest as the only fixed charge
- **D.** it uses an average balance

6. Window dressing a current ratio of 1.5 by paying down payables with cash will: ANSWER

- **A.** raise it
- **B.** lower it
- **C.** leave it unchanged
- **D.** leave working capital changed

True or false

1. Solvency concerns long-run financial viability. T F

2. Loan covenants define default and the remedies available. T F

3. The short-term debt to total debt ratio measures reliance on short-term financing. T F

4. A current ratio of 4:1 is unambiguously better than 2:1. T F

5. Liquidity depends mainly on the level of cash rather than on prospective cash flows. T F

N8 The conventions used when marking THEORY

YOUR TURN

1. Which formula does this course use for the operating profit margin?

ANSWER

- **A.** Net income ÷ sales
- **B.** Operating income ÷ sales
- **C.** NOPAT ÷ sales
- **D.** Gross profit ÷ sales

2. Which denominator belongs in the operating asset turnover?

ANSWER

- **A.** Average total assets
- **B.** Average net operating assets
- **C.** Average equity
- **D.** Year-end total assets

3. Which days convention is used throughout this course?

ANSWER

- **A.** 365
- **B.** 360
- **C.** 252
- **D.** 350

4. Which default tax rate does the lecturer assume when the exercise does not give one?

ANSWER

- **A.** 20%
- **B.** 25%
- **C.** 30%
- **D.** 40%

5. Which items does he treat as financial rather than operating when the balance sheet shows them separately?

ANSWER

- **A.** Inventory and receivables
- **B.** Prepaid expenses
- **C.** Short-term investments, marketable securities and long-term investments
- **D.** Accounts payable

✎ EXTRA SPACE

TYPE N.5

Rapid classification and mixed true or false

Speed drills across the course.

① Worked example → ② N9 → ③ N10

WORKED EXAMPLE · SOLVED

Rapid classification and mixed true or false

METHOD Read the stem, cross out the options that are clearly wrong, choose, then justify in one line — the justification is what makes the answer stick, and on a true-or-false with reasoning it is worth the marks.

Rapid classification

Item	Classification
Income tax payable	**Current liability (operating)**
Patents	**Non-current asset (intangible)**
Share premium	**Equity (contributed capital)**
Allowance for doubtful accounts	**Contra-asset (reduces receivables)**
Bonds due in 8 years	**Non-current liability (financing)**
Prepaid insurance	**Current asset**

Mixed true or false

1. Depreciation is added back under the indirect method because it is a source of cash.

   Answer: **False** — it is added back because it is a non-cash expense already deducted in net income

2. A positive spread means that more leverage raises ROCE.

   Answer: **True** — ROCE = RNOA + LEV × Spread: with Spread > 0, a higher LEV adds to ROCE

✎ EXTRA SPACE

N9 Rapid-fire classification THEORY

YOUR TURN

Classify each item. Allow yourself thirty seconds for the whole list.

Item	Classification
Current portion of long-term borrowings	
Advances received from customers	
Treasury shares	
Accumulated depreciation	
Deferred tax liabilities, long-term	
Goodwill	
Marketable securities	
Notes payable due in eight months	
Construction in progress	
Provisions	
Unearned revenue	
Bank overdraft	

N10 Mixed true or false THEORY

YOUR TURN

True or false

1. Investing equals financing is another statement of the balance sheet identity. T F

2. A gain on the sale of an asset increases cash from operations. T F

3. Index-number trend percentages can be compared between two companies with different base years. T F

4. Operating profit is measured before interest expense. T F

5. If only one balance sheet date is given you should still compute an average. T F

6. A company can report rising net income and negative cash from operations for several years. T F

7. The allowance for doubtful accounts is a liability. T F

8. LIFO raises inventory turnover and lowers the current ratio in a period of rising prices. T F

9. Depreciation expense is a financing activity. T F

10. The net trade cycle equals days in inventory plus days in receivables less days in payables. T F

11. A negative spread means the company is unprofitable. T F

12. Common-size analysis removes the effect of size, so a small and a large company can be compared directly. T F

SECTION X

Mixed papers — exam simulation

Six full papers. Work them under time, with a calculator and nothing else, and mark yourself against the solutions before reading them in full.

TYPE X.1

Foundations and cash flow papers

Time budget, then answers laid out the way the marker reads them.

① Worked example → ② X1 → ③ X2

WORKED EXAMPLE · SOLVED

How to sit a paper — a solved mini paper (30 minutes, 30 marks)

TIME BUDGET Minutes per question = marks ÷ total marks × time allowed. In a 90-minute, 100-mark paper a 25-mark question gets about 22 minutes; keep five minutes at the end to check the totals. Answer the cheap questions first, but in the order printed on your sheet.

Q1 (8 marks) Classify: prepaid rent; bonds due in 2031; treasury shares; customer deposits received. **Q2 (12 marks)** A machine bought on 1 January Year 1 for €120,000, residual value €12,000, 6 years straight-line, is sold at the end of Year 3 for €60,000: plan, gain or loss, and where it goes in the cash flow statement. **Q3 (10 marks)** Net income €50,000; depreciation €15,000; receivables up €8,000; inventory down €4,000; payables down €3,000; equipment bought for €30,000 cash; dividends paid €12,000. Compute cash from operations and free cash flow.

Q1 — one word and one reason each

Item	Answer
Prepaid rent	**Current asset** — service received within twelve months
Bonds due in 2031	**Non-current liability** — financing
Treasury shares	**Contra-equity** — deducted from equity
Customer deposits received	**Current liability** — operating, settled in goods

Q2 — plan, disposal, cash flow

Year	Depreciation	Accumulated	Book value
Year 1	18,000	18,000	102,000
Year 2	18,000	36,000	84,000
Year 3	18,000	54,000	66,000

Annual charge (120,000 − 12,000) ÷ 6 = 18,000. Loss = 60,000 − 66,000 = **€6,000**: added back in operating activities; the €60,000 proceeds are an investing inflow.

Q3 — the indirect method in one line

Cash from operations	50,000 + 15,000 − 8,000 + 4,000 − 3,000	**58,000**
Free cash flow	58,000 − 30,000 − 12,000	**16,000**

Comment

Every answer: formula, numbers, result with its unit, one line of meaning. Marks are lost on missing steps far more than on arithmetic.

X1 Paper 1 — accounting foundations and comparative analysis EXAM

YOUR TURN

Time allowed 90 minutes · Total 100 marks · four questions, all compulsory

Question 1 (20 marks)

Classify each item as a current asset, non-current asset, current liability, non-current liability or equity.

Item	Classification	Item	Classification
Advances received from customers		Goodwill	
Current portion of long-term borrowings		Treasury shares	
Accumulated depreciation		Marketable securities	
Deferred tax liabilities, long-term		Prepaid insurance	
Notes payable due in eight months		Construction in progress	

✎ EXTRA SPACE

Question 2 (25 marks)

The income statement of Prato Srl is presented in T-format. Rewrite it in two-step format.

Left side / debit		Right side / credit	
Cost of goods sold	540	Revenue	870
SG&A expenses	180	Gain on sale of equipment	15
Depreciation expense	60		
Interest expense	24		
Income tax expense	29		

Prato Srl — two-step income statement	€

✎ EXTRA SPACE

Question 3 (30 marks)

On 1 January Year 1 Prato purchased a machine for €240,000, with an estimated useful life of six years and a residual value of €24,000. The company uses the straight-line method.

Required

1. Prepare the depreciation plan for the first four years, showing the annual expense, accumulated depreciation and book value.
2. The machine is sold at the end of Year 4 for €105,000. Compute the gain or loss on disposal.
3. State where the gain or loss and the proceeds appear in the statement of cash flows.

✎ 1.

Year	Annual depreciation	Accumulated depreciation	Book value
Year 1			
Year 2			
Year 3			
Year 4			

✎ 2.

✎ 3.

Question 4 (25 marks)

Prepare a horizontal analysis and a common-size analysis of the following condensed income statements, and comment in no more than six lines.

	2025	2024	Change €	Change %	% 2025	% 2024
Net sales	870	750				
Cost of goods sold	540	450				
Gross profit	330	300				
Operating expenses	240	195				
Operating profit	90	105				

✎ EXTRA SPACE

✎ COMMENT

X2 Paper 2 — cash flow analysis EXAM

YOUR TURN

Time allowed 90 minutes · Total 100 marks · four questions, all compulsory

Question 1 (55 marks)

The comparative statements of financial position of Gallo Spa follow.

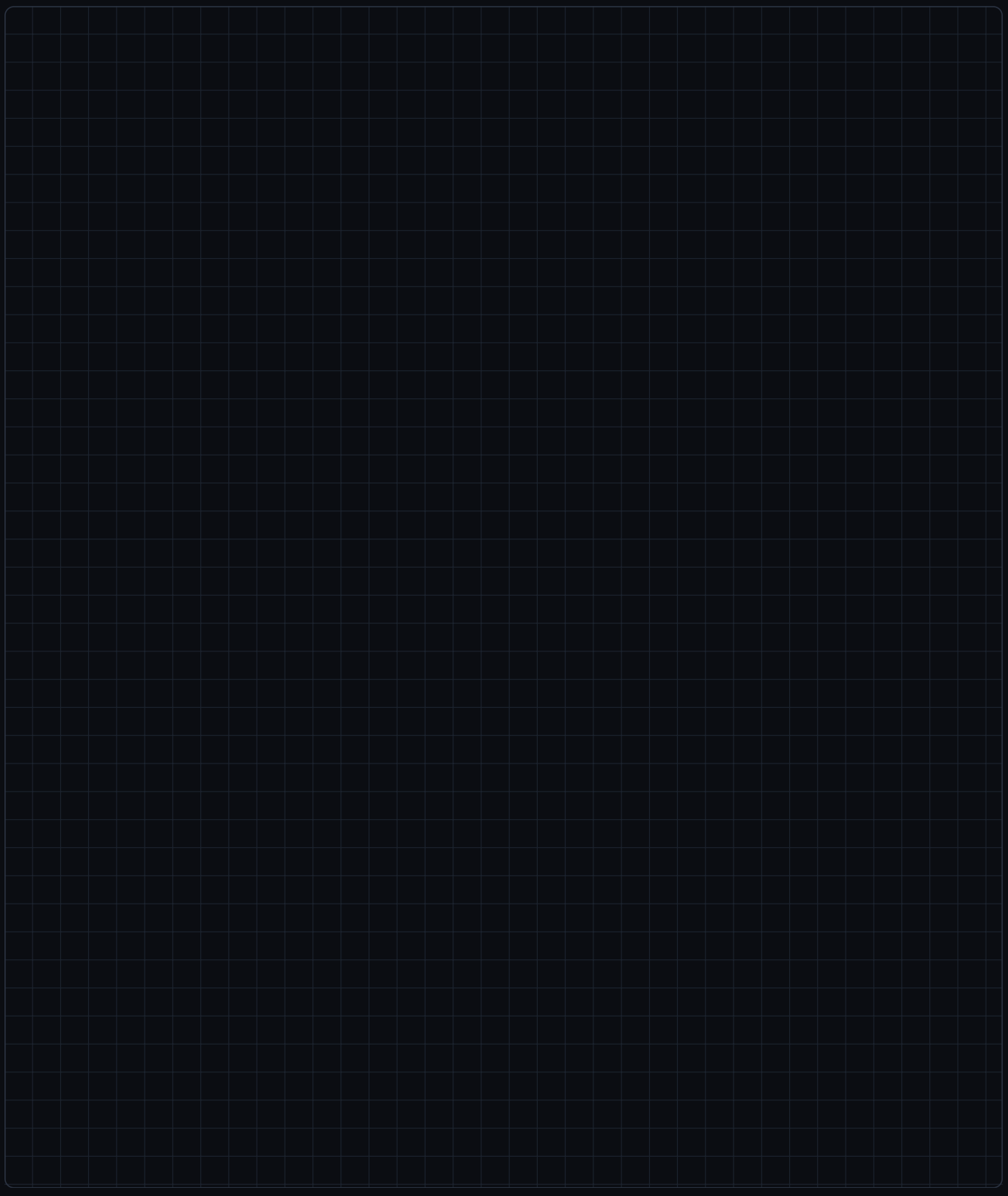
	2025	2024	Change	Classification
Assets				
Cash	74,000	58,000		
Accounts receivable	136,000	118,000		
Inventory	162,000	185,000		
Prepaid expenses	12,000	9,000		
Equipment	640,000	560,000		
Accumulated depreciation	(214,000)	(188,000)		
Total assets	**810,000**	**742,000**		
Equity and liabilities				
Accounts payable	118,000	134,000		
Accrued liabilities	26,000	21,000		
Long-term note payable	160,000	120,000		
Share capital	340,000	310,000		
Retained earnings	166,000	157,000		
Total equity and liabilities	**810,000**	**742,000**		

✎ EXTRA SPACE

Additional information. Net income for 2025 was €69,000. Equipment costing €48,000 with accumulated depreciation of €31,000 was sold for €22,000. Equipment costing €40,000 was acquired by issuing a long-term note to the supplier; all other equipment purchases were for cash. Shares were issued for cash. Dividends were declared and paid.

> **Required**
>
> 1. Prepare the comparative statement with the change and the classification of each difference (columns above).
> 2. Prepare the statement of cash flows for 2025, indirect method, with the schedule of non-cash activities.

✎ GAIN, DEPRECIATION, PURCHASES, DIVIDENDS



Statement of cash flows — indirect method	€
Cash flows from operating activities	
Net cash from operating activities	
Cash flows from investing activities	
Net cash from investing activities	
Cash flows from financing activities	
Net cash from financing activities	
Net increase (decrease) in cash	
Cash at 1 January	
Cash at 31 December	
Non-cash investing and financing activities	

✎ EXTRA SPACE

Question 2 (15 marks)

Compute free cash flow and interpret it in three lines.

✎ WORKING

✎ COMMENT

Question 3 (20 marks)

Net income is €69,000 and net cash provided by operating activities is €112,000. Explain the difference item by item and state what each of the two figures measures.

Item	€
Net income	
Net cash provided by operating activities	

✎ WHAT EACH FIGURE MEASURES

Question 4 (10 marks)

State, with one line of reasoning each, where the following appear in a statement of cash flows.

Item	Where, and why
Interest paid	
Dividends paid	
Dividends received	
Proceeds from issuing bonds	
Conversion of bonds into ordinary shares	

✎ EXTRA SPACE

TYPE X.2

Credit and profitability papers

One set of statements, three families of ratios, one memorandum.

① Worked example → ② X3 → ③ X4

WORKED EXAMPLE · SOLVED

Credit and profitability on one set of statements — a solved mini paper

Corallo Srl. Assume all cost of sales items are purchased, marketable securities are financial, all current liabilities operating, tax 30 per cent; one balance sheet date.

Corallo Srl — income statement	€ 000	Balance sheet	€ 000
Sales	1,000	Cash	50
Cost of goods sold	600	Marketable securities	20
Selling and administrative	250	Accounts receivable	130
Operating income	**150**	Inventory	150
Interest expense	30	Fixed assets, net	650
Income before tax	**120**	**Total assets**	**1,000**
Income tax (30%)	36	Accounts payable	150
Net income	**84**	Income tax payable	50
		Bonds payable	300
		Equity	500

Q1 — Liquidity

Current ratio	350 ÷ 200	**1.75**
Acid test	(50 + 20 + 130) ÷ 200	**1.00**
Cash ratio	(50 + 20) ÷ 200	**0.35**
Days' sales in receivables	130 ÷ (1,000 ÷ 360)	**46.8**
Days' sales in inventory	150 ÷ (600 ÷ 360)	**90.0**
Days' purchases in payables	150 ÷ (600 ÷ 360)	**90.0**
Net trade cycle	46.8 + 90.0 − 90.0	**46.8 days**

Q2 — Solvency

Total debt ratio	500 ÷ 1,000	**50.0%**
Total debt to equity	500 ÷ 500	**1.00**
Long-term debt to equity	300 ÷ 500	**0.60**
Short-term to total debt	200 ÷ 500	**40.0%**
Times interest earned	150 ÷ 30	**5.0x**

Q3 — Returns, with every intermediate figure

NOPAT	150 × 0.70	**105**
NOA	(1,000 − 20) − 200	**780**
NFO / Equity / check	300 − 20 = 280; 280 + 500 = 780	✓
Operating profit margin	105 ÷ 1,000	**10.5%**
Operating asset turnover	1,000 ÷ 780	**1.28x**
RNOA	105 ÷ 780	**13.46%**
NFE / NFR	30 × 0.70 = 21; 21 ÷ 280	**7.50%**
LEV / Spread	280 ÷ 500 = 0.56; 13.46% − 7.50%	**5.96%**
ROCE	13.46% + 0.56 × 5.96% = 84 ÷ 500	**16.80%** ✓

Comment

Leverage starts to destroy value when RNOA falls below the 7.5 per cent net financial rate. Model memo line: *"Liquid enough (acid test 1.0), moderately geared (debt 50 per cent of assets, coverage 5 times) and profitably levered (spread +6 points): approve, with a covenant on times interest earned."*

X3 Paper 3 — credit analysis EXAM

YOUR TURN

Time allowed 90 minutes · Total 100 marks · four questions, all compulsory

You are a lending officer at a commercial bank. Two companies have applied for a three-year term loan. Their statements at 31 December 2025 are reproduced below, in euro.

Income statement	Adria	Tirrena
Sales	2,400,000	1,800,000
Cost of goods sold	1,560,000	1,296,000
Selling and administrative expense	480,000	324,000
Operating income	360,000	180,000
Interest expense	72,000	90,000
Income before taxes	288,000	90,000
Income tax expense	86,400	27,000
Net income	**201,600**	**63,000**

✎ EXTRA SPACE

Balance sheet	Adria	Tirrena
Cash	90,000	36,000
Marketable securities	30,000	0
Accounts receivable, net	300,000	252,000
Inventory	420,000	324,000
Total current assets	840,000	612,000
Fixed assets, net	1,560,000	1,188,000
Total assets	**2,400,000**	**1,800,000**
Accounts payable	360,000	396,000
Income taxes payable	60,000	24,000
Total current liabilities	420,000	420,000
Bonds payable	780,000	720,000
Total liabilities	1,200,000	1,140,000
Total equity	1,200,000	660,000
Total equity and liabilities	**2,400,000**	**1,800,000**

Question 1 (35 marks)

Compute the following liquidity ratios for both companies: current ratio, acid test ratio, cash ratio, days' sales in accounts receivable, days' sales in inventory, days' purchases in accounts payable. Assume all cost of sales items are purchased.

Ratio	Adria	Tirrena
Current ratio		
Acid test ratio		
Cash ratio		
Days' sales in accounts receivable		
Days' sales in inventory		
Days' purchases in accounts payable		

✎ WORKING

Question 2 (30 marks)

Compute the total debt ratio, total debt to equity, long-term debt to equity, short-term debt to total debt and times interest earned for both companies.

Ratio	Adria	Tirrena
Total debt ratio		
Total debt to equity		
Long-term debt to equity		
Short-term debt to total debt		
Times interest earned		

Question 3 (25 marks)

Write the credit memorandum: no more than twelve lines, stating which application you would approve, on what terms, and the two pieces of information you would require first.

✎ CREDIT MEMORANDUM

Question 4 (10 marks)

State whether each is true or false, with one line of reasoning.

1. A current ratio of 4:1 is unambiguously better than one of 2:1. T F

2. Paying a supplier early raises a current ratio that is already above 1. T F

3. Interest paid is classified as a financing outflow. T F

4. Times interest earned treats interest as the only fixed charge. T F

5. A negative net trade cycle means the company is insolvent. T F

X4 Paper 4 — profitability and return on invested capital EXAM

YOUR TURN

Time allowed 90 minutes · Total 100 marks · four questions, all compulsory

Adria Spa, whose 2025 statements appear in Paper 3, reported the following at 31 December 2024: total assets €2,160,000, marketable securities €30,000, current liabilities €360,000, bonds payable €780,000 and total equity €1,020,000. Assume a tax rate of 30 per cent, and that all assets except marketable securities and all current liabilities are operating.

Question 1 (20 marks)

Compute the net operating assets and the net financial obligations at both dates, and show that NOA equals NFO plus equity.

	31 Dec 2025	31 Dec 2024
Operating assets		
− Operating liabilities		
= NOA		
Bonds − marketable securities = NFO		
Equity		
Check: NFO + Equity		

✎ EXTRA SPACE

Question 2 (30 marks)

Compute Adria's profit margin, asset turnover and return on assets for 2025, and verify that the third is the product of the first two.

Measure (name the substitution)	Working	Result
Margin		
Turnover		
Return		
Check: margin × turnover		

Question 3 (35 marks)

Compute the return on ordinary shareholders' equity for 2025 and disaggregate it as ROCE = RNOA + (LEV × Spread), showing every intermediate figure.

Step	Working	Result
NOPAT		
Average NOA		
RNOA		
NFE		
Average NFO		
NFR		
LEV		
Spread		
ROCE = RNOA + LEV × Spread		
Check: Net income ÷ Average equity		

Question 4 (15 marks)

Comment in no more than eight lines on what the disaggregation shows, and state the level of RNOA at which the company's leverage would begin to destroy value for shareholders.

✎ COMMENT

TYPE X.3

The full-length paper and the short answers

Method, arithmetic, meaning — in three lines.

① Worked example → ② X5 → ③ X6

WORKED EXAMPLE · SOLVED

Short answers — one line of method, one of arithmetic, one of meaning

Five short questions of the kind in Paper 6, each solved the way the marker wants to read it.

1. Opening accumulated depreciation €300,000, closing €330,000; disposal of an asset with accumulated depreciation of €45,000. Depreciation expense?

Depreciation	Closing − Opening + Disposals = 330,000 − 300,000 + 45,000	**75,000**

2. Inventory turnover 6.0, COGS €900,000, opening inventory €140,000. Closing inventory and days' sales in inventory?

Average inventory	900,000 ÷ 6.0	**150,000**
Closing inventory	2 × 150,000 − 140,000	**160,000**
Days' sales in inventory	160,000 ÷ (900,000 ÷ 360 = 2,500)	**64 days**

3. A 5-year bond, face €1,000, 5% annual coupon, priced at the end of its third year at a required rate of 7%?

Price, n = 2	$50 \div 1.07 + 1{,}050 \div 1.07^2 = 46.73 + 917.11$	**€963.84**

Below par, as it must be: coupon 5 per cent under the 7 per cent required.

4. RNOA 12.0%, NFR 5.0%, LEV 0.50. ROCE?

ROCE	12.0% + 0.50 × (12.0% − 5.0%)	**15.5%**

Positive spread: leverage adds 3.5 points to the shareholders' return.

5. Why is interest paid an operating cash flow in this course?

Interest expense is deducted in arriving at net income, so the operating section — which starts from net income — already contains it; dividends paid are not an expense but a distribution to owners, hence financing.

X5 Paper 5 — comprehensive case EXAM

YOUR TURN

Time allowed 180 minutes · Total 200 marks · six questions, all compulsory — the full-length paper

Sirio Spa manufactures industrial components. Its statements for 2025 and 2024 follow, in euro. The tax rate is 30 per cent.

Income statement	2025	2024
Net sales	3,600,000	3,000,000
Cost of goods sold	2,232,000	1,800,000
Gross profit	1,368,000	1,200,000
Selling expenses	540,000	420,000
Administrative expenses	288,000	270,000
Depreciation expense	180,000	150,000
Operating profit	360,000	360,000
Interest expense	84,000	90,000
Income before tax	276,000	270,000
Income tax expense	82,800	81,000
Net income	**193,200**	**189,000**

✎ EXTRA SPACE

Balance sheet at 31 December	2025	2024
Cash	120,000	180,000
Marketable securities	60,000	60,000
Accounts receivable, net	540,000	420,000
Inventory	620,000	480,000
Prepaid expenses	40,000	30,000
Total current assets	1,380,000	1,170,000
Property, plant and equipment	2,400,000	2,050,000
Accumulated depreciation	(780,000)	(640,000)
Total assets	**3,000,000**	**2,580,000**
Accounts payable	480,000	390,000
Accrued liabilities	110,000	95,000
Income taxes payable	70,000	55,000
Total current liabilities	660,000	540,000
Bonds payable	840,000	900,000
Total liabilities	1,500,000	1,440,000
Share capital	1,100,000	800,000
Retained earnings	400,000	340,000
Total equity	1,500,000	1,140,000
Total equity and liabilities	**3,000,000**	**2,580,000**

Additional information for 2025. Equipment that had cost €150,000, with accumulated depreciation of €40,000, was sold for €125,000. All other movements in property, plant and equipment were cash purchases. Share capital was issued for cash. Bonds were repaid in cash. Dividends were declared and paid. Marketable securities did not change. All sales are on credit and all cost of sales items are purchased.

✎ EXTRA SPACE

Question 1 (30 marks)

Prepare a horizontal analysis of the income statement for 2025, base 2024, and a common-size income statement for both years.

	Change €	Change %	% 2025	% 2024
Net sales				
Cost of goods sold				
Gross profit				
Selling expenses				
Administrative expenses				
Depreciation expense				
Operating profit				
Interest expense				
Income before tax				
Income tax expense				
Net income				

✎ EXTRA SPACE

Question 2 (45 marks)

Prepare the change-and-classification table and the statement of cash flows for 2025, indirect method. Then compute free cash flow.

✎ CHANGE AND CLASSIFICATION

Account	Change	Classification
Cash		
Marketable securities		
Accounts receivable		
Inventory		
Prepaid expenses		
Property, plant and equipment		
Accumulated depreciation		
Accounts payable		
Accrued liabilities		
Income taxes payable		
Bonds payable		
Share capital		
Retained earnings		

✎ GAIN, PURCHASES, DIVIDENDS

Statement of cash flows — indirect method	€
Cash flows from operating activities	
Net cash from operating activities	
Cash flows from investing activities	
Net cash from investing activities	
Cash flows from financing activities	
Net cash from financing activities	
Net increase (decrease) in cash	
Cash at 1 January	
Cash at 31 December	

✎ FREE CASH FLOW

Question 3 (35 marks)

Compute, for both years: current ratio, acid test ratio, cash ratio, days' sales in accounts receivable, days' sales in inventory, days' purchases in accounts payable and the net trade cycle.

Ratio	2025	2024
Current ratio		
Acid test ratio		
Cash ratio		
Days' sales in accounts receivable		
Days' sales in inventory		
Days' purchases in accounts payable		
Net trade cycle		

✎ WORKING

Question 4 (25 marks)

Compute, for both years: total debt ratio, total debt to equity, long-term debt to equity, short-term debt to total debt and times interest earned.

Ratio	2025	2024
Total debt ratio		
Total debt to equity		
Long-term debt to equity		
Short-term debt to total debt		
Times interest earned		

✎ EXTRA SPACE

Question 5 (45 marks)

For 2025, compute NOPAT, net operating assets at both dates, the operating profit margin, the operating asset turnover, RNOA and ROCE, all on average balances, and disaggregate ROCE as RNOA + (LEV × Spread).

Step	Working	Result
NOPAT		
NOA 31 Dec 2025		
NOA 31 Dec 2024		
Average NOA		
Operating profit margin		
Operating asset turnover		
RNOA		
NFE		
Average NFO		
NFR		
Average equity / LEV		
Spread		
ROCE = RNOA + LEV × Spread		
Check: Net income ÷ Average equity		

Question 6 (20 marks)

Write the conclusion of the analysis: fifteen lines at most, covering profitability, liquidity and solvency, and ending with the single question you would put to management.

✎ CONCLUSION

✎ CONTINUED

X6 Paper 6 — short answers across the course EXAM

YOUR TURN

Time allowed 90 minutes · Total 100 marks · ten short questions, all compulsory — ten marks each

1. A company reports net sales of €1,800,000, cost of goods sold of €1,170,000, operating expenses of €414,000, interest expense of €36,000 and a tax rate of 30 per cent. Compute operating profit, net income and NOPAT.

✎ WORKING

2. Opening accumulated depreciation is €410,000 and closing is €388,000. During the year an asset with accumulated depreciation of €96,000 was disposed of. Compute depreciation expense.

✎ WORKING

3. Inventory turnover is 4.5 times and cost of goods sold is €1,350,000. Opening inventory was €280,000. Compute closing inventory and days' sales in inventory on the closing balance.

✎ WORKING

4. Sales rose 25 per cent to €2,250,000 while cost of goods sold rose 32 per cent from €1,200,000. Compute gross profit for both years and the percentage change.

✎ WORKING

5. A ten-year bond with a face value of €1,000 and a 6 per cent annual coupon is priced at the end of its seventh year when the required rate is 9 per cent. Compute the price.

✎ WORKING

6. Total assets are €5,000,000, total liabilities €3,000,000, current liabilities €1,200,000 and marketable securities €200,000. Compute net operating assets and net financial obligations, assuming marketable securities are financial and all current liabilities are operating.

✎ WORKING

7. RNOA is 9.0 per cent, the net financial rate is 11.0 per cent and leverage is 0.80. Compute ROCE and comment in one line.

✎ WORKING

✎ COMMENT

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8. A company collects €50,000 from a customer, prepays €12,000 of rent and declares a €20,000 dividend. Current assets were €400,000, of which inventory €80,000, and current liabilities €200,000. Compute the current ratio and the acid test ratio after all three events.

✎ EXTRA SPACE

✎ WORKING

9. Explain in three lines why a gain on the disposal of a fixed asset is deducted in the operating section of the cash flow statement.

✎ ANSWER

10. State the three sources of accounting distortions named in chapter 2, and give one example of each.

✎ ANSWER